



Port of Oakland 2015 SEAPORT AIR EMISSIONS INVENTORY Final Report

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GLOSSARY

Adjustment factors: Used to adjust emissions or engine load or other situations for non-standard conditions.

ARB: California Air Resources Board, the state of California's regulatory agency for air pollution.

Assist mode: Period when a tugboat is engaged in assisting a ship to/from the harbor and to/from its berth.

Auxiliary engine: Used to drive on-board electrical generators to provide electric power or to operate equipment on board the vessel.

Auxiliary power: Electric power generated via the auxiliary engines or supplied by shore power and used for non-propulsion equipment.

Barge: A flat-bottomed craft built mainly for water transport of heavy goods and, in this report, dredged material. Most barges are not self-propelled and need to be moved by tugboats towing or towboats pushing them.

Buoy: Sea Buoy, North ('November'), South ('Sierra'), and West ('Whiskey') used to designate shipping lanes to enter the San Francisco Bay.

Bollard pull class: A power measure of the tug's capacity to push or pull ships.

Brake-specific fuel consumption (BSFC): This is the measure of the engines efficiency in terms of the fuel consumption rate (weight of fuel burned per hour) divided by the engine load or output (e.g. kilowatts). For marine engines, a different term, standard fuel oil consumption (SFOC), is sometimes used to describe the identical efficiency measure.

Cargo handling equipment: Equipment used to transfer cargo or containers. Cargo handling equipment is used to move containers from one mode of transportation to another (e.g. from a storage area to a truck chassis) or to reposition containers within a marine terminal or rail yard. Typical cargo handling equipment at the Port of Oakland include yard trucks, RTG cranes, top and side picks, forklifts, and other general industrial equipment.

Clamshell dredge: Equipment used to scoop, lift, and release sediment from berths and channels. It hangs from an onboard crane, or is carried by a hydraulic arm, or is mounted like on a dragline.

CH₄: methane. It is a hydrocarbon species that has a global warming potential.

CO₂: carbon dioxide

CO₂e: greenhouse gas carbon dioxide equivalent is a metric used to estimate all of the emissions from various greenhouse gases based upon their global warming potential relative to carbon dioxide.

CO: carbon monoxide.

Cruise modes: The vessel operation while traveling in the open ocean or in an area without speed restrictions.

Dead weight tonnage (DWT): Dead Weight Tonnage (DWT) is the weight of the ship, all her stores and fuel, pumps and boilers, crews quarters with crew and the cargo. In other words, the amount of water the vessel displaces when loaded.

Deep draft marine vessel: Deep draft vessels are larger vessels typically with draft in excess of 14 feet measured at the highest waterline and the bottom of the vessel. Other works describe this type of vessel as only Ocean-Going Vessels (OGV), but deep draft is used in this report to distinguish and avoid confusion between these larger vessels and smaller ocean-going tugs, supply vessels, and fishing vessels that could also be considered “ocean-going vessels”.

DPF: Diesel particulate filters or traps used to filter particulate matter from engine exhaust.

DPM: Diesel particulate matter

Drayage Truck: An on-road truck used to transport marine and rail intermodal freight (primarily shipping containers) to and from terminals.

Dredging: An excavation activity or operation carried out underwater typically for the purpose of the removal of accreted materials or sediments from the bottom of channels and berths to allow vessels with deep drafts.

Emission estimation: Method by which the quantity of a particular pollutant emission is estimated.

Emission factor: The average emission rate of a given pollutant for a given source, relative to a unit of activity. For example, grams per kilowatt of actual power or grams per hour of engine operation.

Emissions inventory: A listing of all the pollutant emissions included in the study.

g/kW-hr: This is the unit for reporting emission or fuel consumption factors, and means the grams per kilowatt-hour of work performed. Work and energy are used synonymously in this context.

GHG: Greenhouse gases includes CO₂, methane (CH₄), and nitrous oxide (N₂O)

HC: hydrocarbon emissions

Hotelling: On-board activities while a ship is in port and at its berth with similar electrical and other demands when anchored nearby.

Hydrolyze: To add water to a chemical compound.

Hydrated sulfuric acid: sulfuric acid to which water had been added.

Installed power: The engine power available on the vessel. The term most often refers only to the propulsion power available on the vessel, but could incorporate auxiliary engine power as well.

Intermodal site: Terminal or site where cargo is transferred from one form of transportation to another, for example between trucks and an ocean-going vessel or a railroad car.

Knot: A nautical unit of speed meaning one nautical mile per hour and is equal to about 1.15 statute miles per hour.

Lift: Movement of a shipping container (box) on or off a vessel, truck, or rail car.

Link: A defined portion of a vessel's, train's, or truck's travel. For example a link was established extending from the November Buoy to the location where the pilot boards the vessel. A series of links defines all of the movements within a defined area or a trip.

Load: The actual power output of the vessel's engines or generator. The load is typically the rated maximum power of the engine multiplied by the load factor if not measured directly.

Load factor: Average engine load expressed as a fraction or percentage of rated power.

MAQIP: Port's Maritime Air Quality Improvement Plan¹

Maximum power: A power rating usually provided by the engine manufacturer that states the maximum continuous power available for an engine.

Medium speed engine: A 4-stroke engine used for auxiliary power and rarely, for propulsion. Medium speed engines typically have rated speeds of greater than 250 revolutions per minute.

Mode: Defines a specific set of activities, for example, a tug's transit mode includes travel time to/from a port berth while escorting a vessel.

NO_x: nitrogen oxides. Includes all types of nitrogen oxide compounds.

N₂O: nitrous oxide. A nitrogen oxide that has a global warming potential.

Ocean-going vessels (OGV): Vessels equipped for travel across the open oceans. These do not include the vessels used exclusively in the harbor, which are covered in this report under commercial harbor craft. In this report OGV are restricted to the deep draft vessels that carry containers.

Off-Road activity: Activity that occurs off of established roadways. Activity within a marine terminal yard is considered off-road activity.

On-road activity: Activity that occurs on established roadways.

Operation mode: the current mode of operation for a ship – cruise, reduced speed zone (RSZ), maneuver, or berth.

Sea (Pilot) Buoy: used to mark a maritime administrative area to allow boats and ships to navigate safely where the Bay pilot boards and disembarks the ship. This location is 10 nautical miles from the Golden Gate and more than 15 nautical miles from the Port.

PM₁₀: particulate matter emissions less than 10 micrometers in diameter.

PM_{2.5}: particulate matter emissions less than 2.5 micrometers in diameter, and a subset of PM₁₀.

Port of Call (POC): A specified port where a ship docks.

Port berth: A location in a port or harbor used specifically for mooring vessels.

Propulsion engine: Shipboard engine used to propel the ship.

Propulsion power demand: Power used to drive the propeller and the ship.

¹ <http://www.portofoakland.com/community/environmental-stewardship/maritime-air-quality-improvement-plan/>

Rated power: A guideline set by the manufacturer as a maximum power that the engine can produce continuously.

Reefer plug: Plug allowing a refrigerator container to plug into an outlet connected to the ship's power generation.

ROG: reactive organic gas; all hydrocarbon compounds that can assist in producing ozone (smog). Includes HC plus aldehyde and alcohol compounds minus methane.

Roll-on/roll-off (RORO) vessels: Ships designed to carry wheeled cargo such as automobiles, trailers, or railway carriages that drive or are pulled onto the vessels.

RSZ: Reduced speed zone.

RTG Crane: Rubber tired gantry (RTG) crane is sometimes called a straddle crane because the crane 'straddles' a row of containers stored in the terminal yard as it drives up and down the row selecting and repositioning containers or loading them onto truck chassis.

Shoaling: Shoaling is term used in this report to describe subsidence of the shore or other filling of the navigation channel near shore.

Shore Power: Electric power supplied to ships while at berth in place of power generated by the ships' on-board auxiliary diesel engines.

SO₂: Sulfur dioxide.

SO_x: Oxides of sulfur. Interchangeable term with sulfur dioxide but include some other minor forms of sulfur oxides.

Spatial allocation: Areas on a map allocating a specific set of activities.

Spatial scope: A specified area on a map that defines the area covered in study.

Slow speed engine: Typically a 2-stroke engine or an engine that run below 250 rpms.

Standard fuel oil consumption (SFOC): See brake specific fuel consumption (BSFC).

Steam boiler: Boiler used to create steam or hot water using external combustion.

Steam turbines: A mechanical device that extracts thermal energy from pressurized steam, and converts it into useful mechanical work.

Survey boat: A small marine vessel used during dredging to survey the berth and channel depths.

Tender: a utility vessel used to service another type of vessel, for example, to service a clamshell dredge.

TEU: Twenty foot equivalent unit. A 20 foot long container = 1 TEU and a 40 foot long container = 2 TEUs.

Time in mode: The amount of time a vessel remains in a specified mode, for example the amount of time a ship spends in the reduced speed zone.

Tons: Represents short tons (2,000 lbs) unless otherwise noted.

Tonnes: Metric tons (1,000 kg)

Transit mode: The time a tug spends traveling to/from its berth to the pick-up location.

Tug class: A tugboat's Bollard pull class designation.

Two-stroke engine: Engine designed so that it completes the four processes of internal combustion (intake, compression, power, exhaust) in only two strokes of the piston.

EXECUTIVE SUMMARY

The Port of Oakland (the Port) oversees the Oakland seaport and Oakland International Airport. The Port's jurisdiction includes 20 miles of waterfront from the Bay Bridge through Oakland International Airport. The Oakland seaport is the seventh busiest container port in the U.S.; Oakland International Airport is the second largest airport in the San Francisco Bay Area offering over 300 daily passenger and cargo flights; and the Port's real estate includes commercial developments such as Jack London Square and hundreds of acres of public parks and conservation areas. The Port was established in 1927 and is an independent department of the City of Oakland.

The Port of Oakland 2015 Seaport Air Emissions Inventory identifies and quantifies air emissions from the Port's maritime activities, organized by the major source categories:

- Deep-Draft Ocean-Going Vessels (OGV);
- Commercial Harbor Craft (dredging and assist tugs);
- Cargo Handling Equipment (CHE);
- Trucks (container movements);
- Locomotives; and
- Other Off-road Equipment.

The Port voluntarily chose to prepare an air emissions inventory for its seaport operations for calendar year 2005 and decided to periodically update the Port's activity and emission estimates in the subsequent years. The Port's 2005 emissions inventory (ENVIRON 2008) and the 2012 emissions inventory (ENVIRON 2013) are both available on the Port's website². This calendar year 2015 emissions inventory highlights the Port's commitment to meet its goal of reducing emissions and to improve understanding of the nature, location and magnitude of emissions from its maritime-related operations. The Port is committed to conducting its operations in the most sustainable and environmentally sensitive manner possible.

The purpose of this inventory is to better understand the emissions from typical Port activities so the Port can better address its impact on air quality. The inventory:

- Updates changes in Port activity and emissions for the 2015 calendar year;
- Continues to evaluate air pollution control regulations as they are phased in;
- Informs local, state and federal regulatory decision-makers in their efforts to establish regulations and initiatives to reduce air emissions from Port-related sources and improve air quality;
- Provides air quality background information to be used in future environmental documents; and

² <http://www.portofoakland.com/community/environmental-stewardship/seaport-air-emissions-inventory-2005/>
<http://www.portofoakland.com/community/environmental-stewardship/maritime-air-quality-improvement-plan/>

- Provides metrics for emissions reductions as the Port implements projects and programs within its Maritime Air Quality Improvement Plan (MAQIP).

An emissions inventory is best understood as an estimate of the quantity of pollutants that a group of sources produce in a given area, over a prescribed period of time. Emissions inventories should be used with care and in conjunction with other information and tools to evaluate and assess air quality problems.

The inventory provides estimates for emissions of five “criteria” air pollutants, reported as tons per year. The pollutants are:

- Reactive organic gases (ROG);
- Carbon monoxide (CO);
- Nitrogen oxides (NO_x);
- Particulate matter (including diesel) (PM); and
- Sulfur oxides (SO_x).

Particulate matter emissions estimated in this report are primarily diesel particulate matter (DPM). DPM has been designated a toxic air contaminant by the California Air Resources Board (ARB). A fraction of particulate matter emissions come from boilers and LPG-powered engines, and thus are not classified as DPM. Total particulate is divided into two size ranges: PM₁₀ (particles with aerodynamic diameter 10 microns or less) and PM_{2.5} (particles with aerodynamic diameter 2.5 microns or less).

In addition, three greenhouse gas (GHG) components (carbon dioxide [CO₂], methane [CH₄], and nitrous oxide [N₂O]) were estimated. These components were combined in a CO₂ equivalent (CO₂e) estimate using the relative global warming potential of each component.

Introduction, Scope and Coordination

This is an inventory of the air emissions generated by maritime activities conducted by the Port of Oakland tenants and, to a lesser extent, by Port construction activity in the seaport. On the water side, the spatial domain of the inventory includes Port-related marine vessel transit to and from dockside out through the Golden Gate Bridge, to the first outer buoys beyond the Sea Buoy, approximately 30 miles away from the Port. On the landside, the spatial scope of the inventory includes five marine terminals, one rail yard, and the road traffic between those facilities and the nearest freeway interchanges. The Port area was defined approximately by the boundaries of I-80, I-880, and Howard Terminal (Berths 67 and 68) adjacent to Jack London Square. Within this defined geographic domain, operations within three areas were specifically excluded: the privately-owned Schnitzer Steel terminal and Union Pacific rail yard, and the former Oakland Army Base located between Maritime Street and I-880, where redevelopment has yet to include permanent operations. However, construction activity on Port-owned property at the former Oakland Army Base was included in the inventory. Figures 1-1, 2-1 and

2-2 in the body of the report illustrate the spatial scope of the inventory. Other areas were not controlled or operated by the Port of Oakland in 2005, and were therefore not included in the emissions inventory for that year. As this 2015 inventory update is designed in part to examine the trends of air pollutant emissions over time, the same domain and area exclusions were maintained to allow a clear understanding of the changes in emissions over time. Nevertheless, new estimation methods or other factors may affect comparisons with the prior year inventory in some cases. We have noted such changes where appropriate throughout this report.

With the exception of rare roll-on and roll-off activity, the Port of Oakland operated exclusively as a container port in 2015. All of the 1,393 ship calls were by deep-draft vessels designed as container ships. On the land side, Port terminals operated as a collection of intermodal sites where cargo handling equipment transferred containers to and from vessels to truck or rail transportation.

Ramboll Environ prepared the baseline 2005 seaport emissions inventory, and updated the inventory for the 2012 calendar year. This is the 2015 emissions inventory update for the Port. We assembled the emissions inventory by analyzing the time-in-mode, load or speed, and engine characteristics of the marine vessels and other equipment used to transport container cargo. Assigning emissions by time-in-mode allows for emissions to be defined by location. Data from previous studies, literature reviews, and ARB input data or models were used when port-specific estimates of activity (hours per year), load factors, or equipment model year from survey data were not available.

Technical Approach to Major Source Categories

Emissions were estimated for the five source categories as described below; a summary of the emission results are presented in Table ES-1.

Ocean-going Marine Vessels (OGVs). Ocean-going vessel emissions were estimated in each operating mode: cruising in the open ocean, cruising in the reduced speed zone (RSZ) inside the Bay, maneuvering (low speed operation between the Bay Bridge and Port berths in the Inner or Outer Harbors), and hotelling (vessels at berth being loaded or offloaded and at anchor in the Bay). Separate mode estimates are important for distinguishing the location of emissions, especially the proximity to on-shore areas like West Oakland, which is the neighborhood east and northeast of the seaport. OGV Emissions sources included the vessels' main propulsion engines, auxiliary engines, and small auxiliary boilers. Except for the boilers, air emission sources from OGVs are diesel engines.

Harbor Craft. Smaller marine vessels are included in a category described as "Commercial Harbor Craft". Vessels in this category are associated with Port maritime operations and consist primarily of assist and tow tugs and a few smaller boats that support maintenance dredging. On average about two tugs assist large vessels during the maneuvering mode as they enter and leave the Port. While six tug companies provide assist services, one is a subsidiary of another, and one pair of firms has a joint operating agreement. Information from several data

sources was used to characterize the tug fleets and installed equipment. The inventory includes tug emissions estimates in two operating modes: 1) vessel assist, and 2) transit to and from the tug's home location and the vessel assist point located near the port. Emissions sources include tug main propulsion and auxiliary diesel engines.

The inventory also addresses emissions from operation and maintenance dredging, which occurs annually to maintain safe depths in Federal channels and at Port berths. Reported maintenance dredging activity was much higher in 2015 and 2012 as compared to 2005 because maintenance dredging activity was incorporated in the channel and berth deepening project in 2005, which used zero emission equipment and disposed of the dredged material near the port. Emissions were estimated from dredges, dredge tenders, survey boats, and tugs that push barges containing dredged material to disposal or reuse areas. Dredging equipment is typically powered by diesel engines, though in 2005 some maintenance dredged material was removed by an electric-powered dredge as part of the deepening project. Survey boats in support of dredging operations use outboard gasoline engines.

Cargo Handling Equipment. Ramboll Environ collected specific activity information about cargo handling equipment used at the Port in 2015 to move containers within maritime and rail yards. We determined annual emissions for each piece of equipment according to engine characteristics (model year, rated power, and equipment type) and equipment operation (hours of operation and fuel consumption rates). Yard trucks (sometimes called hostlers), side picks and top picks were the most prevalent types of equipment. Other equipment included rubber tired gantry cranes, forklifts, and tractors. Most of the equipment was powered by diesel engines and many units had been retrofitted with emissions control devices or repowered to comply with California ARB regulations. Some of the cargo handling units were fueled by liquid petroleum gas (LPG) or gasoline. Equipment that solely uses electric power was not included.

Drayage Trucking. Maritime operations create a demand for a significant number of truck trips, including short trips within the Port moving containers from marine terminals to other locations. Trucks that pick up and deliver cargo containers are known as drayage trucks, and are subject to ARB drayage truck regulations. Trucks arrive at the Port terminals primarily via freeway interchanges or rail yards, and leave through the same general exits. Even if trucks arrive via surface streets, the trips mostly pass through the same intersections that serve the primary freeway interchanges. The spatial scope of the truck emissions inventory was therefore defined to include truck routes from the marine terminals to each of three freeway interchanges and the two rail yards serving the Port. To ensure consistency with the 2005 and 2012 inventories, this inventory does not include emissions from Port trucks operating on freeways.

Ramboll Environ's general approach to estimate truck emissions was to determine resulting truck travel by estimating the number of truck trips to and from the marine terminals, the trip mileage to and from the terminals, and the average speed by link and in-terminal trip speed. To estimate the number of truck trips, the Port conducted an in-depth survey with the terminal

operators to determine the gate counts by truck configuration (tractor only or tractor with a trailer) at the entrance and exit to the terminals. We then estimated truck trips from truck gate count data and container lift data provided by the Port.

Emissions from trucks depend on the age distribution and emission control devices of the transport trucks as well as site-specific conditions. State regulations, Port incentives and a Port ban on non-compliant trucks have led truck owners to use new technology trucks or retrofit relatively new trucks to meet emission standards. As of 2014, all drayage trucks were required to have 2007 or later engines, with diesel particulate filters (DPFs) and NOx reduction. In contrast, the 2005 seaport emission inventory noted that Port trucks tended to be older than average in that calendar year. By 2012, all drayage trucks with engines that had model years of 2003 or earlier had been retrofitted with DPFs that reduced diesel emissions by at least 85%, and trucks had been upgraded to 2007 or later engines also used DPFs.

Ramboll Environ estimated emissions for four truck operating modes: idling at terminal queues, in-terminal idling, in-terminal driving, and over-the-road driving to and from the rail yard and freeway exits. We used the most recent version of ARB's EMFAC model to estimate emission rates for the various modes. In-terminal activities on a per-truck basis were estimated from results of survey data collected for previous years. However, longer than usual wait times associated with a slow down at the Port that was likely due to the reported labor/terminal operator dispute may have occurred at times in 2015, resulting in increased truck idling. While data on terminal wait times was unavailable during this period, a sensitivity analysis was conducted to evaluate the potential impact of longer idling times on truck emission estimates (see Appendix B).

Locomotives. The Oakland International Gateway (OIG) rail yard is a Port of Oakland terminal operated under a lease by Burlington Northern Santa Fe (BNSF) railway. BNSF uses the OIG as a near dock transfer point for the Port maritime traffic and only Port containers are handled at the yard. Locomotives and trains enter the general port area from the north via the Union Pacific (UP) lines, and leave in the same direction via tracks going north through Richmond and onto BNSF lines out of the Bay Area. The Union Pacific rail yard (UP Railport) that sits adjacent to the Port terminals serves as an intermodal yard for freight movements through the port, but it is not included in the Port's emissions inventory because it is independently operated and may handle non-Port cargo. UP has in the past provided the ARB with an independent analysis of emissions from its Oakland facility.

Because different locomotive and engine models have different emission characteristics, it was important to characterize the types and models of the locomotives that arrive/depart and are serviced at OIG. BNSF provided the locomotive fleet fractions of different locomotive types and models based on train arrival and departure records. One switching engine is usually assigned to the OIG yard at any one time; these switch locomotives typically all have very similar engine models.

Other Off-road Equipment. Off-road equipment included general industrial and construction equipment that are most often used for sporadic maintenance and construction activity occurring at the Port. They are not to be confused with cargo-handling equipment (CHE), which is primarily used to transfer shipping containers or intermodal freight cargo. Three sources of off-road equipment were considered: (1) facility maintenance and construction at each terminal, (2) Port of Oakland general maintenance, and (3) construction at the Oakland Army Base.

To estimate the annual 2015 off-road equipment emissions, a list of equipment including engine characteristics (model year, rated power, and equipment type) and equipment operation (hours of usage and fuel consumption rates) were collected from terminal operators and the Port. The terminal operators' equipment population and operation estimates by terminal were derived from surveys conducted by the Port of Oakland. Fleet data for the Port's general maintenance equipment and equipment used for Oakland Army Base construction were provided by the Port. Where there were missing data, default input estimates were obtained from applicable ARB inventory guidance documentation. A combination of OFFROAD 2007 and OFFROAD 2011 models were used to estimate emissions for diesel and other fuel types for all reported equipment.

Emissions Inventory Results

Results of this Port of Oakland 2015 Seaport Emissions Inventory update, the 2005 Inventory and the percent changes between these inventories are summarized in Table ES-1. Note that the 2005 inventory did not distinguish between PM_{10} and $PM_{2.5}$; only total PM and DPM emissions were reported. For emission sources found at the Port, total PM in the 2005 inventory can be considered equivalent to PM_{10} . Also shown in Table ES-1 are the emission estimates for 2012 showing recent progress in reducing emissions. The results in Table ES-1 are also shown graphically in Figures ES-1 and ES-2.

As shown in Table ES-1, the comparisons of 2015 with 2005 emissions show large reductions in emissions of all pollutants except ROG, mostly due to the use of more modern engines, retrofits and cleaner fuels. Notably, the DPM and SO_x emissions are substantially lower in 2015 for all source categories. Changes to emission factors for ROG resulted in increases in estimated OGV and harbor craft ROG emissions. A small increase in the OGV NO_x emissions between 2005 and 2015 is a result of relatively more OGVs with diesel engines and fewer steamship calls in 2015, which was only slightly offset by minor reductions of NO_x from incorporation of newer engines in the fleet and the use of cleaner fuels. Emission reductions have continued from 2012, and further reductions are expected in future years.

OGVs constitute the largest source category for all pollutants. In 2015, OGVs produced 82% of estimated particulate matter emissions and the major portion of other pollutants within the scope of this inventory which extends as far as 30 miles seaward of the Port. Harbor craft and cargo handling equipment together produced nearly 16% of the estimated Port-related DPM emissions in 2015. Trucks, locomotives from the OIG rail yard, and other off-road equipment

used for construction and maintenance work contributed a small fraction of the total emissions. The implementation of the Port's truck registry, under which only upgraded trucks complying with the Drayage Truck Regulation are allowed at the Port, has significantly reduced the DPM impact from this source category.

During most of 2015, there was a slow down at the Port likely due to the reported labor/terminal operator³ dispute increasing at-berth times and forcing some ships to anchor in the San Francisco Bay. After the end of the dispute, it may have required several months to reduce the backlog of ships, and resume normal operations. As a result, emissions calculated using 2015 activity data are higher than would otherwise be expected and are not fully representative of normal port operations. Most significantly impacted were ocean going vessels at berth (longer berthing times) and at anchor (more vessels at anchor for longer periods of time). See Appendix A, "Effect on Berthing, Shifts, and Anchorage Emissions Due to Unusual Activity Slowdown at the Port during 2015" for a comparison of 2015 average berthing emissions to those from ship calls during the latter portion of 2015 and in the first half of 2016 when Port operations were closer to normal. Truck idling times also increased during the dispute due to longer waiting times, although the waits were not routinely measured during 2015. See Appendix B, "Potential Effect on Truck Idling Emissions of Unusual Activity Slowdown at the Port during 2015" for a discussion of the effect of multiplying normal idling emissions by a factor of 10 to represent a worst case idling scenario.

³ CNN, "West Coast Port Goes Back to Work," March 15, 2015, <http://money.cnn.com/2015/03/12/news/economy/port-labor-dispute/>

Table ES-1. Port of Oakland 2015, 2012, and 2005 air emissions inventory comparisons (tons per year).

2015 Inventory^a	ROG	CO	NO_x	PM₁₀	DPM	SO_x
Ocean-going vessels	182	259	2,715	58.7	51.8	141.2
Harbor craft	23	97	166	6.6	6.2	0.1
CHE	43	253	332	3.9	3.7	0.6
Trucks	6.0	20	87 ^b	1.2	0.4	0.2
Locomotives	0.2	2	14	0.2	0.2	0.01
Other Off-road Equipment	1.0	12	11	0.6	0.6	0.01
Total	255	642	3,324	71	63	142
2012 Inventory	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	176	232	2,591	67.0	57.4	289
Harbor craft	25	95	235	9.3	9.3	0.2
CHE	35	207	413	8.0	7.9	0.6
Trucks	13	49	135	3.0	2.0	0.2
Locomotives	1.2	2	19	0.5	0.5	0.01
Other Off-road Equipment	0.6	4	4	0.3	0.3	0.004
Total	250	589	3,398	88	77	290
2005 Inventory	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	117	235	2,484	219.5	208.5	1,413
Harbor craft	22	83	345	13.4	13.4	2.85
CHE	53	408	766	21.7	21.2	7
Trucks	49	149	334	15.9	15.9	2.2
Locomotives	7	11	76	2.0	2.0	2
Other Off-road Equipment	N/A	N/A	N/A	N/A	N/A	N/A
Total	248	886	4,005	272	261	1,427
% Change from 2005	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	56% ^c	10%	9% ^d	-73%	-75%	-90%
Harbor craft	4% ^e	17% ^e	-52%	-51%	-53%	-95%
CHE	-19%	-38%	-57%	-82%	-82%	-92%
Trucks	-88%	-87%	-74%	-92%	-98%	-91%
Locomotives	-97%	-85%	-82%	-89%	-89%	-100%
Other Off-road Equipment	N/A	N/A	N/A	N/A	N/A	N/A
Total	3%	-28%	-17%	-74%	-76%	-90%

^a Where emissions are reported to only one significant digit, more precise figures can be found in the body of the report.

^b See Appendix B for a sensitivity analysis of potential impacts of longer truck idling times due to delays associated with operation disruptions during 2015 (see also Sec. 5.4).

^c OGV ROG increase due to change in emissions factor (see Sec. 8.2).

^d OGV NO_x increase due to lower fraction of calls by steamships in 2015 and increased berthing and anchorage time (see Sec. 8.2).

^e Harbor craft ROG and CO increase due to increased dredging activity included in inventory and a change in emissions factor since 2005 (see Sec. 8.3).

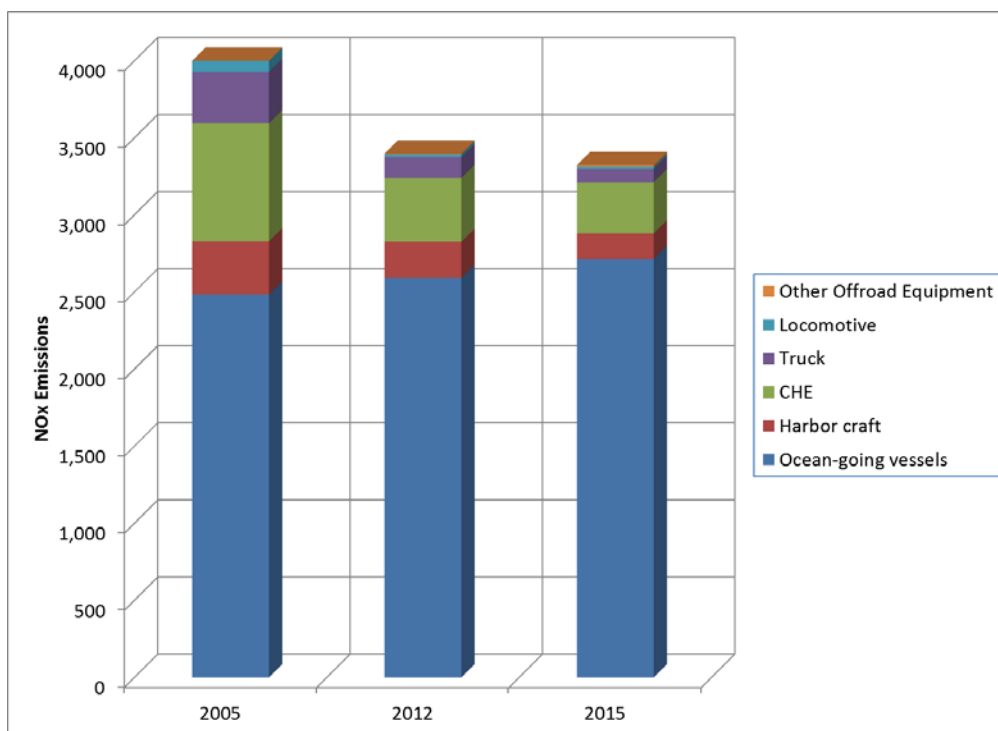


Figure ES-1. Port of Oakland 2015, 2012, and 2005 NOx air emissions inventory comparisons (tons per year; see explanatory notes in Table ES-1).

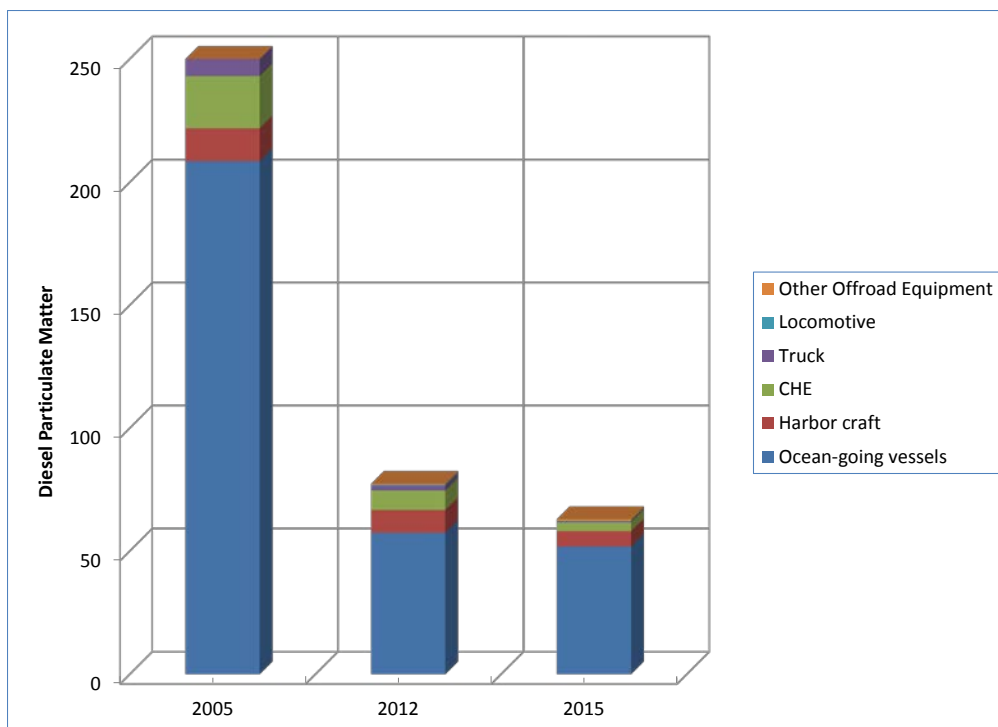


Figure ES-2. Port of Oakland 2015, 2012, and 2005 DPM air emissions inventory comparisons (tons per year; see explanatory notes in Table ES-1).

Table ES-2 shows a more detailed assessment of OGV emissions by mode of operation. In 2015, a reported labor/terminal operator dispute⁴ at the Port could have been responsible for delays found at the Port. This slowdown could have been responsible for longer berthing time (44 hours per call in 2015 compared with 21 hours for 2005 calls), more shifts between berths and anchorage (339 in 2015 compared with 19 in 2005), and more hotelling at anchorage (307 calls averaging 57 hours each in 2015 compared with 99 averaging 15 hours each in 2005). As a result, these activity modes were more significant emission sources in 2015 compared with previous years.

Shore power usage resulted in the avoidance of a significant amount of emissions in 2015 as shown in the last two rows of Table ES-2: there were 636 calls averaging 33.1 hours each on shore power representing about a 34% reduction in auxiliary engine operating hours at berth overall.

Table ES-2. OGV emissions summary by mode, using ARB-specified activity emission factors – tons in 2015.

2015 Inventory	ROG	CO	NO_x	PM₁₀	PM_{2.5}	DPM	SO_x	CO₂	CH₄	N₂O	CO₂e
OGV – Cruise	25.25	36.10	502.00	8.51	7.84	8.13	31.00	20,844	2.56	0.61	21,091
OGV – RSZ	32.06	44.52	509.79	9.27	8.52	8.66	32.95	22,421	2.62	0.63	22,673
OGV – Maneuver	61.39	52.45	416.07	8.93	8.24	8.57	10.18	14,971	1.79	0.38	15,129
OGV – Berth	41.31	86.26	890.16	22.61	21.05	18.05	48.72	81,357	7.59	1.74	82,067
OGV – Shifts	4.91	3.58	27.49	0.54	0.50	0.52	0.47	631	0.49	0.03	653
OGV – Anchorage	17.19	36.13	369.51	8.86	8.15	7.88	17.93	28,520	2.98	0.66	28,792
OGV Subtotal	182.11	259.04	2,715.02	58.72	54.29	51.81	141.24	168,745	18.03	4.06	170,405
Emissions avoided due to shore power (tons)	24.17	51.13	501.23	11.62	10.69	11.62	18.59	32,068	4.18	0.84	32,422
Berthing % reduction	36.9%	37.2%	36.0%	33.9%	33.7%	39.2%	27.6%	28.3%	35.5%	32.6%	28.3%

It is important to keep in mind that location of emissions is often as significant as the total quantity: the greater the distance between the emission source and the affected area, the lower the pollutant concentration and resulting exposures in the affected area. Thus, emissions generated close to community receptors will have a greater effect on human health risk on a per ton basis. In other words, the impact of the various source categories on West Oakland air quality is not directly proportional to the magnitude of their total emissions. For example, the particulate matter emissions from ocean-going vessels in cruising mode, which occurs outside the Golden Gate, will have less impact on sensitive receptors in Oakland on a per-ton basis than emissions that occur closer to shore during the maneuvering or hotelling modes according to an ARB assessment⁵ of the 2005 emissions. Diesel particulate matter emissions from trucks had

⁴ CNN, “West Coast Port Goes Back to Work,” March 15, 2015, <http://money.cnn.com/2015/03/12/news/economy/port-labor-dispute/>

⁵ ARB 2008. “West Oakland Study,” <https://www.arb.ca.gov/ch/communities/ra/westoakland/westoakland.htm>

the greatest impact on health risk from sensitive receptors in Oakland on a per ton basis due to the proximity of emissions to receptors, and in 2015, emissions from trucking were 98% lower than in 2005.

Each of the emission source categories in Table ES-1 should experience continued emission reductions as a result of the introduction of cleaner engines for all fleets and increased use of shore power for ocean-going vessels while at berth. Accelerated implementation of cleaner fleets has been required by ARB regulations for harbor craft, cargo handling equipment, and drayage truck fleets. Ocean-going vessels, locomotives and other off-road equipment will experience turnover through normal attrition. Shore power use for ocean-going vessels at berth is expected to nearly double by 2020 from the level used in 2015, thus reducing at-berth emissions.

The GHG emissions for each source category are shown in Table ES-3 compared with those estimates for 2012; GHG emissions were not estimated in 2005. Development of the GHG emission inventory followed the same approach used to estimate criteria pollutants and used the same spatial domain and activity data as for the criteria pollutant inventory. GHG emissions are expressed in units of carbon dioxide equivalent (CO₂e) global warming potential where methane (CH₄) has 25 times and nitrous oxide (N₂O) 298 times the global warming potential of CO₂. Increases in GHG noted for 2015 include those from OGVs, which were adversely affected by the Port slowdown during 2015, and a small amount from other off-road equipment, which reflects the level of maintenance and new construction at the Port.

Table ES-3. Summary GHG emission inventory by source category (tons).

2015 Inventory	CO₂	CH₄	N₂O	CO₂e
Ocean-going vessels ^a	168,745	18.03	4.06	170,405
Harbor craft	16,837	3.26	0.50	17,069
CHE	32,606	4.28	-	32,713
Trucks ^b	24,729	0.35	0.72	24,954
Locomotives	639	0.04	0.02	645
Other Off-road Equipment	1,155	0.01	0.12	1,191
Total	244,711	25.97	5.42	246,976
2012 Inventory	CO₂	CH₄	N₂O	CO₂e
Ocean-going vessels	133,005	14.17	3.32	134,332
Harbor craft	20,134	3.57	0.54	20,377
CHE	38,556	5.32	0.00	38,667
Trucks	27,942	0.74	0.78	28,198
Locomotives	926	0.06	0.02	935
Other Off-road Equipment	368	0.08	0.003	370
Total	220,930	23.94	4.67	222,880

^a Includes increases in emissions from extended berthing times and anchoring associated with unusual Port operating conditions (slowdown) during 2015.

^b See Appendix B for a sensitivity analysis of potential impacts of longer truck idling times due to delays associated with operation disruptions during 2015 (see also Sec. 5.4).

1.0 INTRODUCTION

1.1 Purpose and Background

The Port of Oakland (Port) has prepared this 2015 Seaport Air Emissions Inventory (emissions inventory) for the purpose of identifying and quantifying the air quality impacts from the maritime operations of the Port and its tenants. This emissions inventory updates the 2005 and 2012 Seaport Air Emissions Inventories (ENVIRON, 2008 and 2013) for the major categories of maritime equipment:

- Deep-Draft Ocean-Going Vessels (OGV);
- Harbor Craft (dredging and assist tugs);
- Cargo Handling Equipment (CHE);
- Trucks (container movements);
- Locomotives; and
- Other Off-Road Equipment.

The Port voluntarily chose to prepare the original and periodic inventory updates to help in air quality planning and to meet its commitment to develop and implement an emissions reduction program. Because annual emissions from operations vary over time due to changes in cargo volume, implementation of regulations, and other factors, this study was undertaken to provide an updated inventory for 2015 for comparison with the calendar year 2005 baseline inventory.

This emissions inventory highlights the Port's commitment to improve understanding of the nature, location and magnitude of emissions from its maritime-related operations. An emissions inventory is best understood as an estimate of the quantity of pollutants that a group of sources produce in a given area (or domain), over a prescribed period of time. Emissions inventories should be used in context with proper interpretation and in conjunction with other information and tools to evaluate and assess air quality problems.

1.2 Considerations When Using Emissions Inventories

Emissions inventories are used for multiple purposes: to analyze air quality, to develop pollutant control strategies or plans, and to track and communicate progress toward air quality goals. Emissions inventories are essential tools, but they have some inherent shortcomings that are often overlooked and lead to misconceptions about their use and value. The term "inventory" is something of a misnomer because it implies greater precision in "counting" emissions than is really the case. An emissions inventory is better understood as an estimate of the quantity of pollutants that a group of sources produce in a given area, over a prescribed period of time. The methods of making estimates are usually very technical in nature, a characteristic that makes the limitations of emissions inventories less transparent to the general public.

The accuracy of emissions estimates varies due to a number of factors. Even a well-conducted, detailed and carefully constructed inventory, such as this one, does not have access to direct emissions measurements from the specific, individual sources being studied. As a result, it is necessary to rely on surrogate information to characterize sources, describe source activities, and specify pollutant emission rates.

Emissions estimation methodologies are continuously in flux, changing and evolving over time as better and more accurate information becomes available. Historically, emissions inventory updates have revealed previously overlooked information about sources and source activity that has substantially changed overall emissions estimates. For example, because of new information made available, such as that provided in the 2005 Seaport Air Emission Inventory, the California Air Resources Board (ARB) updated the ocean-going vessel auxiliary boiler activity rates. As a result, emissions inventories conducted even a few years apart may not be directly comparable.

Another important consideration in interpreting emissions inventories is the somewhat counter-intuitive fact that there can be a poor correlation between the magnitude of emissions and an air quality impact. The importance of a given ton of emissions may differ from another ton because of the location at which it is emitted, because of the meteorological conditions that affect its dispersion, and in some cases because of the chemical reactions that occur in the atmosphere. Emissions inventories should be used with care and in conjunction with other information and tools to evaluate and assess air quality problems.

1.3 Important Features of the Port of Oakland Seaport Air Emissions Inventory

Some key features of the Port emissions inventory that should be kept in mind when reviewing this report are described below.

1.3.1 Scope

The inventory estimates emissions from the Port's tenants' and other maritime operations that occurred in the calendar year 2015 using the same geographic scope as the 2005 and 2012 inventories. It is not intended to represent emissions in other years, or emissions outside the geographic domains identified for each major source category, as described under "Technical Approach." Port tenants for which emissions were estimated include marine terminal operators and the rail yard operator. Non-tenant maritime operations include sources for which the Port has no direct leasing arrangements; these emissions sources include shipping lines, trucks, dredges and other assist vessels, and some of the construction equipment emissions.

1.3.2 Sources

The inventory focuses on the largest sources of air emissions from maritime operations, which, except for ship boilers, ship engines while plugged into electric shore power, and various gasoline and compressed gas-fueled off-road equipment, are all powered by diesel engines. Source categories include ocean-going vessels, harbor craft assisting those vessels, vessels

performing or assisting in dredging, cargo handling equipment at marine terminals and the Oakland International Gateway rail yard, locomotives and trucks engaged in transport of maritime cargo containers, and construction and maintenance equipment. The inventory does not address other sources, such as gasoline-powered light-duty vehicles, that operated at the Port.

1.4 Criteria Air Pollutants

The inventory provides estimates for emissions of five “criteria” air pollutants described here, reported as tons per year.⁶

Reactive Organic Gases	Generally colorless gases that are emitted during combustion or through evaporation. They react with other chemicals in the ambient air to form ozone or particulate matter, both of which can have adverse health effects at higher concentrations
Carbon Monoxide	Colorless gas that is a product of incomplete combustion. Has an adverse health effect at higher concentrations.
Nitrogen Oxides	Nitrogen oxides include nitric oxide and nitrogen dioxide. Nitrogen dioxide is a light brown gas formed during combustion from reactions with nitrogen in the fuel or the combustion air. Nitrogen dioxide has adverse health effects at higher concentrations. Both nitrogen dioxide and nitric oxide participate in the formation of ozone and particulate matter in the ambient air.
Particulate Matter	Solid or liquid particles that form from a variety of chemical reactions during the combustion process. Solid particulate may also be emitted from activities that involve abrasion or friction, such as brake and tire wear. Have adverse health effects at higher concentrations. Particulates are divided into those less than 10 microns, PM ₁₀ , and those less than 2.5 microns, PM _{2.5} aerodynamic diameter. Diesel particulate matter (DPM) is defined as particulates from diesel engine exhaust.
Sulfur Oxides	Sulfur bearing gases, primarily SO ₂ , that form during combustion of a fuel that contains sulfur. Has adverse health effects at higher concentrations and participates in the formation of sulfate particulate matter in the ambient air.

1.4.1 Particulate Matter

The particulate matter estimated in this report is primarily diesel particulate matter (DPM), which is defined as a toxic air contaminant by the ARB. Some, primarily older, ocean-going vessels calling at the Port were designed to use boilers to supply steam power for propulsion

⁶ The term “criteria” pollutant is applied to pollutants for which an ambient air quality standard has been set, or which are chemical precursors to pollutants for which an ambient air quality standard has been set.

engines, and all vessels operate auxiliary boilers for hot water on board. In addition, some particulate emissions were from non-diesel gasoline or LPG-fueled cargo handling equipment, as noted in Section 4. The particulate emissions were estimated from emission factors as PM_{10} ; $PM_{2.5}$ was calculated as a fraction of PM_{10} which varies by source category.

1.5 Greenhouse Gases

The greenhouse gas emission inventory includes estimates of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) emissions from each source category. Fuel combustion is the source of CO_2 , while CH_4 results from incomplete combustion and N_2O is generated during the high temperature combustion. A combined carbon dioxide equivalent (CO_2e) estimate was prepared by adding 25 times the CH_4 and 298 times N_2O emissions to the CO_2 emissions to account for the greater greenhouse gas potential of these two emissions. (IPCC, 2007).

1.6 Technical Approach

This report outlines the maritime emissions inventory from mobile sources at the Port of Oakland in 2015 and includes the input data and methodology used in estimating emissions. The emissions inventory includes the following major source categories:

- Deep-Draft Ocean-Going Vessels (OGV);
- Commercial Harbor Craft (dredging and assist tugs);
- Cargo Handling Equipment (CHE);
- Trucks (container movements);
- Locomotives; and
- Other Off-road Equipment.

This is an inventory of the air emissions generated by maritime activities conducted by the Port of Oakland's tenants and seaport customers. On the water side, the spatial domain of the inventory includes Port-related marine vessel transit from dockside out through the Golden Gate Bridge, to the first outer buoys beyond the Sea Buoy approximately 30 miles away from the Port. On the landside, the spatial scope of the inventory includes five marine terminals, one rail yard, and the road traffic between those facilities and the nearest freeway interchanges. The Port area was defined approximately by the boundaries of I-80, I-880, and the Howard Terminal (Berths 67 and 68) adjacent to Jack London Square although the Howard Terminal was lightly used during 2015. Within this defined geographic domain, three areas were specifically excluded: the Schnitzer Steel terminal, the Union Pacific rail yard and interim operations at the former Oakland Army Base located between Maritime Street and I-880. Construction activities on the on the former Oakland Army Base are included in the inventory. These areas were not controlled or operated by the Port of Oakland in 2005, and are therefore not included in this update. Figures 1-1 and 2-1 illustrate the spatial scope of the inventory.



Figure 1-1. Port of Oakland maritime facilities – 2015.

This inventory was prepared by Ramboll Environ by analyzing all maritime activity in 2015 including the time in different modes of operation, the load or speed, and the engine characteristics of all equipment and vessels used in the Port's maritime operations. To obtain this data, Port, State, and terminal and rail operator records were used. Previous studies and literature reviews, and ARB input data or model estimates were used when more precise estimates could not be generated during the period of this study.

Emissions estimates included in this report are based on ARB inputs and methodologies.

1.7 Report Organization

This emissions inventory report is organized into an Executive Summary and nine sections, and two appendices.

- The Executive Summary briefly describes the methodologies used to estimate air emissions for all Port activities, and includes a summary of the results (Tables ES-1, ES-2, and ES-3)
- Section 1 contains this introduction to the report.

- Section 2 describes deep-draft ocean-going marine vessels.
- Section 3 describes operation and maintenance dredging activity and tug assists.
- Section 4 describes cargo handling equipment.
- Section 5 describes the Port of Oakland on-road truck activity associated with container movements.
- Section 6 describes locomotive emissions.
- Section 7 describes other off-road equipment emissions.
- Section 8 contains the summary and results of the report and comparisons with the 2005 and 2012 seaport emission inventories.
- Section 9 provides the references used in developing the emissions inventory.
- A glossary defines the technical terms used in the report.
- Appendix A compares 2015 average berthing emissions to those from ship calls later in 2015 and in the first half of 2016 to demonstrate the impact on emissions of the terminal slowdown in early 2015.
- Appendix B describes a sensitivity analysis that evaluated the potential impact on emissions of longer truck idling times in 2015.

2.0 DEEP-DRAFT OCEAN-GOING VESSELS (OGV)

2.1 Deep-Draft Ocean-Going Vessel Activity and Inventory

This section documents the emission estimation methods and results for large deep-draft ocean-going vessels (OGV) calling at Port of Oakland terminals in 2015. Ramboll Environ followed the latest California Air Resources Board (ARB) emission estimation methodology for ocean-going vessels (ARB, 2011a, 2016).

OGV use propulsion engines for transiting, auxiliary engines for on-board electrical power and small boilers to meet steam and hot water needs. Each vessel has unique characteristics of design speed, engine type and power that affect the estimate of time and engine load for each vessel call.

Data on vessel calls to Port of Oakland berths were provided by the Marine Exchange of the San Francisco Bay Region (SFMX) and included the berth and date and time stamps at the beginning and end of each movement, allowing a calculation of time at berth and at anchor. SFMX identified the vessels by the IMO identification number, allowing for cross reference to the vessel characteristics.

Ramboll Environ excluded from this inventory the 20 vessel calls at the privately owned Schnitzer Steel operation, which lies within the boundaries of the Port of Oakland terminals and generally sees bulk carriers calling for scrap steel. Vessel calls at Schnitzer Steel are not included because the Schnitzer facility is not owned or controlled by the Port of Oakland.

In addition, there were five calls to Berth 68 for the purposes of long term (greater than a month) cold ironing, and those vessels later shifted to other Port Berths thereby initiating a new vessel call record, which were included in the inventory. The time at Berth 68 was assumed to use no auxiliary engines or boilers because those vessels were not expected to be staffed during this period. One call only to anchorage was excluded because that vessel eventually moved to Redwood City and Richmond but not to the Port of Oakland. There was one additional call at the Port by an oil tanker barge; emissions from this call are included with the harbor craft emissions described in Chapter 3.

Of the remaining 1,393 deep draft vessel calls to the Port of Oakland in 2015, all were container ships or ships converted to carry containers.

Ship sizes were defined by three different methods:

- Dead weight tonnage (DWT),
- Container capacity in twenty-foot equivalent units (TEU), or
- Length.

Each of these size measurements may affect one emission source or another. Table 2-1 describes general ship characteristics using three size measurements for vessels calling at the

Port of Oakland in 2015. Some of the ships calling in 2015 are significantly larger than in previous years including vessels exceeding 1100 feet and with carrying capacity over 12,000 TEUs.

Table 2-1. Ocean-Going Vessels – 2015 Port of Oakland vessel calls by three different ship size measures.

Dead Weight Tonnage	Calls	TEU	Calls	Length	Calls
<20,000	12	<1,000	11	<750 feet	161
<40,000	201	- <2,000	62	750 – 1100	1195
<60,000	258	- <3,000	89	>1100	37
<80,000	388	- <4,000	148		
<100,000	165	- <5,000	304		
<120,000	222	- <6,000	159		
<140,000	126	- <7,000	159		
140,000+	21	- <8,000	77		
		- <10,000	231		
		- <12,000	126		
		12,000+	27		
All	1,393	All	1,393	All	1,393

Vessels call at both regular and irregular frequencies. Many vessels follow regular routes between ports in Hawaii, New Zealand, the South Pacific or Asia, and the Port of Oakland, while others make infrequent calls. Vessels calling between 4 and 10 times in 2015 accounted for 69% of total calls, and 9% of calls were from vessels calling 11 or more times during 2015. Table 2-2 lists the distribution of Port of Oakland call counts by individual ships in 2015.

Table 2-2. Ocean-Going Vessel - Port of Oakland vessel call counts in 2015.

Number of Calls in 2015	Ship Count	Cumulative Calls	Number of Calls in 2015	Ship Count	Cumulative Calls
1	93	93	13	1	1313
2	46	185	14	1	1327
3	41	308	15	0	1327
4	45	488	16	0	1327
5	33	653	17	0	1327
6	44	917	18	0	1327
7	33	1148	19	0	1327
8	9	1220	20	1	1347
9	4	1256	21	0	1347
10	1	1266	22	1	1369
11	2	1288	23	0	1369
12	1	1300	24	1	1393

The age distribution of the vessels calling at the Port in 2015 is shown in Table 2-3. Most were relatively new with 82% of calls by vessels built since 2000, but there were several frequently calling vessels older than 30 years. The median age of vessels calling the Port in 2015 was 8

years. The age distribution is important because the international emission standards limit NO_x emissions from newer marine engines: Tier I emission standards started with model year 2000 vessels, Tier II started with model year 2011. Tier III standards will take effect with model year 2016. In 2015, 21% of calls were by vessels required to comply with Tier II standards while 61% of calls were by vessels required to comply with Tier I standards.

Steamships (ships powered by propulsion boilers) are among the oldest vessels calling at the Port. Steamships that were not originally designed for operation on marine distillate fuel or natural gas are exempt from the North American ECA fuel sulfur requirements as per International Maritime Organization resolution MEPC.202(62) until at least 2020. However, Matson (a primary operator of steamships in Hawaiian service) has a surcharge for low sulfur fuel, so we have assumed the use of low sulfur fuel in steamships.^{7, 8} Steamship propulsion boilers are also exempt from the ARB fuel sulfur requirements. Auxiliary boilers, however, are not exempt from the ARB fuel requirements. NO_x emission limits from the IMO emission standards do not affect steamship propulsion boilers which have low NO_x emission rates even without any additional controls.

Table 2-3. Ocean-Going Vessels – Port of Oakland vessel age distribution in 2015.

Model Year	Count of Calls	Individual % of Calls	Model Year	Count of Calls	Individual % of Calls
2015	0	0%	1992	2	0%
2014	118	8%	1991	0	0%
2013	63	5%	1990	1	0%
2012	55	4%	1989	5	0%
2011	54	4%	1988	1	0%
2010	97	7%	1987	0	0%
2009	59	4%	1986	0	0%
2008	150	11%	1985	0	0%
2007	151	11%	1984	0	0%
2006	109	8%	1983	0	0%
2005	44	3%	1982	0	0%
2004	36	3%	1981	32	2%
2003	29	2%	1980	37	3%
2002	80	6%	1979	12	1%
2001	44	3%	1978	19	1%
2000	48	3%	1977	26	2%
1999	22	2%	1976	0	0%
1998	4	0%	1975	0	0%
1997	10	1%	1974	0	0%
1996	6	0%	1973	17	1%
1995	44	3%	1972	1	0%
1994	8	1%	1971	9	1%
1993	0	0%			

Source: Marine Exchange of the San Francisco Bay Region (2016) and IHS Fairplay (2015)

⁷ http://www.imo.org/blast/blastDataHelper.asp?data_id=30761&filename=202%2862%29.pdf

⁸ <http://www.hawaiifreepress.com/ArticlesMain/tabid/56/ID/11963/Matson-Raising-Fuel-Surcharge-5.aspx>

The spatial domain of this inventory includes vessel transit activity between the outer buoys that lie beyond the Sea Buoy outside the Golden Gate and berths at the Port as shown in Figures 2-1 and 2-2. This domain is the same as was used in the 2005 emission inventory (ENVIRON, 2008) although a slightly shorter route was used for all inbound vessels based on recent traffic route samples obtained from the Coast Guard's Vessel Traffic Service and interviews with the San Francisco Bar Pilots.

Based on discussions with the Marine Exchange of the San Francisco Bay Region (2013), and San Francisco (SF) Bar Pilots (2013), a schematic description of the transit activity for vessels calling at the Port of Oakland in 2015 is shown in Table 2-4. Entries in Table 2-4 correspond to the schematic link descriptions shown in Figures 2-1 and 2-2. Links listed in Table 2-4 are used to specify activity applicable to each portion of the vessel's transit.

Generally, vessel activity is classified into four modes of operation: cruise, reduced speed zone (RSZ), maneuvering, and hotelling as follows:

- Cruise mode occurs in the open ocean where there are fewer navigational challenges and where ships typically operate at their design speed.
- RSZ mode occurs where ships are required to slow down and stay within prescribed lanes as shown on Figure 2-1 and 2-2. For ships arriving in the SF Bay, the RSZ mode occurs after a SF bar pilot boards and takes command of the vessel at the Sea Buoy until the vessel slows to a very low maneuvering speed near the Port, defined for the purposes of this inventory as starting at the Bay Bridge. The RSZ mode for departing ships is the inverse of that for arriving ships.
- Maneuvering mode is defined as occurring between the Bay Bridge and the berth.
- Lastly, the hotelling or 'at berth' mode occurs when the vessel is stopped at berth or lying at anchor south of the Bay Bridge.

Two additional modes were added to account for any time spent at anchorage in the South Bay before or after calling at the Port and for vessel shifts to or from anchorage or between berths at the Port. Vessel emissions were calculated for each operating mode by multiplying the engine operating time for the mode by the engines' rated power, the engine load factor and the appropriate emission factor. The number of vessel calls multiplied by the time per call spent in each of the four operational modes constituted the total time in mode.

The time in mode and load for propulsion engines was calculated based on vessel speed and the distance (length) of each transit mode. The SF Bar Pilots (2013) estimated the RSZ average speed and typical maneuvering mode times as listed in Table 2-4. An average RSZ mode speed of 13.5 knots was chosen to account for an average compliance margin relative to the legal requirement for vessels to "Not exceed a speed of 15 knots through the water" in the regulated navigation areas (RNAs) included in Coast Guard (2014) regulations. The cruise speed was designated as the design speed reported for each vessel by the IHS Fairplay (2009, 2015)

database. The time in mode from the speed and distance along each link was used to estimate the propulsion and auxiliary engine activity for cruise and RSZ modes.

The Marine Exchange provided the vessel call data for the Port of Oakland, and for each call provided date and time stamps for the ‘first line on’ to the port and ‘last line off’ records or the shift beginning and end time.

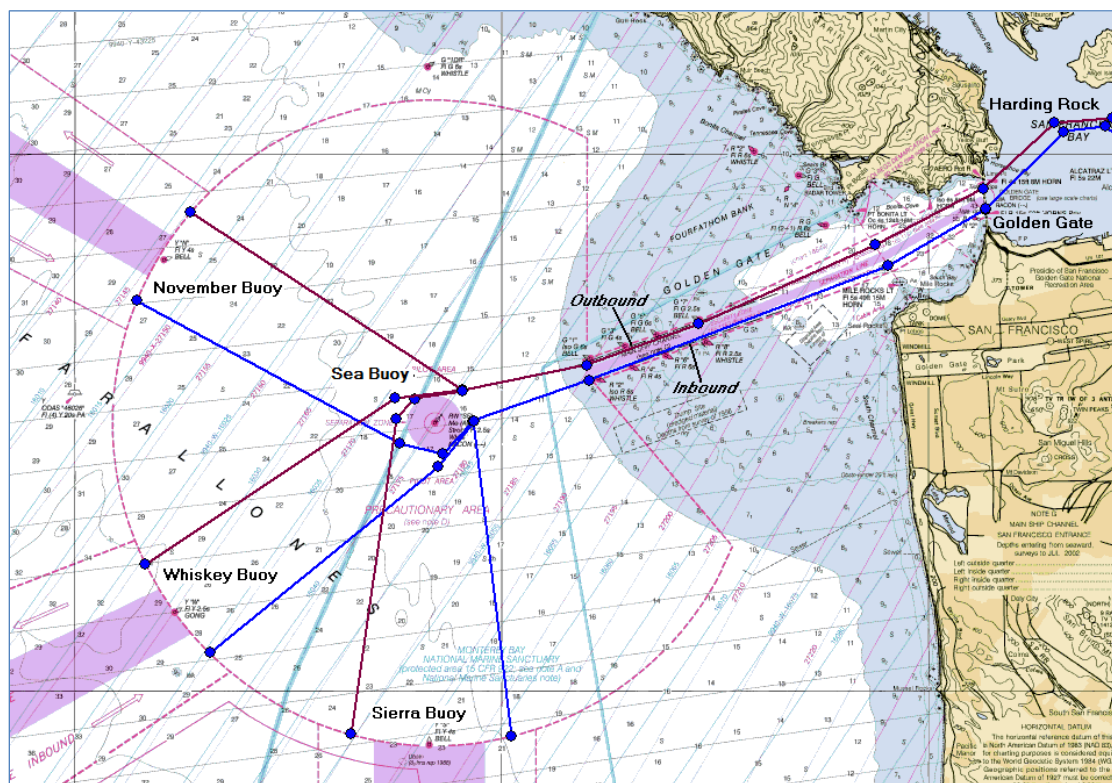


Figure 2-1. Link descriptions outside of the Golden Gate.

There are two potential transit routes between the Golden Gate and Bay bridges shown in Figure 2-2. Ships follow one or the other depending on a number of factors including weather and scheduling of public events on the Bay. The SF Bar Pilots (2013) indicated that the primary or exclusive in-bound transit route between the Golden Gate and Bay Bridges is south of Alcatraz unless the vessel is drawing more than 45 feet in which case it should use the deep water route north of Harding Rock. Only three vessels exceeding 45 foot draft called on the Port; the maximum draft was 47 feet. The northern route may also occasionally be used by other vessels under unusual weather conditions or if public events interfere with the southern route. Because insufficient data was available to describe each call’s specific route, the typical (and shorter) route south of Alcatraz was assumed for all inbound transits. The outbound transit must use the deep water route north of Harding Rock if the vessel draft exceeds 28 feet, and the southern route may not be available due to traffic concerns. Because almost all ships outbound from Oakland draw more than 28 feet and the route south of Alcatraz is rarely

available for outbound transit, all vessels were assumed to use the route north of Harding Rock for outbound transit. Alternative inbound and outbound transit routes are shown and described in Figure 2-2 and Table 2-4, but these alternatives were not used in the emission estimation.

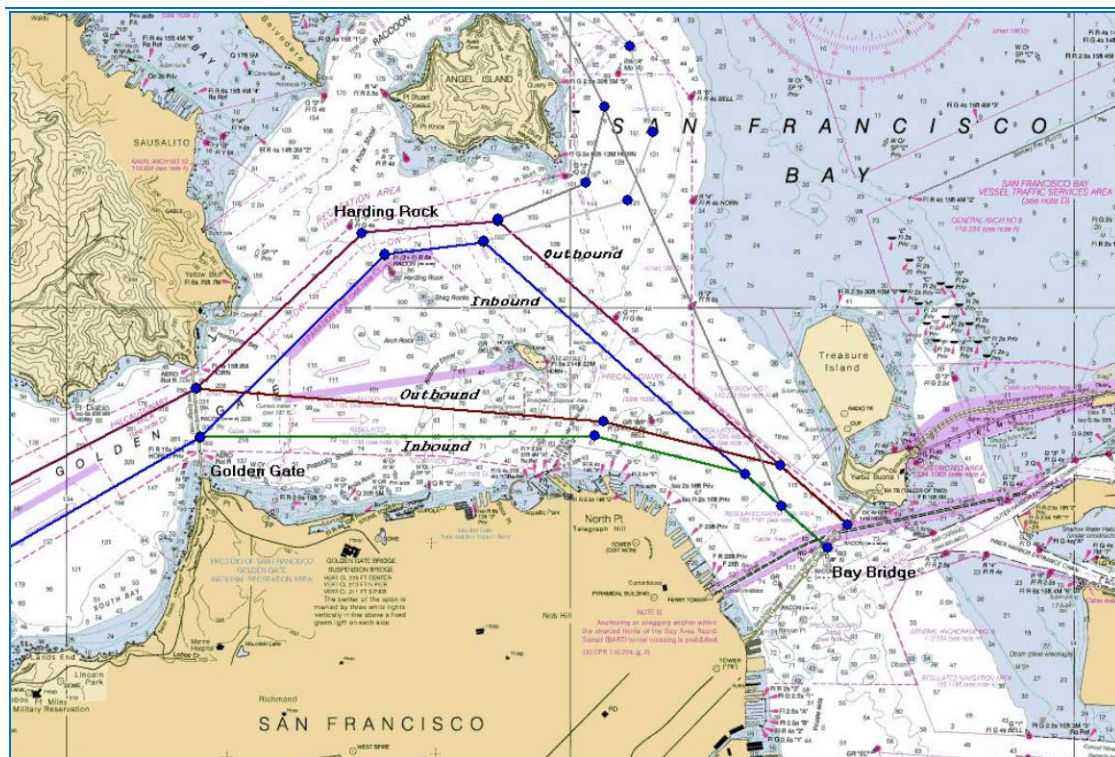


Figure 2-2. Transit link descriptions in San Francisco Bay (direct route primarily used inbound and less direct route outbound).

Vessels are assumed to be in maneuvering mode while moving between the Bay Bridge and the berths. This mode consists of a short low speed transit, turn at the berth or in the turning basin, and propulsion engine start and stop at the berth with tug assist. Based on the SF Bar Pilots' (2013) best judgment, the maneuvering time is longer for the Inner Harbor berths and for larger vessels, defined here as two types of longer vessels, one greater than 750 foot and another greater than 1100 feet in length. The larger ships require more time to turn and can only turn in prescribed areas, such as the Inner Harbor and Outer Harbor turning basins. Therefore, as shown in Table 2-4, the SF Bar Pilots (2013) estimated the maneuvering time for larger ships to be longer than for smaller ships. Also, maneuvering time is shorter for the Outer Harbor terminal calls than the Inner Harbor terminal calls because of the shorter distance from the Bay Bridge and proximity of the Outer Harbor turning basin to the Outer Harbor berths.

Table 2-4. Ocean Going Vessels – Transit link descriptions.

Transit into Port				
Direction	Link Start	Link End	Distance (nautical miles)	Speed (knots)
In – From Asia or Northern Ports	North Buoy	Pilot Boards	7.4	Cruise
In – From Hawaii and points west	West Buoy	Pilot Boards	6.7	Cruise
In – From Southern Ports	South Buoy	Pilot Boards	6.0	Cruise
In – All	Pilot Boards	Sea Buoy	1.5 ²	9
In – All	Sea Buoy	Golden Gate	8.7	13.5
In – All (alternative route) ¹	Golden Gate ¹	Harding Rock	2.0	13.5
In – All (alternative route) ¹	Harding Rock ¹	Bay Bridge	4.5	13.5
In – All ¹	Golden Gate	Bay Bridge	5.3	13.5
Maneuvering Modes				
Direction	Link Start	Link End	Time (hrs)	Load
In/Out – Inner Harbor Terminals (≤ 750 foot Ships)	Bay Bridge	Dock	0.833 / 0.833	2%
In/Out – Inner Harbor Terminals (>1100 or >750 foot Ships – Turning Basin)	Bay Bridge	Dock	2.09 or 1.42 / 0.833	2%
In/Out – Outer Harbor Terminals (≤ 750 foot Ships)	Bay Bridge	Dock	0.75 / 0.75	2%
In/Out – Outer Harbor Terminals (>750 foot Ships – Turning Basin)	Bay Bridge	Dock	1.33 / 0.75	2%
Shifts (small number of calls have shifts from one terminal to another)	Oakland	Oakland	0.75	2%
Transit Out of Port				
Direction	Link Start	Link End	Distance (nautical miles)	Speed (knots)
Out – All ¹	Bay Bridge ¹	Harding Rock	4.8	13.5
Out – All ¹	Harding Rock ¹	Golden Gate	1.8	13.5
Out – All (alternative route) ¹	Bay Bridge ¹	Golden Gate	5.5	13.5
Out – All	Golden Gate	Sea Buoy	8.9	13.5
Out – All	Sea Buoy	Pilot Departs	1.5 ²	9
Out – To Asia or Northern Ports	Pilot Departs	North Buoy	6.1	Cruise
Out – To Hawaii	Pilot Departs	West Buoy	6.8	Cruise
Out – To Southern Ports	Pilot Departs	South Buoy	7.3	Cruise

¹ SF Bar Pilots (2013) reported that ships with drafts greater than 45 feet must use the Deep Water Traffic Lane north of the Harding Rock Buoy, though other ships under certain conditions (such as occurrence of special events) may also take northern route. For transit out of the Bay, ships with drafts greater than 28 feet must use the Deep Water Traffic Lane.

² Assumes 10 minutes at 9 knots for the pilot to board and depart safely. Distance in this mode was subtracted from the cruise mode. Distances were measured from east of Sea Buoy.

Total activity was estimated along each link using the number of vessel movements along each link for the 2015 vessel calls. The purpose of defining these links was to provide emissions that were accurately spatially allocated and to estimate the time in mode based on vessel speed and the distance along each link. Determining the total vessel movements for most segments is straightforward because each call required transiting along a set route as described in Table 2-4. Vessel movements between the Sea Buoy and the South ('Sierra'), West ('Whiskey') and North ('November') outer buoys as shown in Figure 2-1 were determined on the basis of the

vessel's previous or next port of call as listed in the San Francisco Marine Exchange data. Table 2-5 lists the resulting number of inbound and outbound transits in each direction outside of the Sea Buoy for vessel calls to the Port in 2015. Maneuvering mode movements inside the Bay Bridge were determined based on which berth the vessel called and the vessel length as described above.

Table 2-5. Ocean-Going Vessels – Port direction from the San Francisco Bay.

Last or Next Port of Call	Direction	Trips In or (Out)
US northern continental ports including Alaska, Canada, and all Asian ports	N	209 (975)
US Hawaii, Guam, New Zealand, Fiji, Tahiti	W	107 (90)
US Southern continental ports (e.g. Los Angeles and Long Beach), Mexico, Panama, Chile and other South American ports, and Caribbean and European ports through the Panama canal	S	1078 (329)

Emissions were determined for each link using the equation below, accounting for the engine rated power, typical load factor, and time at that load. The rated power is the maximum power that the engine can produce. The load factor is the fraction of the actual to the rated power that the engine operates for a given mode. Emissions were calculated separately for propulsion and auxiliary engines, and for boilers, using emission factors from ARB (2011a).

$$\text{Emissions per vessel/mode} = (\text{Rated Power}) \times (\text{Load Factor}) \times (\text{Time}) \times (\text{Emission Factor})$$

$$\text{Emissions total} = \sum \{\text{All vessel calls and modes}\}$$

The time in each link was calculated from the link length and estimated speed. The load factor was calculated on the basis of the vessel's maximum speed and the actual vessel speed in each mode as described in Section 2.2.

2.2 Input Data and Use

The basic input data for calculating emissions from OGVs include the number of vessel calls in 2015, vessel installed power and speed, and estimates of load and time during each operation mode.

- 1) Vessel Port Calls
- 2) Vessel Type
- 3) Vessel Characteristics
 - a) Cruise speed (knots)
 - b) Auxiliary Power (kW)
- 4) Engine Characteristics
 - a) Rated Power
 - b) Engine Type (slow 2-stroke, medium 4-stroke, or steam)

- 5) Model Year
- 6) Berthing Time (with and without adjustment for shorepower use specific to each call)
- 7) Anchorage Vessel Calls and Time
- 8) Route between Sea Buoy and Outer buoys (north, west, or south)

2.3 Vessel Calls

Data on vessel calls to the Port of Oakland in 2015 were provided by the Marine Exchange of the San Francisco Bay Region (2016), and the Port of Oakland (2016) provided date and time stamps for the shorepower at the Port's berths.

For each ship call, the Marine Exchange data identified the ship by name and number and date and time stamps and position at the beginning and end of each movement within the Bay. For example, the Marine Exchange data identified the time and berth for the first line on and last line off for each ship call. It also identified those vessel calls for which the ship went to the anchorage area before or after calling to the Port or shifts between berths at the Port. For those vessels that went to an anchorage area, the Marine Exchange data included the arrival and departure time at the anchorage position in the south Bay.

The Port data was recorded at the terminal and included the shorepower on and off times, thereby allowing calculation of the shorepower time for those vessel calls.

2.3.1 Propulsion Power and Load

Propulsion power and vessel speed were derived from the Fairplay (2009) database, which reports design features for each vessel. To obtain estimates of maximum power and speed, Lloyds main engine power and Lloyds vessel design speed were used directly, consistent with ARB's methodology (ARB, 2011a). The vessel design speed was assumed to be the cruise speed.

The load factors for the propulsion power over any given link were determined from the classic Stokes Law cubic relationship for speed and load. The proportional relationship of load to the vessel speed can be expressed as in the following equation where the 100% load factor would correspond to the vessel operating at its maximum speed.

$$\text{Load Factor} = (\text{Vessel Speed} / \text{Vessel Maximum Speed})^3$$

The design speed of the vessel was estimated to be 93.7% of the maximum speed. Thus the load factor at the cruise speed is 0.823. For other transiting modes the load was calculated directly from the equation shown above and is unique to each vessel's reported design speed.

2.3.2 Auxiliary Power and Load

As described in Port of Oakland Seaport Emissions Inventory for 2005, the auxiliary power was primarily derived from auxiliary generator capacity taken from the Lloyds database and supplemented by other available data and estimates. The load factors shown in Table 2-6

describe the vessel activity in the Port of Oakland for container ships. These load factors were taken from ARB (2011a).

Table 2-6. Ocean Going Vessels – Auxiliary engine load factors assumptions.

Ship-Type	Cruise	Reduced Speed Zone (RSZ)	Maneuvering	Hotel
Container Ship	13%	13%	50%	18%

Source: ARB, 2011a.

For each vessel call, the time when the auxiliary engine was running was estimated and used in the emission calculations. For those calls without shoreside power, the hotelling time was set equal to the time at berth. The average berthing time in 2015 was 44.0 hours per call which is much higher than in 2012 (average of 21.4 hours per call) or in 2005 (average of 20.8 hours per call). The increase in berthing time was possibly due to the reported labor/terminal operator dispute⁹ at the Port during the early part of 2015.

2.3.3 Shore Power

Emissions avoided as a result of shore power usage were addressed in the calculation of hotelling emissions by subtracting the time when shore power was used from the berthing time. There were 636 calls averaging 33.1 hours each on shore power representing about a 34% reduction in auxiliary engine operating hours at berth overall. Shore power use is increasing, and Appendix A shows how shore power has increased throughout 2015 and into 2016.

2.4 Emission Factors

Emission factors depend on the type of engine and fuel used in the vessel for propulsion or auxiliary engines. Three types of engines were used for propulsion power on ships; slow speed engines (2-stroke and typically lower than 200 rpm), medium speed engines (4-stroke), and steam turbines coupled with steam boilers. Ramboll determined from Fairplay data (IHS Fairplay, 2009) that the primary propulsion engines used on vessels calling at the Port of Oakland were slow speed engines (1,296 vessel calls), steam engines powered by boilers (85 calls), and medium speed engines (12 calls). Emission factors for these engines are shown in Table 2-7 (ARB 2011a).

⁹ CNN, "West Coast Port Goes Back to Work," March 15, 2015, <http://money.cnn.com/2015/03/12/news/economy/port-labor-dispute/>

Table 2-7. Ocean Going Vessels – Emission factors (g/kW-hr) for Precontrol (<2000), Tier I (2000 – 2010), Tier II (2011 – 2015), and Tier III (2016+) engines as noted. (Source: ARB 2016).

Engine Type	Fuel Type	ROG	CO	NO _x	PM ₁₀	PM _{2.5}
Slow Speed	Marine Distillate (0.1% S)	0.78	1.10	17.0 Pre-controlled 17.0 Tier I 14.4 Tier II 3.4 Tier III	0.25	0.23
Medium Speed	Marine Distillate (0.1% S)	0.65	1.10	13.2 Pre-controlled 13.2 Tier I 10.9 Tier II 2.675 Tier III	0.25	0.23
Steam	Marine Distillate (2.7% S)	0.11	0.2	2.1	0.8	0.78
Auxiliary	Marine Distillate (0.1% S)	0.52	1.10	13.9 Pre-controlled 11.54 Tier I 9.20 Tier II 2.31 Tier III	0.25	0.23
Auxiliary Boiler	Marine Distillate (0.1% S)	0.11	0.2	1.995	0.133	0.122

NO_x emissions from marine engines are regulated by model year with Tier I beginning with the 2000 model year, Tier II for model year 2011 and Tier III with model year 2016 (for vessels operating in the North American Emission Control Area). Minimum marine engine emission standards for foreign flagged vessels are specified in MARPOL Annex 13 which defines the model year as, “Ships constructed means ships the keels of which are laid or which are at a similar stage of construction.” Though not all of the ships have ‘keel laid’ as an entry in the Fairplay database, all ships have a date of delivery listed. This date was used together with the average time from the keel laid to delivery date for container ships calling the Port (where both dates were provided) of 158 days to estimate the model year of the vessel. Tier I, II, and III NO_x emission rates were derived from ARB (2016).

Emission rates assuming 0.1% fuel sulfur content were used based on the ARB fuel regulation. The fuel sulfur level and the fuel consumption of the engines and boilers are used to estimate the SO_x emissions assuming all sulfur is emitted as SO₂.

2.4.1 Low Load Adjustment Factors

Emission factors for OGV engines were derived from data collected at high operational loads. Adjustment factors are applied to obtain emission factors applicable to operation at very low loads where the engine does not operate as efficiently. As recommended by ARB (see ENVIRON, 2008), low load adjustment factors for propulsion engines were applied and the adjustment factors were consistent with those used in the calendar year 2008 Port of Los Angeles emission inventory (Starcrest, 2009) for HC, CO, NO_x and SO_x. These adjustment factors are listed in Table 2-8. Low load adjustment factors for PM listed in Table 2-8 are from ARB (2006).

Table 2-8. Ocean Going Vessels – Low load adjustment factors for propulsion engines.

Load %	HC	CO	NO _x	SO _x	PM
1	N/A	N/A	N/A	N/A	9.82
2	21.18	9.68	4.63	1.00	5.60
3	11.68	6.46	2.92	1.00	4.03
4	7.71	4.86	2.21	1.00	3.19
5	5.61	3.89	1.83	1.00	2.66
6	4.35	3.25	1.60	1.00	2.29
7	3.52	2.79	1.45	1.00	2.02
8	2.95	2.45	1.35	1.00	1.82
9	2.52	2.18	1.27	1.00	1.65
10	2.18	1.96	1.22	1.00	1.52
11	1.96	1.79	1.17	1.00	1.40
12	1.76	1.64	1.14	1.00	1.31
13	1.60	1.52	1.11	1.00	1.22
14	1.47	1.41	1.08	1.00	1.15
15	1.36	1.32	1.06	1.00	1.09
16	1.26	1.24	1.05	1.00	1.03
17	1.18	1.17	1.03	1.00	1.00
18	1.11	1.11	1.02	1.00	1.00
19	1.05	1.05	1.01	1.00	1.00
20	1.00	1	1.00	1.00	1.00

Source: Table 3.8 from Starcrest (2009) except PM.

There have been recent reassessments of the load adjustment factors to emission factors for propulsion engines. Starcrest (2015) have provided alternative load adjustment factors for emission factors for slide-valve and non-slide-valve MAN 2-stroke slow speed engines. In addition, ARB (2011a) does not adjust emission factors for propulsion engines during maneuvering modes occurring when ships operate in port areas. The low load adjustment factors used in the previous Port of Oakland inventories have not been changed in this emission inventory effort because the revised load adjustment factors have not been fully vetted. This also serves to maintain consistency with previous emission inventories prepared for the Port.

The low load adjustments in Table 2-8 were applied to propulsion engines when in the RSZ and maneuvering modes. Low load adjustment factors only affect propulsion engine emissions because no single auxiliary engine operates below 20% load at any time. Typically, each vessel has a set of three or more auxiliary engines to provide auxiliary power, so individual engines are shut-down when the load decreases thereby leaving the remaining working engines operating above 20% load.

A 2% average propulsion engine load was assumed for the maneuvering mode (accounting for activity between the Bay Bridge and berth). For the RSZ mode (between the Bay Bridge and the Sea Buoy), a load factor was calculated specifically for each vessel as the cube root of the ratio of the assumed RSZ mode speed (13.5 knots) to the maximum speed of the vessel. Of all vessels calling at Oakland, the maximum speed of the fastest vessel was estimated to be 28.4 knots, so the load factor was 11% within the RSZ mode. For slower vessels, the RSZ load factor was higher than the 11% load calculated for the fastest vessel.

2.5 Boiler Emissions

In-use boiler power estimates of 380 kW for container ships and 109 kW for bulk cargo vessels were assumed based on ARB (2011). Boiler emission factors shown in Table 2-9 were used; these are consistent with emission factors used in ARB (2011).

Table 2-9. Auxiliary boiler emission rates (g/kW-hr).

Fuel Type	ROG	CO	NO _x	PM ₁₀	SO _x	CO ₂	CH ₄	N ₂ O
0.1% Sulfur	0.11	0.20	1.995	0.133	0.58	921.5	0.032	0.013

Source: ARB, 2011a

2.6 Anchorage Emissions

For 2015, there were 307 calls averaging 57.2 hours each at anchorage either before or after calling at a Port berth. The anchorage activity was significantly higher (30 times the total vessel hours) than previous years, including the 2012 Port of Oakland Seaport Emission Inventory when there were 37 calls averaging 13.9 hours at anchor and 99 calls averaging 15.2 hours each in 2005. The increase in anchorage was possibly due to the reported labor/terminal operator dispute¹⁰ at the Port during the early part of 2015. Emissions at anchorage were estimated in the same way as at-berth emissions but with no adjustment for shore power use. Appendix A demonstrates how in the latter part of 2015 anchorage activity returned to previous levels.

2.7 Shift Emissions

For 2015, there were 32 calls where a berth shift occurred in addition to the 307 anchorage calls that necessitated a shift to or from the anchorage and the Port. A shift occurs when a ship moves from one berth to another and is considered an additional maneuvering mode for those calls. The time for the beginning and the end of the maneuvering mode was provided by the SFMX data to which 15 minutes was added for ship propulsion engine time to start up and shut down. The 0.75 hours per shift from anchorage to berth (or vice versa) was also included in the shift activity and emissions estimates.

2.8 Emission Results

Estimated total emissions from the Port of Oakland ocean-going vessels are presented in Table 2-10 by operating mode (cruise, RSZ, maneuvering, and berthing). RSZ mode includes all transit between the Bay Bridge and the location where the Bar Pilot boards or disembarks. All vessels calling at the Port of Oakland were assumed to operate small auxiliary boilers on-board. Emissions from propulsion and auxiliary engines and boilers are included in Table 2-10. Because the ARB considers diesel particulate emissions to differ in toxicity from boiler particulate emission, total diesel particulate matter (DPM) emissions from the main and auxiliary diesel engines are provided in Table 2-10. All auxiliary engine PM emissions are DPM

¹⁰ CNN, "West Coast Port Goes Back to Work," March 15, 2015, <http://money.cnn.com/2015/03/12/news/economy/port-labor-dispute/>

because all auxiliary engines are diesel engines. Propulsion steam and auxiliary boiler particulate emissions are not included in the DPM total. The Shore Power emission reductions represent the reduced emissions due to shore power that would have otherwise been included in the OGV – Berth emissions totals.

Table 2-10. Emissions totals for OGV calling at the Port of Oakland in 2015 by mode for main and auxiliary engines and boilers – tons.

2015 Inventory	ROG	CO	NO_x	PM₁₀	PM_{2.5}	DPM	SO_x	CO₂	CH₄	N₂O	CO₂e
OGV – Cruise	25.25	36.10	502.00	8.51	7.84	8.13	31.00	20,844	2.56	0.61	21,091
OGV – RSZ	32.06	44.52	509.79	9.27	8.52	8.66	32.95	22,421	2.62	0.63	22,673
OGV – Maneuver	61.39	52.45	416.07	8.93	8.24	8.57	10.18	14,971	1.79	0.38	15,129
OGV – Berth	41.31	86.26	890.16	22.61	21.05	18.05	48.72	81,357	7.59	1.74	82,067
OGV – Shifts	4.91	3.58	27.49	0.54	0.50	0.52	0.47	631	0.49	0.03	653
OGV – Anchorage	17.19	36.13	369.51	8.86	8.15	7.88	17.93	28,520	2.98	0.66	28,792
OGV Subtotal	182.11	259.04	2,715.02	58.72	54.29	51.81	141.24	168,745	18.03	4.06	170,405
Emissions avoided due to shore power (tons)	24.17	51.13	501.23	11.62	10.69	11.62	18.59	32,068	4.18	0.84	32,422
Berthing % reduction	36.9%	37.2%	36.0%	33.9%	33.7%	39.2%	27.6%	28.3%	35.5%	32.6%	28.3%

For comparison, Figure 2-2 and Figure 2-3 show the changes to the OGV NO_x and DPM emissions inventory for 2005, 2012, and 2015. The NO_x emissions increased due to fewer low-NO_x steam ship calls replaced by diesel motorship calls, and an increase berthing, shifts, and anchorage activity in 2015 due to a slowdown during the early part of the year. NO_x emission reductions include those from the use of shore power and fleet turnover to ship engines designed to meet lower NO_x emission standards. The DPM reductions are dramatic primarily because of the use of a low sulfur fuel in 2012 and further sulfur reductions in 2015 as well as the use of shore power. Increased DPM from a greater fraction of diesel motorship calls and the general slow down at the Port during 2015 obscure progress made in reducing emissions.

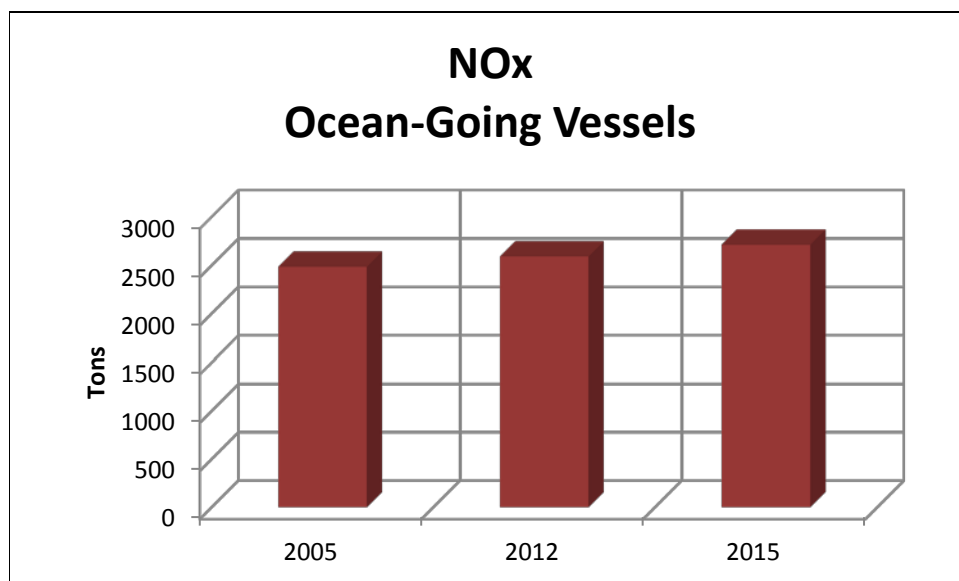


Figure 2-3. OGV NO_x emissions inventories for 2005, 2012 and 2015.

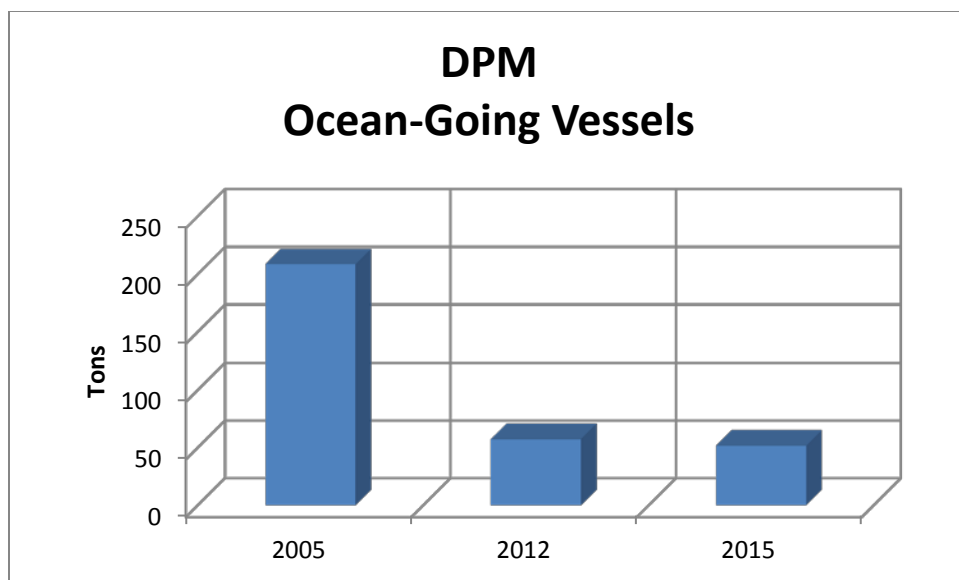


Figure 2-4. OGV DPM emissions inventories for 2005, 2012 and 2015.

3.0 COMMERCIAL HARBOR CRAFT (DREDGING AND ASSIST TUGS)

This section describes the emissions estimation methodologies and results for two regularly occurring activities at the Port of Oakland: 1) operation and maintenance dredging and disposal, and 2) container vessel assists. Other than a few small work boats that assist dredging operations and the dredges themselves, tugs are the primary category of commercial harbor craft that are a part of the Port's maritime emissions inventory. This inventory does not include dredging and vessel assist activities at the privately owned Schnitzer Steel bulk terminal berths or emissions from harbor pilot boats based in San Francisco.

3.1 Operation and Maintenance Dredging and Disposal

3.1.1 Background and Limitations

Operation and maintenance (O&M) dredging is conducted annually at the Port of Oakland to maintain the depth of channels and berths and to ensure safe navigation. O&M dredging removes material that is deposited into the Bay by stream and urban runoff throughout the Sacramento-San Joaquin River Delta area extending east to the Sierras, and eliminates shallow areas created by the redistribution of bottom sediments through a process known as "shoaling." To protect sensitive species, such as the endangered California Least Tern, and fisheries (Pacific herring), O&M dredging is conducted during a limited seasonal period. The channel dredging was conducted from October through April, while berth dredging was conducted in August and October during 2015.

The Port and the US Army Corps of Engineers (USACE) contract separately for O&M dredging at the Port's berths and in the Federal channels serving the Port, respectively. During 2015, dredging was conducted by a diesel-powered derrick barge (clamshell) dredge, accompanied by tender tugs and survey boats.

Dredged material is transferred into scows (barges) which are then pushed or towed by a diesel-powered tug to a disposal or reuse site. After the barge is emptied, the tug returns with the empty barge to pick up a new load.

The contractor working for both the Port in 2015 and the USACE in 2015 and 2016 removed 148,791 cubic yards of material from the Port's berths and 696,030 cubic yards of material from the channel (Dutra, 2016). All of the material excavated from Port berths was disposed of at the Montezuma wetlands, a privately owned and operated site located in the Sacramento-San Joaquin River delta adjacent to Montezuma Slough in Solano County. The material removed from channel for the USACE was sent to Montezuma, Winter Island (near Montezuma) or the San Francisco Deep Ocean Disposal Site (SF-DODS). SF-DODS is an open water site located approximately 49 nautical miles (nm) west of the Golden Gate.

3.1.2 Methodology

To estimate emissions, O&M dredging and disposal activities were treated as two separate activities: 1) dredging (operation of the clamshell dredge and associated support vessels), and

2) disposal (transport of dredge materials from the dredging area to disposal sites). Emissions from these activities were summed to form the final total emissions estimate.

3.1.2.1 Dredging

Dutra Construction, the contractor responsible for the 2015 POAK berth dredging project and the 2015 calendar year federal channel dredging, has provided a list of equipment used for O&M and channel deepening dredging; they include:

- A clamshell dredge with
 - two diesel engines; or
 - separate hoist and generator engines,
- Dredge tenders with two diesel engines,
- Survey boats with two outboard gasoline engines,
- An unpowered scow into which the dredged material was loaded for disposal or reuse.

The basic equation used to calculate emissions from each of the engines involved in dredging is:

$$Equip_{Emiss} = \frac{EF \times Time_{hrs} \times Engine_{bhp} \times LF_{wt}}{(453.6 \times 2000)}$$

Where:

$Equip_{Emiss}$ is the engine's emissions in tons per year,
 EF is the engine emission factor in grams per brake horsepower-hour,
 $Time_{hrs}$ is the annual operating hours,
 $Engine_{bhp}$ is the brake horsepower rating of the engine,
 LF_{wt} is the time weighted engine load factor (fraction of full load), based on different engine operating modes during a round trip, and
 (453.6×2000) is the conversion factor from grams to tons.

3.1.2.2 Dredged Materials Disposal

In 2015 all dredged material was disposed of by removing it to an offsite disposal area. In a typical operation, a diesel powered tug pushes or tows the loaded scow to its destination and, after unloading, pushes the empty barge back to the dredge. The tow boat tug has two main propulsion engines and one or two auxiliary engines.

The basic equation used to calculate main propulsion and auxiliary engine emissions from the tug is:

$$Tug_{emiss} = \frac{EF \times Engine_{bhp} \times Time_{hrs} \times LF_{wt} \times Trips}{(453.6 \times 2000)}$$

Where:

Tug_{emiss} is the tug emissions in tons per year,
EF is the tug main propulsion or auxiliary engine emission factor in grams per brake horsepower-hour,
Engine_{bhp} is the combined brake horsepower rating of a tug's main propulsion engines and the brake horsepower rating of the auxiliary engines,
Time_{hrs} is the tug operating time per round trip in hours,
LF_{wt} is the time weighted engine load factor (fraction of full load), based on different engine operating modes during a round trip,
Trips is the annual number of round trips per tug, and
(453.6 x 2000) is the conversion factor from grams to tons.

Once it reaches the disposal area, a barge or scow is unloaded either by gravity or mechanically. Unloading at the ocean disposal site SF-DODS was accomplished by gravity - that is, by opening the bottom of the scow and allowing material to flow out. At land-based sites the scows are mechanically unloaded for redistribution ashore. At Montezuma and Winter Island, a dedicated shore powered electric "off-loader" was used to draw the wet material out of the barge and pump it upland for distribution.

Dredging performed by the USACE in the federal channel serving the Port during 2015 was all associated with normal maintenance activity and is therefore included in the Port's 2015 emission inventory.

3.1.3 Input Data and Emissions

Key input data for estimating dredging emissions include the physical characteristics of the vessels and equipment used by the Port and USACE contractor, equipment emission factors, engine load factors, the volume of material removed, and the hours of operation. The dredging contractor Dutra (2016) provided the engine characteristic data and activity in hours of use. ARB vessel emission, deterioration, fuel correction, and load factors were used to estimate emissions for all engines used on the dredging and support vessels. (ARB 2011)

The 2015 berth dredging at the Port occurred in August and October, and all of the dredged material was sent to Montezuma Wetlands Restoration Project. The USACE dredging occurred from October 2015 through April 2016. Dredged material from the CY2015-16 project was sent to Montezuma, Winter Island (disposal site located near but south of Montezuma), and the deep ocean disposal site, SF-DODS. Dutra (2016) provided the collected dredging volume and tug and barge trip log data. A one-way trip between the Port of Oakland and Montezuma wetlands was estimated at approximately 44.5 nautical miles (45.4 nm to Winter Island), as shown in Figure 3-1 below. Because the SF-DODS disposal site is outside the geographic scope of the Port of Oakland emission inventory, only the portion of the USACE disposal trips between the Bay Bridge and the West Buoy near the Farallon Islands was included in the inventory calculations; this one-way distance measures approximately 22.2 nautical miles. Dutra confirmed the Port of Oakland 2012 Seaport Emission Inventory (ENVIRON, 2013) estimate of 8 knots as a representative average speed for the tug and barge trips.

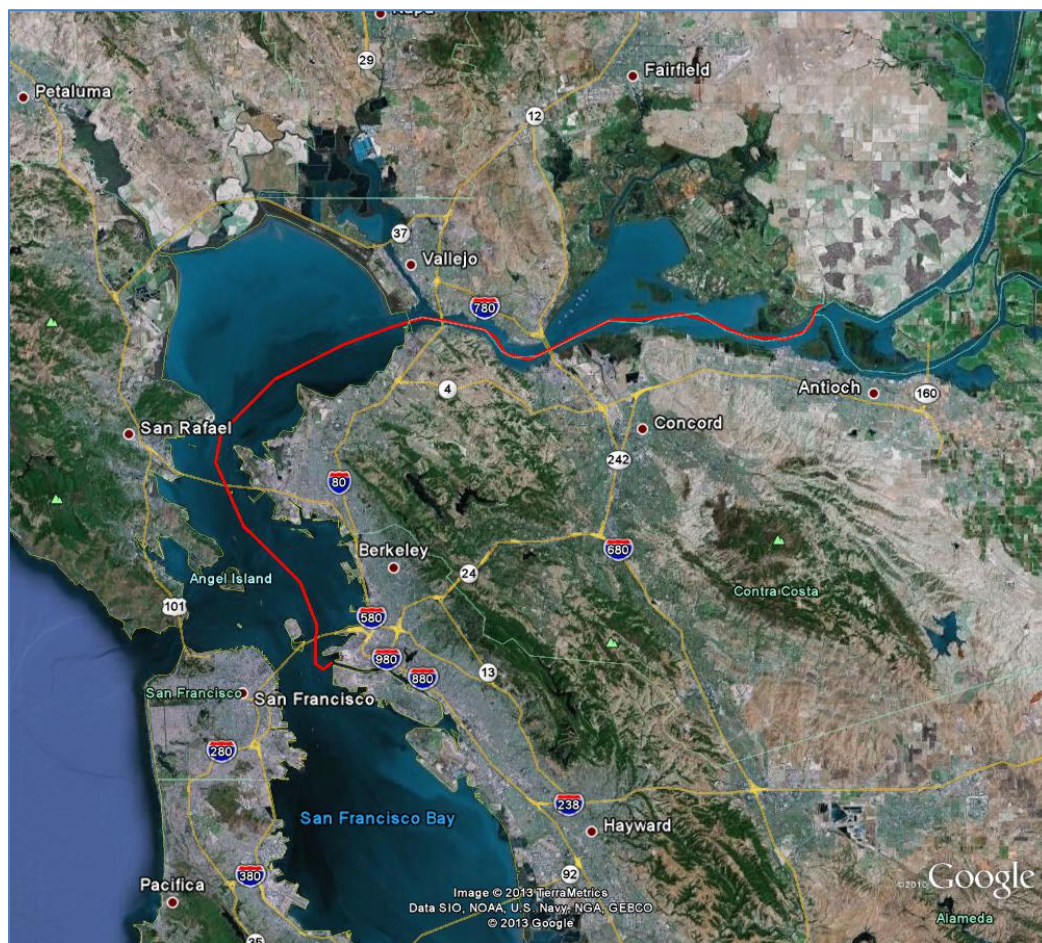


Figure 3-1. Approximate transit route for barge tugs between the Port of Oakland and the Montezuma site. Source: Google Earth (Winter Island is adjacent and due south of the Montezuma site).

The tugs Arthur Brusco and Sarah Reed were used to transport most of the materials dredged from the Port berth maintenance dredge project in 2015. No specific tugs were identified for the USACE contracted Federal Channel dredging projects. Therefore, the characteristics of the A. Brusco and S. Reed were used to represent the tug boats used to transport materials from all dredging projects. Emptying of loaded barges at the SF-DODS (performed via a gravity method) and at the Montezuma Slough and Winter Island (performed via use of an electric powered off-loader) did not result in any additional emissions.

Dredging was accomplished using barge mounted derricks which are positioned using tender tugs. The Dutra (2016) dredgers resemble clamshell excavators or cranes and are not self-propelled. Survey boats are small maneuverable boats used to monitor the dredging progress.

3.1.3.1 Dredging

Input data and assumptions for dredging are summarized in Table 3-1(a), and emission factors associated with each type of equipment are summarized in Table 3-1(b). Emission factors for dredgers and survey boats (using outboard motors) were derived from the OFFROAD model incorporating the model year and age of equipment in 2015, while the diesel engines used in tugs and tenders used the load, zero-hour emissions and deterioration factors available in the ARB harbor craft emission inventory database tool. Table 3-2 presents the resulting emissions.

Table 3-1(a). Operation & maintenance dredging - key data and variables.

Vessel/Equipment	Use	Model Year	Total Power (hp)	Load Factor	Hours	
					POAK	USACE
DB Beaver	Dredge	2002	810	0.51	936	624
DB 24 Hoist	Dredge	2002	425	0.31	0	792
DB Gen. Set	Dredge	2006	325	0.75	0	792
Phyllis T ¹	Tender	2000	430	0.38	312	208
Becky T ¹	Tender	2000	430	0.38	0	264
Survey 3 and 4 ²	Survey	2014	300	0.38	156	255

¹ The tender boats use twin diesel engines

² Survey boats use 2014 model year twin 150 hp outboard gasoline engines

Table 3-1(b). Operation & maintenance dredging – emission factors.

Equipment	Adjusted Emission Factors in g/bhp-hr								
	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O
DB Beaver	0.52	1.16	5.64	0.006	0.162	0.157	580	0.047	0.000
DB 24 Hoist	0.49	1.13	4.97	0.006	0.145	0.141	566	0.044	0.000
DB Gen. Set	0.41	1.07	2.54	0.005	0.128	0.124	548	0.037	0.000
Phyllis T	0.95	2.14	7.45	0.006	0.357	0.346	587	0.085	0.020
Becky T	0.95	2.14	7.45	0.006	0.357	0.346	587	0.085	0.020
Survey 3 and 4	9.34	21.17	2.91	0.012	7.123 ¹	6.909	756	0.58	0.13

¹ Survey boats use 2014 model year twin 150 hp outboard gasoline engines which do not emit diesel particulate matter

Table 3-2. Operation & maintenance dredging emissions - 2015 (tons/yr).

	Equipment	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
POAK	Dredger	0.22	0.49	2.40	0.003	0.069	0.067	247	0.02	0.000	248
	Tender	0.83	1.98	0.42	0.000	0.020	0.020	33	0.07	0.001	35
	Survey Boat	0.18	0.42	0.06	0.000	0.140	0.135	15	0.01	0.003	16
	Annual Tons	1.23	2.89	2.88	0.003	0.229	0.222	295	0.11	0.004	299
USACE	Dredger	0.29	0.69	2.71	0.003	0.090	0.087	346	0.03	0.000	347
	Tender	1.66	3.96	0.63	0.001	0.030	0.029	50	0.15	0.002	54
	Survey Boat	0.30	0.68	0.09	0.000	0.228	0.221	24	0.02	0.004	26
	Annual Tons	2.25	5.32	3.44	0.004	0.348	0.338	420	0.19	0.01	427
Total		3.48	8.21	6.32	0.01	0.58	0.56	715	0.30	0.01	726

3.1.3.2 Dredge Materials Disposal

Tables 3-3 and 3-4 summarize the key input data and assumptions used to calculate emissions from dredge materials disposal activities. Emissions are summarized in Table 3-5.

Table 3-3. Dredged material transport tug engine characteristics and emission factors (tugs used were the Arthur Brusco, 2008 model year main engines 2460 hp total, and 282 hp auxiliary; and the Sarah Reed, 2008 model year main engines 1700 hp total, and 132 hp auxiliary)

Engine	Load Factor	2008 Model Year Adjusted Emission Factors in g/bhp-hr								
		ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O
Main	0.68	0.76	3.98	5.53	0.006	0.189	0.183	587	0.068	0.02
Auxiliary	0.43	0.91	3.98	5.11	0.006	0.208	0.202	587	0.081	0.02

Table 3-4. Dredged material transport activities.

	Destination	Distance (nautical miles)	Speed (knot)	Time (hours)	Trips
USACE	Montezuma	44.5	8	5.56	211
	Winter Island	45.4	8	5.68	29
	SF-DODS ¹	22.2	8	2.78	58
POAK	Montezuma	44.5	8	5.56	70

¹ The location of SF-DODS is out of the scope of this inventory; the distance shown here and the emissions modeled reflect that from the Bay Bridge to the West outer buoy.

Table 3-5. Dredged material disposal emissions in 2015 (tons per year).

	Engine	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
POAK	Main	0.85	4.46	6.20	0.01	0.211	0.205	657	0.08	0.02	666
	Aux	0.06	0.25	0.32	0.00	0.012	0.011	43	0.01	0.00	44
	POAK Total	0.91	4.71	6.52	0.01	0.223	0.216	700	0.08	0.02	709
USACE	Main	2.91	15.21	21.16	0.02	0.722	0.700	2,243	0.26	0.08	2,273
	Aux	0.16	0.73	0.94	0.00	0.028	0.027	126	0.01	0.00	127
	USACE Total	3.07	15.94	22.10	0.02	0.750	0.727	2,369	0.28	0.08	2,400
Total		3.98	20.65	28.62	0.03	0.973	0.944	3,069	0.36	0.10	3,109

3.1.4 Dredging Emission Summary Results

Total emissions from Table 3-2 (dredging) and Table 3-5 (dredged material disposal) combined are listed in Table 3-6.

Table 3-6. Summary of operation & maintenance dredging emissions in 2015 (tons per year).

Activity	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	DPM	CO ₂	CH ₄	N ₂ O	CO ₂ e
Dredging	3.48	8.21	6.32	0.01	0.58	0.56	0.21	715	0.30	0.01	726
Disposal	3.98	20.65	28.62	0.03	0.97	0.94	0.97	3,069	0.36	0.10	3,109
Total	7.46	28.86	34.94	0.04	1.56	1.50	1.18	3,785	0.66	0.11	3,835

3.2 Assist Tugs

3.2.1 Background

This section describes the emissions estimation methods and results for operation of tugs that assisted cargo vessel movements upon arrival and departure from the Port. Assist tug operations include two modes: the actual vessel assist operation and the transit trips the tugs make to and from their various berthing bases to conduct the assists.

The role of the assist tugs is to ensure safe navigation, which is particularly important in windy weather and when vessels turn to reverse direction near the Inner or Outer Harbor berths. As discussed in Section 2, cargo vessels operating in the San Francisco Bay have Bar Pilots on board to guide each vessel to and from its destination. On average, more than two tugs were used for each cargo vessel inbound or outbound between berths at the Port and the Federal Channel near the Bay Bridge.

Tugs perform a variety of services around the Bay including vessel escort, berthing and departure assists at Bay Area ports and refineries; and towing or pushing a wide variety of barges and other equipment. Not all tugs are equipped or certified to provide assist services to container vessels calling at the Port. Cargo vessels vary greatly in size, length, and maneuverability, and the tugs that assist them have different power levels, rudders and other equipment. To ensure safe navigation, it is important that tugs be properly powered and equipped to handle the vessels they are assisting. As might be expected, larger vessels require more tugs (up to five) and the tugs might be larger and more powerful.

Tugs assigned to ships calling at the Port of Oakland are operated by six companies; AMNAV, Foss Marine (AMNAV and Foss are part of Foss Marine Holdings), Starlight Marine (part of Harley Marine), Crowley, BayDelta, and Westar. AMNAV and Crowley based their tugs at or near Berth 9 on the Outer Harbor of the Port, and Starlight tugs are based on the Alameda side of the Inner Harbor Turning Basin for the Port. The other three companies combine for less than 10% of the assist tug activity at the Port. Tugs from all six companies also operate elsewhere in the Bay, but the activity estimated in this study included only activity during transiting and assisting for the Port of Oakland ship calls.

Vessel call data specific to the Port of Oakland was provided by the Marine Exchange as described in Section 2. This data set included the number of tugs by tug operator that performed each vessel assist, but did not identify the individual tugs that provided the assist.

3.2.2 Methodology

Ramboll Environ closely followed the emissions estimation methodology that was developed for ARB's Commercial Harbor Craft Emission Inventory Database (ARB, 2011b). The ARB methodology provides emission factors that are specific to main propulsion and auxiliary engine model year, and applies both an engine emissions deterioration rate and a fuel correction factor.

The basic equation used to calculate emissions from each group of assist tugs is the following:

$$\text{Assist Tug}_{\text{Emiss}} = \frac{AEF \times \text{Time}_{\text{hrs}} \times \text{Engine}_{\text{bhp}} \times LF_{\text{wt}}}{(453.6 * 2000)}$$

Where:

- Assist Tug_{Emiss}* are the assist tug emissions in tons per year;
- AEF* is the main engine or auxiliary engine emission factor in grams per brake horsepower-hour, adjusted for model year, deterioration rate and fuel, and averaged by tug class;
- Time_{hrs}* is the annual operating hours for the tugs in each group, based on the number of vessel calls, the average maneuvering time per call, and the average number of tugs assigned to each inbound and outbound assist;
- Engine_{bhp}* is the weighted average main propulsion and/or auxiliary engine brake horsepower rating of the engines in each tug group;
- LF_{wt}* is the time weighted load factor for the maneuvering phase for the main engine and/or auxiliary engine, taken from the literature or the ARB methodology, stated as a fraction of full load; and
- (453.6 * 2000)* is the conversion of grams to tons.

3.2.3 Input Data and Emissions

There are a number of variables that affect actual tug emissions during an assist event. Among the most important are the following:

- The number of tugs assisting a vessel,
- The horsepower ratings of assist tug propulsion engines (which vary from tug to tug),
- The load carried by the tug's main propulsion engines (which varies substantially during the assist),
- The time required to complete the assist operation (which varies depending on where the vessel is berthing or departing), and
- The model year of the engines used on the vessel.

In the absence of a central record that identified individual assist tugs and their activities, Ramboll created a list of the fleet of tugs that were operating in 2015. The individual tugs and their relevant characteristics are listed in Table 3-7. Data were obtained from tug operator websites (AMNAV, Starlight, BayDelta, Crowley, Foss, and Westar, 2016) and from publicly available sources as noted in Table 3-7. Average auxiliary engine horsepower ratings were based on data from tugs for which auxiliary engine installed power was provided. For the assist tug providers, Ramboll Environ distributed the assists for each company amongst the tugs listed.

Table 3-7. Assist tug fleet characteristics and other fleet vessels. (Highlighted cells indicate assumed auxiliary engine power based on average value)

Company	Name	Engine	Engines	MY	HP total	Auxiliary	Aux. kW	Notes	Ref
AMNAV Maritime Services	Patricia Ann	Cat 3512B	2	2008	5850		210	http://www.tugboatinformation.com/tug.cfm?id=3954	
AMNAV Maritime Services	Delta Lindsey	Cat 3512B	2	2010	6850		210		
AMNAV Maritime Services	Independence	Cat 3512B	2	2007	5080		210		
AMNAV Maritime Services	Revolution	Cat 3512B	2	2006	5080		210		
AMNAV Maritime Services	Sandra Hughes	Cat 3512B	2	2007	5080		210		
AMNAV Maritime Services	Liberty	Cat 3512B	2	2008	3400		210	Rebuilt; original 1978; 4000 HP	http://www.workboat.com/newsdetail.aspx?id=14814
Starlight Marine Services	Ahbra Franco	cat C-175	2	2013	6850	(2) Caterpillar C6.6	218		
Starlight Marine Services	Z-3	3516B	2	1999	4000	3406C and 3304 DIT	204		
Starlight Marine Services	Z-4	3516B	2	1999	4000	3406C and 3304 DIT	204		
Starlight Marine Services	Z-5	3516B	2	1999	4000	3406C and 3304 DIT	204		
BayDelta	Delta Billie	3516C	2	2009	6712		215		
BayDelta	Delta Cathryn	3516C	2	2009	6712		215		
BayDelta	Delta Audrey	3516C	2	2014	6712		215		
Crowley (BayDelta)	Valor	Cat 3516C	2	2007	6772		215	http://www.tugboatinformation.com/tug.cfm?id=5195	
Crowley (BayDelta)	Veteran		2	2014	6772	(2) Caterpillar C6.6	218		
Foss (AMNAV)	Keegan Foss		2	1998	3900	John Deere 6068T	198		
Foss (AMNAV)	Lynn Marie		2	2001	6250		210		
Westar Marine	Sagittarian	Cummins KTA38M2	2	2001	2400		210	http://magazines.marinelink.com/Magazines/MaritimeReporter/200107/page/42	
Westar Marine	Apollo	Cat 3512B	2	2001	3000		210		
Westar Marine	Orion	EMD 12-645-E6	2	2005	3000		210	http://www.workboat.com/component/content/article?id=3659	

Ramboll Environ used Port of Oakland specific data to estimate the time tugs spent in the assist mode by assuming that the assist operation coincides with the vessel maneuvering mode. While assists generally start and end near the Bay Bridge, the time required for ships to maneuver between this location and each berth varies between the Inner and Outer Harbor as described for ocean-going vessel maneuvering time in Section 2. Ramboll Environ estimated a specific maneuvering time for each vessel call based on berth location (Inner or Outer Harbor) and vessel length.

Ramboll Environ estimated the time transiting to and from assists for each tug operator using the distances from each operator's home base to various assist destinations, and assuming the transit trips were made at an average speed of 8 knots. Occasionally, tugs may 'lay up' near their next assignment (such as at Berth 38-Nutter Terminal nearest the Bay Bridge or at the berth for the next outbound ship), but no adjustment was made for this circumstance, so assuming a return to base for each assist may result in an overestimate of emissions associated with tug transiting. Transit trips included the following links:

- Base to incoming vessel pickup point (about 3.25 nautical miles from Berth 9, and 4 nautical miles from the Inner Harbor turning basin),
- Return trip to base from the Inner and Outer Harbor berths,
- Trip from base to Inner and Outer Harbor berths to begin outbound vessel assist, and
- Return to base from the outbound vessel assist.

In summary, Ramboll Environ estimated the tug assist activity during the assist phase of their operation at the Port of Oakland as follows:

- Allocated annual assists by tug operator, based on the information contained in the Marine Exchange report described above.
- Developed a database that described the key characteristics of the fleet of the tugs that the six tug companies operate at the Port of Oakland.
- Assigned the number of tugs to incoming and outgoing vessel calls based on the Marine Exchange (2016) report, which showed an average of 2.13 tugs per movement.
- Estimated the time that assist tugs operate on Port of Oakland vessel maneuvering
 - While engaged in maneuvering ships inbound and outbound from the Port and
 - While transiting to and from maneuvering assists.

Ramboll Environ used zero hour emission factors, engine emissions deterioration factors and fuel correction factors for both main propulsion and auxiliary engines from ARB's database emission inventory tool (ARB, 2011b). However, the main engine load factor was estimated to be 0.31, and the auxiliary engines load factor was estimated to be 0.43. These load factors correspond to values used in the Port of Oakland 2005 and 2012 Seaport Air Emissions Inventory (ENVIRON, 2008) and the Port of Los Angeles Inventory of Air Emissions (POLA, 2012).

Table 3-8 summarizes the 2015 activity factors for both the assist and transit modes; emission estimates for assist tugs are shown in Table 3-9.

Table 3-8. Assist tug activity levels for 2015.

# of Outer Harbor Assists		Assist Inbound	Assist Outbound	Transit Hours	Total Hours
Inbound	Outbound	Hours	Hours		
3136	3094	4,219	2,380	3,425	10,024

Table 3-9. Tug assist emissions (tons per year)^a.

Engine	ROG	CO	NO _x	SO _x	PM ₁₀ ^b	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
Main	14.44	63.03	122.92	0.10	4.77	4.63	12,149	1	0	12,290
Auxiliary	1.20	4.90	7.77	0.01	0.29	0.28	897	1	0	937
Total	15.64	67.92	130.70	0.11	5.06	4.91	13,046	3	0	13,227

^a Includes both assist and transit modes

^b All PM₁₀ emissions are DPM

3.3 Oil Tanker Barge Tow Boats

3.3.1 Activities and Emissions

In addition to assist tug services, Ramboll Environ estimated emissions from one call of an oil tanker barge (OTB), which arrived at Berth 30 during 2015 from Richmond's Kinder Morgan terminal estimated at approximately 10.6 nm from the Port. The transit time was estimated using an average speed of 8 knots for the round trip to the Port from the oil terminal. The barge was the Lovel Briere, which was owned by Harley Marine. Of the Harley Marine tow boats, we chose the Emery Zidell tug shown in Table 3-10a and applied load factors representative of a tow boat obtained from the ARB harbor craft database (ARB, 2011b). We assumed the tow boat was hotelling where the OTB was berthed, and the auxiliary engine was running during this time. The activities of the tow boat and emissions factors are summarized in Table 3-10(a) and Table 3-10(b), respectively.

Table 3-10(a). Tow boat characteristics and key data input.

Name	Model Year	Power (hp)		Load Factor		Distance (nautical mi)	Time (hours)	
		Main	Aux.	Main	Aux.		Main Engine Transit	Auxiliary Engine At Berth
Emery Zidell	2014	4000	266	0.68	0.43	21.2	2.65	22.33

Table 3-10(b). Tow boat emission factors with deterioration included (g/hp-hr).

Engine	ROG	CO	NO _x	PM ₁₀	PM _{2.5}	Fuel	SO _x	CO ₂	CH ₄	N ₂ O
Main	0.69	3.77	4.41	0.10	0.97* PM ₁₀	184.16	0.000015*Fuel	44*Fuel/13.8	0.09*ROG	0.02
Auxiliary	0.83	3.77	4.41	0.10	0.97* PM ₁₀	184.16	0.000015*Fuel	44*Fuel/13.8	0.09*ROG	0.02

Emissions associated with the OTB tow boat are summarized in Table 3-11. All emissions are from diesel engines so all of the PM₁₀ is DPM.

Table 3-11. OTB tow boat emissions (tons)

Name	ROG	CO	NO _x	SO _x	PM ₁₀ ^a	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
Emery Zidell	0.0078	0.0406	0.0475	0.0001	0.0011	0.0011	6.32	0.0007	0.0001	6.38

^a All PM₁₀ emissions are DPM

3.4 Summary of Commercial Harbor Craft Emissions

Table 3-12 summarizes the emissions of harbor craft engaged in both O&M dredging and vessel assists. All of the PM₁₀ emissions listed here come from diesel engines and are therefore DPM except for the dredging survey vessels, which used outboard gasoline engines.

Table 3-12. Total harbor craft & dredge emissions, 2015 (tons).

Harbor Craft	ROG	CO	NO _x	PM ₁₀	PM _{2.5}	DPM	SO _x	CO ₂	CH ₄	N ₂ O	CO ₂ e
O&M Dredging	7.46	28.86	34.94	1.55	1.50	1.18	0.04	3,785	0.66	0.11	3,835
Assist Tug	15.64	67.92	130.70	5.06	4.91	5.06	0.11	13,046	2.59	0.39	13,227
OTB Tow Boat	0.01	0.04	0.05	0.00	0.00	0.00	0.00	6	0.00	0.00	6
Total Emissions	23.11	96.82	165.69	6.61	6.42	6.25	0.14	16,837	3.26	0.50	17,069

The harbor craft NO_x and DPM emissions estimates for 2005, 2012, and 2015 are shown in Figures 3-2 and 3-3. The emissions have been progressively lower as the vessels fleets have turned over and are now using lower emitting engines.

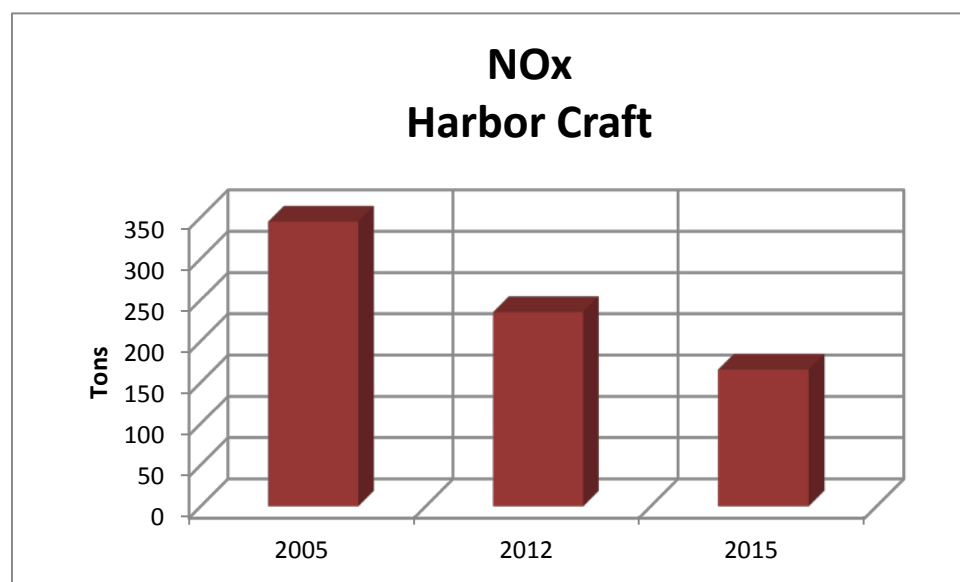


Figure 3-2. Harbor Craft NO_x emissions estimates for 2005, 2012, and 2015.

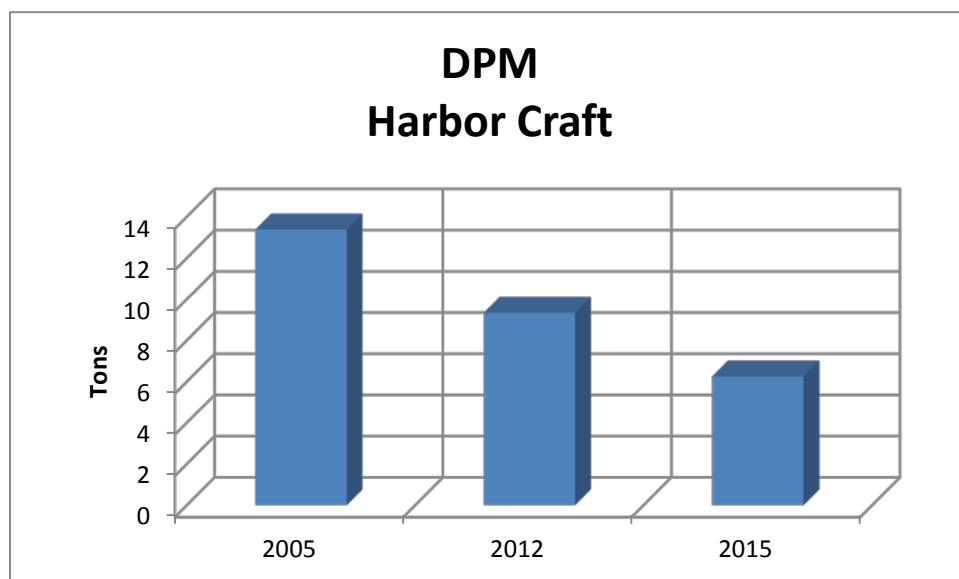


Figure 3-3. Harbor Craft DPM emissions estimates for 2005, 2012, and 2015.

4.0 CARGO HANDLING EQUIPMENT

This section documents the emission estimation methods and results for cargo handling equipment (CHE) operated at Port of Oakland terminals and the rail yard. This inventory does not include CHE at the Schnitzer facility and Union Pacific rail yard because those privately owned facilities are not part of the Port. This inventory includes only equipment listed as CHE; emissions estimates from other off-road equipment used at terminals are provided in Section 7.

4.1 Background

CHE is primarily used to transfer freight between modes of transportation, such as between marine vessels and trucks or between trains and trucks. CHE are used in many types of operations, but at the Port of Oakland, CHE is used almost exclusively to transfer shipping containers. As such, the types of CHE at the Port are limited to yard trucks (hostlers), rubber-tired gantry (RTG) cranes, top or side handlers (also called picks), and forklifts. Other types of equipment used as CHE for transfer of bulk materials were not found at the Port. Some general purpose equipment types including sweepers, construction, and other off-road equipment used for facility maintenance and construction, are included in the other off-road equipment category (see Section 7) and not in the CHE category.

4.2 Emission Calculation Methodology

The approach used to estimate CHE emissions was to determine annual 2015 emissions for each piece of equipment by terminal according to engine characteristics (equipment type, model year, rated power, and after-treatment retrofit control) and equipment operation (hours of operation and fuel consumption rates). The equipment population and operation estimates were derived from terminal and rail yard surveys conducted during the first part of 2016 by the Port of Oakland.

Operating firms at the Ben E. Nutter and the Outer Harbor terminals changed between 2015 and 2016, and the 2015 operators were unavailable to gather CHE survey data. Where there were missing data from the surveys, equipment numbers and engine characteristics were derived from the results of the equipment survey conducted for 2012 and from lists of equipment disbursed from discontinued operations at these terminals. Equipment-hours were scaled to the number of lifts at each of these two terminals for which 2015 survey data were unavailable.

Per ARB (2011c) guidance, the following types of equipment were used to categorize CHE:

- Cranes (including rubber tire gantry cranes);
- Forklifts;
- Container Handling Equipment (top or side handlers); and
- Yard Trucks.

At the Port, equipment listed as industrial or construction equipment is used primarily for sporadic maintenance and construction activity and is not considered to be CHE. The equipment activity and emissions for these equipment types are included in this emission inventory as described in Section 7.

CHE emissions were calculated using the following equation:

$$Equip_{emiss} = \frac{(EF_{zh} + dr \times CHrs) \times Engine_{bhp} \times FCF \times LF_{wt} \times CF \times Time_{hrs} \times Pop}{(453.6 \times 2000)}$$

Where:

Equip_{emiss} is the annual emissions in tons per year,
EF_{zh} is the zero-hour emission factor in grams per brake horsepower-hour,
dr is the deterioration rate or the increase in zero-hour emissions as the equipment is used (grams/bhp-hr²),
CHrs is the cumulative hours or total number of hours accumulated on the equipment,
FCF is the fuel control factor (% reduction) used to correct for emission reductions due to California diesel fuel,
LF_{wt} is the weighted load factor (average load expressed as a % of rated power),
CF is the control factor (% reduction) associated with use of emission control technologies where applicable,
Time_{hrs} is the annual operating hours of the equipment,
Pop is the population number of the equipment, and
(453.6 x 2000) is a conversion from grams to tons.

4.3 Input Data and Use

Surveys were sent out to the Port of Oakland terminals and rail yards requesting the following detailed information for each piece of CHE. This information was used as input for the emissions calculations.

1. Equipment Type
2. Numbers of Similar Equipment
3. Engine Model
4. Engine Model Year
5. Aftertreatment Retrofit Type
6. Chassis Make / Model
7. Chassis Model Year
8. Fuel Type
9. Annual hours of operation
10. Engine Rated horsepower

Surveys were returned for four facilities - two terminal operators were not available to provide data. For equipment specific operation and characteristics that were not provided, lists of equipment from the 2012 survey or equipment sold from discontinued operations were assumed to have retrofits as needed to comply with ARB rules. The total equipment hours by equipment type were scaled from 2012 results using the relative number of lifts at each terminal during 2012 and 2015.

For diesel-powered equipment, the zero-hour emission factors, deterioration rates, fuel correction factors, and emission control factors for HC, CO, NO_x, and PM were obtained from ARB's Cargo Handling Equipment Inventory (CHEI) model (ARB, 2012). Because the current version of the CHEI model does not support emission estimates for other pollutants or other fuel types, emission factors for gasoline and propane powered equipment, and for SO_x and CO₂, were obtained from ARB's 2011 CHE Calculator, (ARB 2011c) following methodologies described in the 2005 Mobile CHE at Ports and Intermodal Rail Yards original rulemaking (ARB, 2005). Emissions factors for greenhouse gases CO₂, CH₄ and N₂O were estimated using OFFROAD 2007 because they were unavailable from either the CHEI or the CHE Calculator. Note that the OFFROAD 2007 model reports N₂O emissions as zero for all of the equipment included in this inventory.

CHE were grouped into equipment type categories as defined by ARB (2011c). The resulting populations by equipment type for the Port of Oakland are summarized in Table 4-1. Out of 486 total pieces of cargo handling equipment, 433 were diesel powered, 28 were gasoline powered, and 25 were LPG (liquid petroleum gas) powered.

Table 4-1. Cargo handling equipment - population by type.

Equipment Type	Equipment Population	% Total
Container Handling Equipment	119	24%
Forklift	56	12%
RTG Crane	31	6%
Yard Tractor	239	49%
Yard Tractor On-road	41	8%
Total	486	100%

Table 4-2 summarizes the average horsepower and annual use by equipment type and power range. Actual annual hours of operation for each piece of equipment were used to estimate emissions.

Table 4-2. Cargo handling equipment - Average horsepower and actual hours of operation by equipment type and horsepower range.

Equipment Type	HP Bin	Equipment Population	Average HP	Average Annual Operation (Hours)
Container Handling Equipment	175	5	164	3,451
	300	40	235	1,274
	600	80	336	2,451
Forklift	100	31	99	505
	175	14	156	1,098
	300	11	211	660
RTG Crane Forklift	175	8	139	2,684
	600	7	315	891
	750	3	625	325
	9999	13	1,005	1,500
Yard Tractor	175	236	172	3,931
	300	3	178	4,177
Yard Tractor On-road	175	4	160	976
	300	37	224	2,050

4.4 Cargo Handling Equipment Emission Results

Table 4-3 and Table 4-4 present emission results for the CHE by equipment type and by fuel type, respectively, based on the survey data. All PM₁₀ from diesel engines listed in Table 4-4 is DPM. PM_{2.5} emissions were calculated as a fraction of PM₁₀ based on fuel type using factors provided by ARB (2013).

Table 4-3. 2015 Port of Oakland CHE emissions by equipment type (tons per year).

Equipment Type	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
Container Handling Equipment	21.50	65.69	176.42	0.27	2.29	2.11	14,079	1.58	0	14,118
Forklift	0.84	10.12	5.69	0.01	0.06	0.06	787	0.99	0	812
RTG Crane	2.39	6.77	35.62	0.03	0.20	0.18	2,177	0.48	0	2,189
Yard Tractor	16.84	159.74	104.07	0.19	1.18	1.09	12,448	0.93	0	12,471
Yard Tractor On-road	1.12	10.38	10.23	0.05	0.13	0.12	3,116	0.30	0	3,123
Total	42.68	252.69	332.02	0.55	3.86	3.55	32,606	4.28	0	32,713

Table 4-4. 2015 Port of Oakland CHE emissions by fuel type (tons per year).

Fuel Type	ROG	CO	NO _x	SO _x	PM ₁₀	PM _{2.5}	CO ₂	CH ₄	N ₂ O	CO ₂ e
Diesel	40.72	195.59	323.44	0.53	3.72 ^a	3.42	32,411	3.35	0.00	32,494
Gasoline	1.37	49.32	6.09	0.02	0.12	0.11	0.00	0.00	0.00	0
Propane	0.59	7.79	2.49	0.00	0.02	0.02	196	0.93	0.00	219
Total	42.68	252.69	332.02	0.55	3.86	3.55	32,606	4.28	0.00	32,713

^a All diesel PM₁₀ emissions are DPM.

Figures 4-1 and 4-2 show the CHE NOx and DPM emission estimates for 2005, 2012, and 2015. There has been a year by year reduction in both NOx and DPM as the fleet has turned over to lower emitting engines. Further emission reductions are expected, especially for DPM, because as the regulatory exemptions from the use of DPF on some older equipment expire, Tier 4 engines with DPFs will be installed.

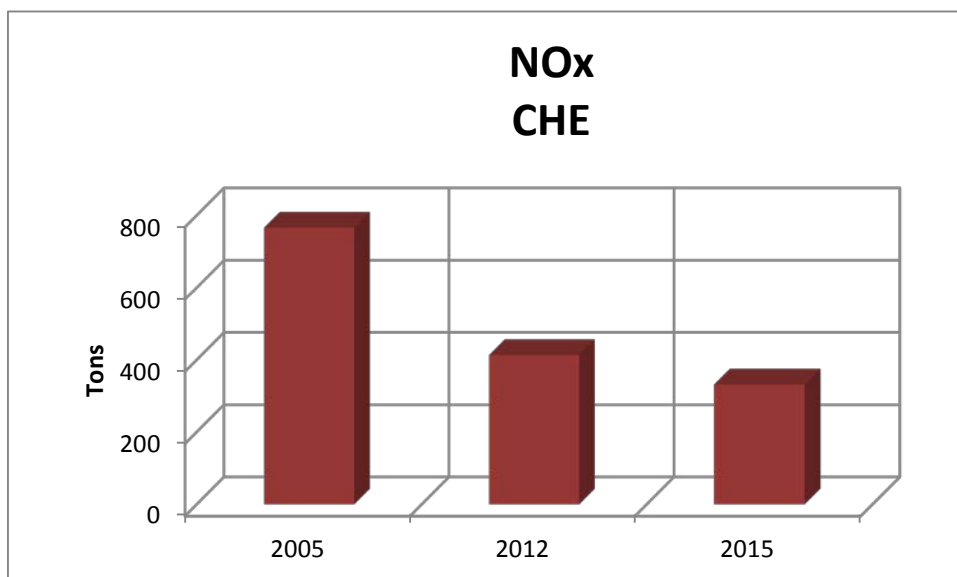


Figure 4-1. Cargo Handling Equipment NOx emissions estimates for 2005, 2012, and 2015.

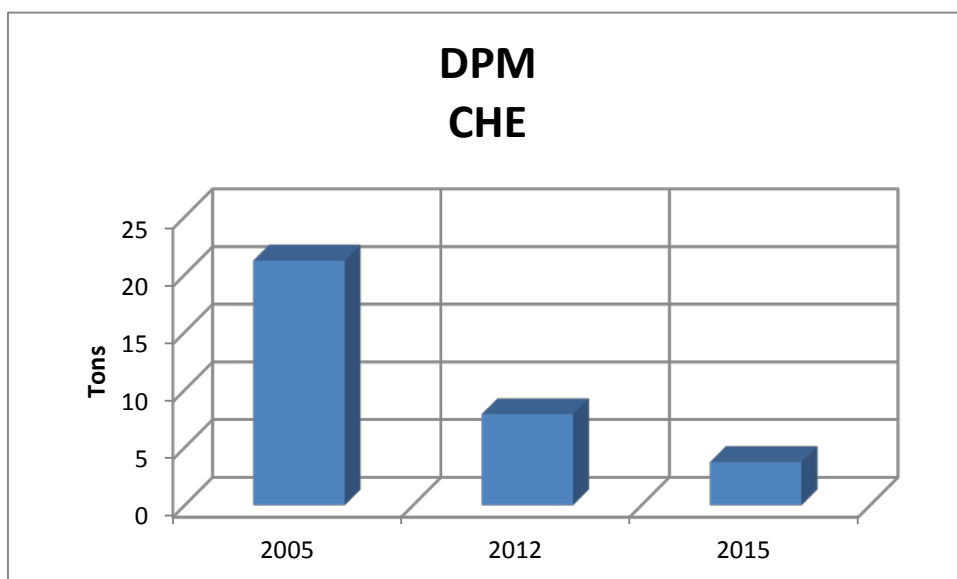


Figure 4-2. Cargo Handling Equipment DPM emissions estimates for 2005, 2012, and 2015.

5.0 ON-ROAD HEAVY-DUTY TRUCKS

Operations at the Port of Oakland create a demand for truck trips to transport containers between marine terminals, freeway interchanges, and nearby rail yards. Historically, emissions from on-road trucks servicing the Port (drayage trucks) have been an important component of diesel exhaust emissions at the Port. Prior to implementation of the California Air Resources Board's Drayage Truck Regulation, the average drayage truck was older than that of the general on-road truck fleet, resulting in higher emission rates. In addition, drayage trucks generally follow driving patterns consisting of shorter trips, lower average speeds and more stop-and-go driving which generally tend to result in higher emissions per mile traveled.

In 2009, the State of California instituted the Drayage Truck Regulation (ARB, 2009) in an effort to reduce emissions from the relatively old drayage truck fleet at that time. Under this regulation, by December 31, 2013, all drayage trucks engines were required to meet or exceed emission standards for 2007 model year engines. Different emission standards and compliance dates apply to non-drayage trucks.

The geographical boundaries of this Port of Oakland air emissions inventory include truck routes between the marine terminals and three nearby freeway interchanges and the two port area rail yards. Trucks must arrive at or depart from the Port area via the three freeway interchanges: Maritime/West Grand Street, Seventh Street, and Adeline/Market Street. Even if trucks arrive by surface streets, they must pass through one of these access points to enter the Port area. The Port emissions inventory also includes truck trips that move intermodal cargo containers between marine terminals and the two rail yards in the Port area: the Port's Oakland International Gateway (OIG) operated by BNSF and the Union Pacific rail yard.

The following sections describe the activity and emissions calculation methods for the 2015 drayage truck emission inventory, including the equations, assumptions, and the underlying truck activity data and emission factors. Truck activity in terms of trips to and from the Port's terminals were combined with emission factors from the ARB's on-road emissions factor model (EMFAC2014¹¹) to estimate emissions from the drayage trucks moving and idling within the Port area. A summary of the 2015 Port of Oakland truck emission inventory is provided at the end of this chapter.

5.1 Emission Calculation Methodology

Operating modes were separated into four categories: (1) idling inside marine terminals, (2) idling at gate queues, (3) driving within marine terminals, and (4) driving on surface streets between terminals and freeway interchanges or rail yards. For each of these modes, the average time and speed define the emissions for each trip.

Emissions per trip were calculated by multiplying the appropriate emission factor (idling or by speed) by the activity level indicator (idling time or trip distance). As expressed in the following

¹¹ <http://www.arb.ca.gov/msei/categories.htm>

equation, emissions are the product of the number of trips, distance per trip, and emission rate per mile traveled. For the idling calculation, the emissions are the product of number of trips, average idling time per trip, and emission rate per hour of idling.

$$E_p = (n_{truck\ trip})(miles_{trip})(EF_{p,trip})$$

Where:

- E_p = emissions of pollutant p ,
- n_p = number of trips,
- $miles_{trip}$ = trip mileage or hours at idle,
- $EF_{p, trip}$ = trip emission factor (grams/mile) or for idle (grams/hour) for pollutant p
(Requires trip-based EFs defined on the basis of individual link speeds as described below).

A “link” is a term used by transportation planners to describe a segment of roadway. A “trip” for this analysis refers to one-way travel along a multiple links pieced end-to-end. For example, one-way travel from the freeway interchange of I-880 at Adeline Street to Hanjin terminal is defined as one trip made up of seven links. Truck speeds differ by link, due to link-specific variables such as posted speed limits, traffic lights, and stop signs.

Inputs to the emissions calculations are:

1. Number of truck trips, traveling between
 - a. Marine terminal and freeway
 - b. Marine terminal and rail yard
 - c. Rail yard and freeway
2. Trip mileage
 - a. Outside terminals and rail yards
 - b. Within terminals and rail yards
3. Truck idling time
 - a. Entrance queues at terminals and rail yards
 - b. Within terminals and rail yards
4. Emission Factors derived from the EMFAC2014 model based on
 - a. Age distribution
 - b. Individual link speeds comprising a trip
 - c. Idle emission rate

5.2 Truck Trip Counts

Two data sources were used to estimate the number of truck trips: 1) a survey of gate counts and 2) container lifts. A gate count refers to the terminal recordkeeping of the number of trucks entering a marine terminal. Container lifts (i.e., the number of containers moved onto or off of a ship) provide a second data source by which to estimate the number of truck trips. Container lift data are considered to be reliable because payments to operators are based on the number of lifts. However, trucks may move a container in and out on a single terminal entry or move no containers at all when repositioning empty chassis or for other reasons.

The 2015 truck trip counts for the marine terminals were derived from gate counts provided by the Port or the terminal operators. For the OIG rail terminal, the reported number of lifts was doubled to estimate the sum of inbound and outbound truck trips. Table 5-1 summarizes the resulting estimated total number of truck trips for the Port area in 2015 and compares this with the number of lifts (defined as movement of one container whether a 20-foot or 40-foot container). If each truck carries one container either to or from the terminal, and no containers on the opposite leg, then the number of trips would be equal to twice the number of lifts. The fact that the number of trips is less than twice the number of lifts indicates that trucks are often moving more than one container during their visit to the terminal.

Table 5-1. On-road trucking – estimated truck trips in 2015.

Terminal Type	Reported 2015 truck trips	Lifts
Marine	1,884,558	1,294,532
Rail ¹	196,572	98,286

¹ Rail results are only reported for the rail yard located within the Port boundary (BNSF-operated OIG). Trips to the Union Pacific rail yard were assumed to be twice the number to the OIG rail yard.

5.3 Truck Trip Definitions

This section defines trip routes and link speeds for trucks traveling outside the marine terminals, between marine terminals and rail yards or any of the three freeway interchanges. In-terminal driving is discussed separately. The scope of this study precluded identifying the precise routes of individual trips. Instead, a simple but accurate method to capture the VMT and estimate trip speeds was determined using typical routes to and from each marine terminal.

As previously mentioned, one-way trips can occur between any marine terminal and any freeway interchange or rail yard as listed in Table 5-2. Trips to truck parking areas in the Port area (Roundhouse and Howard Terminal in 2015) are not included, since trips to and from the parking areas replace trips to and from freeway interchanges at the beginning and end of the day, or are short-term stopovers during the day. The emissions impacts are expected to be negligible.

Table 5-2. On-road trucking – list of marine terminals, freeway interchanges, and rail yards.

Berths	Terminal	Freeway Interchange
B 20-23, 24-26	PortsAmerica	Adeline/Market Street
B 30, 32	Trapac	7th Street
B 35, 37	Nutter	Grand/Maritime Street
B 55-56	OICT	
B 57-59	OICT	
B 60-63	Matson	
B 67-68	Howard	
		Rail yard
		OIG (BNSF)
		Union Pacific

These locations are shown on the Port of Oakland map in Figure 5-1. Roadway links numbered 0 through 33, which make up potential truck routes, are also labeled.



Figure 5-1. On-road trucking – roadway links within the Port of Oakland (2012 Terminal Configuration).

While the precise routes for truck trips between terminals and the highway are not known, geographic proximity influences which highway interchange truck drivers will prefer—Adeline/Market Street, 7th Street, or Grand/Maritime Street. The distribution of truck trips between freeway and Port terminals is shown in Table 5-3. This trip distribution is based on historic surveys conducted at the port (CCS, 2003) and the subsequent analysis of the data for the Port’s 2005 emission inventory (ENVIRON, 2008).

Table 5-3. On-road trucking distribution of truck trips between freeway and Port Terminals.

Berths	Terminal	Fraction of Traffic		
		Adeline/Market	7 th Street	West Grand/ Maritime
B 20-26	PortsAmerica	0%	30%	70%
B 30-32	Trapac	0%	65%	35%
B 35-37	Nutter	0%	65%	35%
B 55-56	OICT	0%	65%	35%
B 57-59	OICT	5%	65%	30%
B 60-63	Matson	40%	40%	20%
B 67-68	Howard (idled during 2015)	100%	0%	0%

Based on the preferred routes indicated in Table 5-3, individual links were combined to create realistic trip routes to assign to the total trip counts. Table 5-4 lists all possible constructed trips, their constituent links, total distance, and average speed. The trip distances are summed over individual links that comprise the trip. Reported average speeds are the VMT-weighted averages of the links by trip. The same link-level speeds were determined from a previous study performed for the 2005 and 2012 calendar year inventories.

Table 5-4. On-road trucking – trip IDs, constituent link IDs, total distance, and average speeds.

Trip ID	Terminal	Berth	Trip Beginning/ End	Road Link Segments, One-way	One-way Trip Length (feet)	Average Speed (mph)
T1	PortsAmerica	B 20-23	West Grand	0, 28	3,193	30
T2	PortsAmerica	B 20-23	7th	0, 1, 9, 31, 15	6,780	32
T3	PortsAmerica	B 20-23	Adeline	0, 1, 9, 31, 16, 21, 13, 19, 24, 33, 25	15,635	31
T4	PortsAmerica	B 20-23	BNSF	0, 1, 9, 31, 16, 17	8,816	29
T5	PortsAmerica	B 20-23	Union Pacific	0, 1, 9, 31, 16, 21, 13, 19	12,189	32
T6	PortsAmerica	B 24-26	West Grand	2, 1, 28	6,401	34
T7	PortsAmerica	B 24-26	7th	2, 9, 31, 15	4,580	26
T8	PortsAmerica	B 24-26	Adeline	2, 9, 31, 16, 21, 13, 19, 24, 33, 25	13,435	29
T9	PortsAmerica	B 24-26	BNSF	2, 9, 31, 16, 17	6,616	24
T10	PortsAmerica	B 24-26	Union Pacific	2, 9, 31, 16, 21, 13, 19	9,989	29
T11	Trapac	B 30	West Grand	5, 4, 3, 29, 9, 1, 28	9,888	33
T12	Trapac	B 30	7th	5, 4, 3, 30, 15	6,280	30
T13	Trapac	B 30	Adeline	5, 4, 11, 20, 13, 19, 24, 33, 25	13,462	34
T14	Trapac	B 30	BNSF	5, 4, 3, 30, 16, 17	8,316	27
T15	Trapac	B 30	Union Pacific	5, 4, 11, 20, 13, 19	10,016	36
T16	Trapac	B 32-33	West Grand	6, 7, 4, 3, 29, 9, 1, 28	11,301	32
T17	Trapac	B 32-33	7th	6, 7, 4, 3, 30, 15	7,693	29
T18	Trapac	B 32-33	Adeline	6, 7, 4, 11, 20, 13, 19, 24, 33, 25	14,875	34
T19	Trapac	B 32-33	BNSF	6, 7, 4, 3, 30, 16, 17	9,729	28
T20	Trapac	B 32-33	Union Pacific	6, 7, 4, 11, 20, 13, 19	11,429	35
T21	Nutter	B 34-35, 37-38	West Grand	8, 7, 4, 3, 29, 9, 1, 28	12,474	34
T22	Nutter	B 34-35, 37-38	7th	8, 7, 4, 3, 30, 15	8,866	33
T23	Nutter	B 34-35, 37-38	Adeline	8, 7, 4, 11, 20, 13, 19, 24, 33, 25	16,048	35
T24	Nutter	B 34-35, 37-38	BNSF	8, 7, 4, 3, 30, 16, 17	10,902	30
T25	Nutter	B 34-35, 37-38	Union Pacific	8, 7, 4, 11, 20, 13, 19	12,602	37
T26	OICT	B 55-56	West Grand	10,11, 3, 29, 9, 1, 28	11,201	33
T27	OICT	B 55-56	7th	10,11, 3, 30, 15	7,593	30
T28	OICT	B 55-56	Adeline	10, 20, 13, 19, 24, 33, 25	11,555	32
T29	OICT	B 55-56	BNSF	10, 20, 21, 17	7,068	32
T30	OICT	B 55-56	Union Pacific	10, 20, 13, 19	8,109	34
T31	OICT	B 57-59	West Grand	18, 21, 16, 31, 9, 1, 28	11,849	32
T32	OICT	B 57-59	7th	18, 21, 16, 15	7,534	28
T33	OICT	B 57-59	Adeline	18, 13, 19, 24, 33, 25	8,307	27
T34	OICT	B 57-59	BNSF	18, 21, 17	3,820	21
T35	OICT	B 57-59	Union Pacific	18, 13, 19	4,861	26
T36	Matson	B 60-63	West Grand	22, 19, 13, 21, 16, 31, 9, 1, 28	15,632	31
T37	Matson	B 60-63	7th	22, 19, 13, 21, 16, 15	11,317	28
T38	Matson	B 60-63	Adeline	22, 24, 33, 25	5,214	25
T39	Matson	B 60-63	BNSF	22, 19, 13, 21, 17	7,603	25
T40	Matson	B 60-63	Union Pacific	22	1,768	15
T41	Howard	B 67-68	West Grand	27, 26, 32, 24, 19, 13, 21, 16, 31, 9, 1, 28	19,074	32
T42	Howard	B 67-68	7th	27, 26, 32, 24, 19, 13, 21, 16, 15	14,759	30
T43	Howard	B 67-68	Adeline	27, 26, 32, 33, 25	3,720	23
T44	Howard	B 67-68	BNSF	27, 26, 32, 24, 19, 13, 21, 17	11,045	28
T45	Howard	B 67-68	Union Pacific	27, 26, 32, 24	5,210	28

5.4 Truck Idling and VMT inside Terminals

The vehicle miles traveled (VMT) within marine and rail terminals is limited to driving between the terminal gates and container storage areas. Previously, the Port conducted surveys of terminal operators to determine in-terminal VMT and average speed.

These previous survey data were used to estimate 2015 activity (per-truck speed, distance, and idling time). Note that idling may have increased at times during 2015 due to the reported slowdown at the marine terminals resulting in longer wait times (see text box).

Potential Impact of Longer Terminal Wait Times in 2015: Normal port operations were affected during a portion of 2015 by a labor dispute which at times resulted in longer wait times and increased truck idling at the marine terminals. A sensitivity analysis was performed to evaluate the potential impact on emissions of longer truck idling times in 2015 – see Appendix B.

Table 5-5 below shows the activity summary for the average truck idling at gates, idling in terminal, and driving in-terminal along with average speed in-terminal.

Table 5-5. On-road trucking – average in-terminal activity parameters.

Mode	Average estimate (based on 2012 and 2005 survey data)
Idling at gate (hrs)	0.17
Idling in terminal (hrs)	0.34
Distance traveled (mi)	2.59
Speed (mph)	13.3

5.5 Emission Factors and Age Distribution

Ramboll Environ used the California ARB's on-road emission factor model EMFAC2014 to calculate emission factors for trucks idling and moving in the Port area. Emission factors from on-road trucks depend on the age distribution of the trucks and site conditions such as temperature, humidity, and especially average speeds. The age distribution is particularly important because of ARB's drayage truck regulations that affect specific model years, causing steep declines in NO_x and PM in 2003 as shown in Figure 5-2. The EMFAC2014 model accounts for the benefits of the drayage truck regulations applicable to calendar year 2015, including:

1. Model years 2008-2010 meet 2007 engine emission standards for NO_x and PM.
2. Model years 2010 and newer meet 2010 engine emission standards for NO_x.

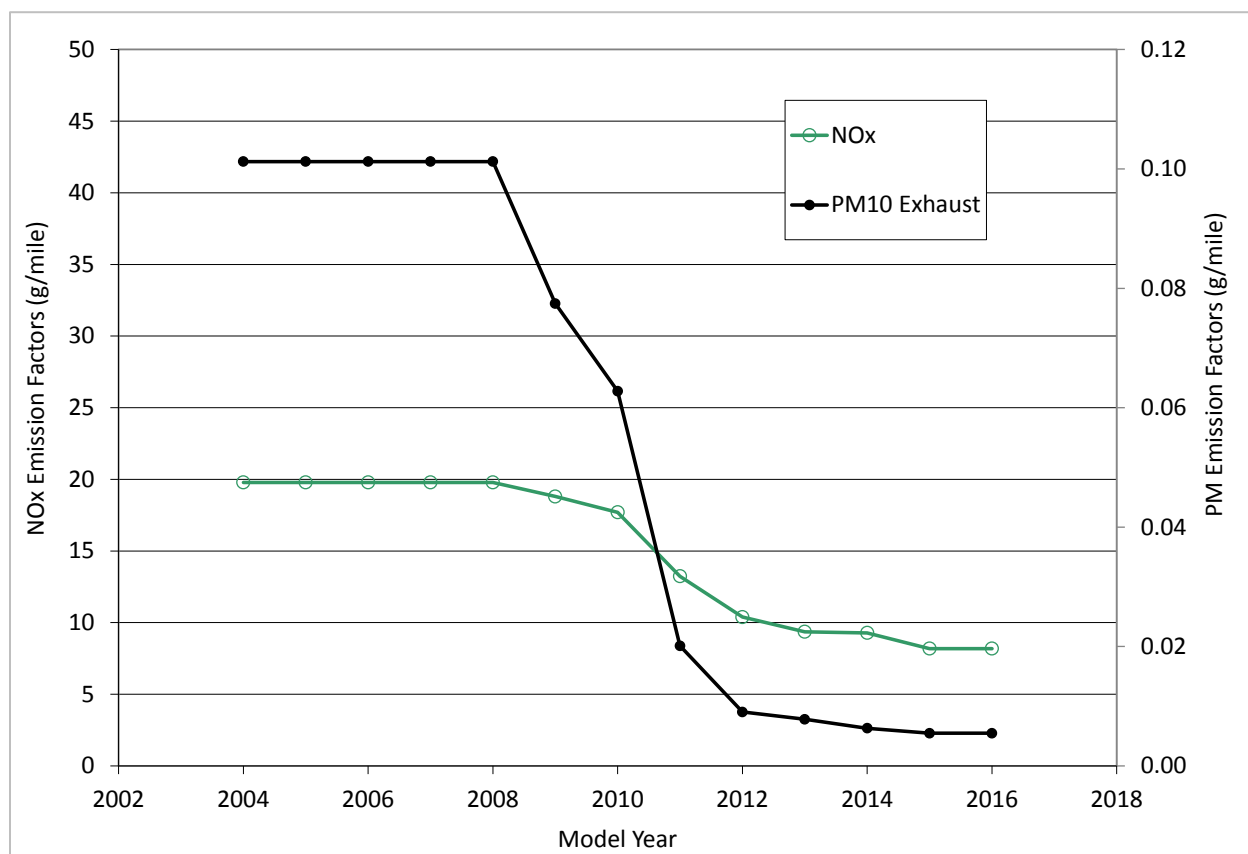


Figure 5-2. 2015 calendar year drayage truck EMFAC2014 emission factors by model year for PM and NO_x at 10 mph.

The truck age distribution used in this analysis was developed from end of year 2015 registration data¹² collected by the Port under the Secure Truck Enrollment Program (STEP). Approximately 8,000 trucks are registered for STEP, although not all of the registered trucks currently work at the Port. This truck fleet included a significant numbers of trucks of older model years which would not originally have 2007 engines and thus would have been prohibited from performing drayage under the ARB rule. We therefore assumed that these older trucks would have been repowered with a 2007 engine and considered to be equivalent to the 2008 model year trucks. The resulting age distribution is shown in Figure 5-3, along with emission factors for multiple pollutants by model year. Emission factors shown in Figure 5-3 represent emissions per mile for an average speed of 10 miles per hour (mph).

¹² <http://www.portofoakland.com/port/seaport/comprehensive-truck-management-program/registry/>

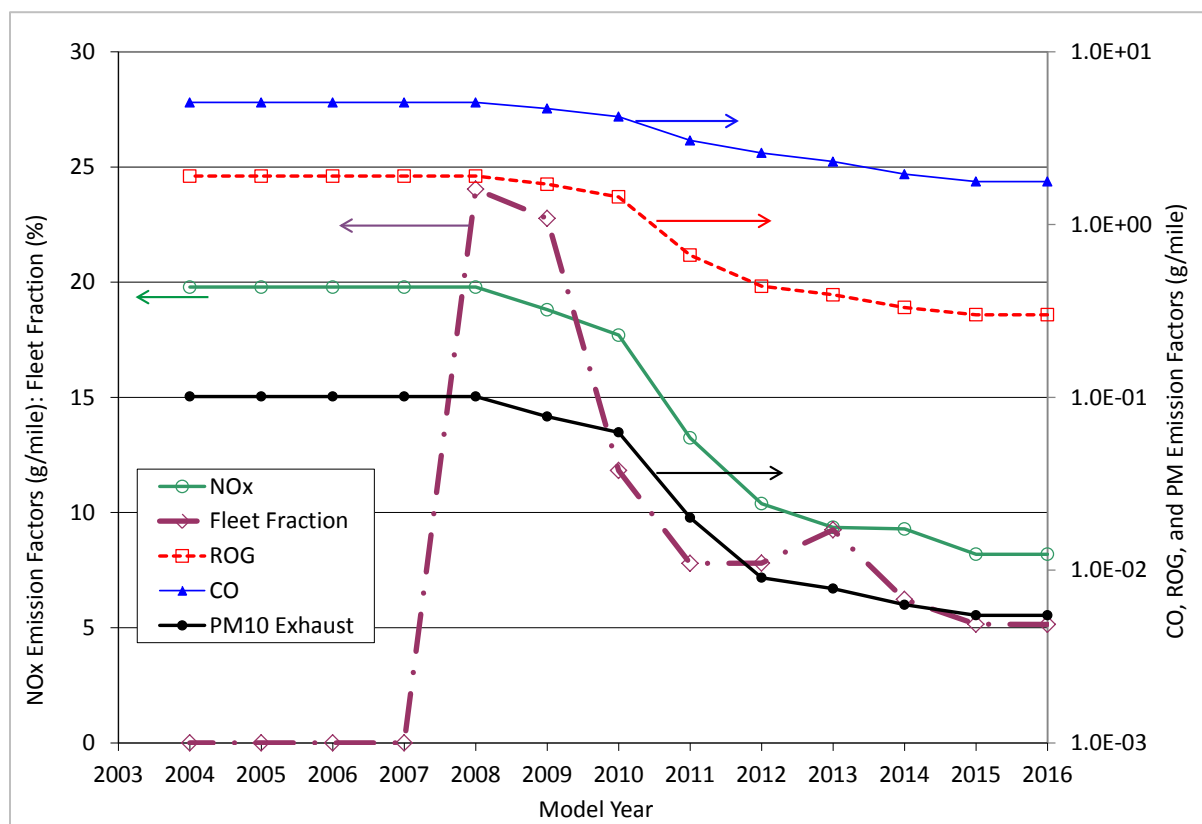


Figure 5-3. Drayage truck emission factors (at 10 mph) and age distribution.

The age distribution (fleet fraction) shown in Figure 5-3 indicates that model years 2008, 2009, and 2010 (using engine model years 2007, 2008, and 2009 respectively) comprised the largest percentage (59% together) of the Port's truck fleet in 2015. These truck engines produce higher emissions than 2011 and later. All trucks must have 2007 and later model year engines to enter Port terminals and rail yards.

Table 5-6 lists all emission factors for the Port's truck fleet in 2015, including idling (grams/hour) and driving (grams/mile) by speed.

Table 5-6. Port of Oakland specific average drayage truck emission factors in 2015.

Speed	ROG	CO	NO _x	PM ₁₀ Total	PM ₁₀ Exhaust	PM _{2.5} Total	Unit
Idle	1.78	7.15	49.50	0.01	0.01	0.01	g/hour
5 mph	1.48	4.64	18.16	0.16	0.06	0.09	g/mile
10 mph	1.19	3.75	15.26	0.15	0.05	0.09	g/mile
15 mph	0.84	2.64	11.60	0.14	0.05	0.08	g/mile
20 mph	0.60	1.90	9.20	0.14	0.04	0.07	g/mile
25 mph	0.44	1.40	7.88	0.13	0.04	0.07	g/mile
30 mph	0.33	1.03	7.15	0.13	0.03	0.07	g/mile
35 mph	0.24	0.76	6.63	0.13	0.03	0.07	g/mile
40 mph	0.18	0.57	6.24	0.13	0.03	0.06	g/mile
45 mph	0.13	0.42	5.93	0.13	0.03	0.06	g/mile

5.6 Drayage Truck Emissions Results

Drayage trucks that provided service to the Port of Oakland marine terminals and rail yards emitted approximately 87 tons of NO_x and less than 0.4 ton of diesel PM (DPM) within the Port area during 2015 as shown in Table 5-7. All trucks use diesel engines, so the PM₁₀ exhaust emissions are DPM emissions but total PM₁₀ and total PM_{2.5} also include non-diesel PM (i.e., brake and tire wear). Trucks traveling on surface roads represented the largest source of emissions of NO_x, PM, and SO_x. For the pollutants ROG and CO, the largest contributors were from in-terminal driving. This demonstrates the relative importance of each source area for different pollutants. Idling and slow speed driving produces higher emission rates for all pollutants, but for some pollutants the difference is more extreme. For example, CO has much higher emission rates during idling than during driving (refer back to Table 5-6), relative to the other pollutants. For PM and SO_x, idling contributed a relatively minor amount to total emissions. The PM_{2.5} size fraction estimate was derived from ARB (2013) estimates for diesel exhaust, tire and brake wear.

Table 5-7. 2015 total emissions by trucks within the terminal and outside the terminal to the nearest freeway entrance (tons per year).

Emission Category	Emissions (tons/year)										
	ROG	CO	NO _x	PM ₁₀ Total	PM ₁₀ Exhaust ^a	PM _{2.5} Total	SO _x	CO ₂	CH ₄	N ₂ O	CO ₂ e
Surface roads	2.21	7.02	46.93	0.83	0.22	0.43	0.11	12,265	0.13	0.39	12,385
Gate idling in queue	0.42	1.68	11.61	0.002	0.002	0.002	0.02	1,819	0.02	0.006	1,838
In terminal idling	0.71	2.84	19.66	0.004	0.004	0.004	0.03	3,081	0.04	0.10	3,113
In terminal driving	2.68	8.45	35.85	0.40	0.13	0.23	0.05	7,564	0.16	0.17	7,619
Truck totals	6.02	19.98	86.65	1.24	0.36	0.662	0.21	24,729	0.35	0.72	24,954

^a PM₁₀ exhaust emissions are DPM.

The drayage truck NO_x and DPM emission estimates for 2005, 2012, and 2015 are shown in Figures 5-4 and 5-5. Due to restrictions on older trucks and fleet turnover, NO_x emissions decreased 60% between 2005 and 2012 and decreased an additional 36% between 2012 and 2015, for an overall NO_x emission reduction of 74% from 2005 to 2015. Similarly, DPM emissions decreased 88% between 2005 and 2012 and decreased an additional 82% between 2012 and 2015, for an overall DPM emission reduction of 98% from 2005 to 2015. All trucks now use aftertreatment devices to control emissions.

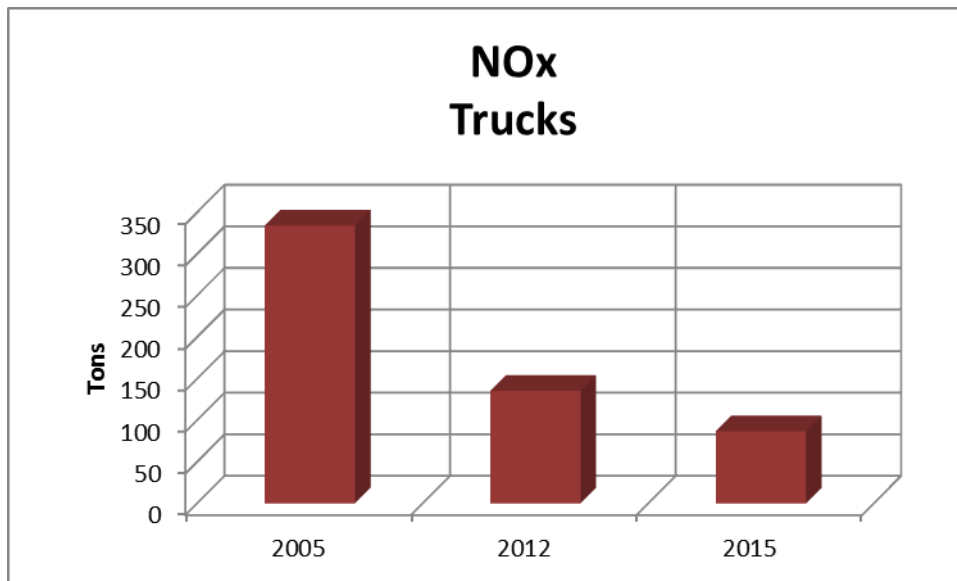


Figure 5-4. Drayage truck NOx emission estimates for 2005, 2012, and 2015.

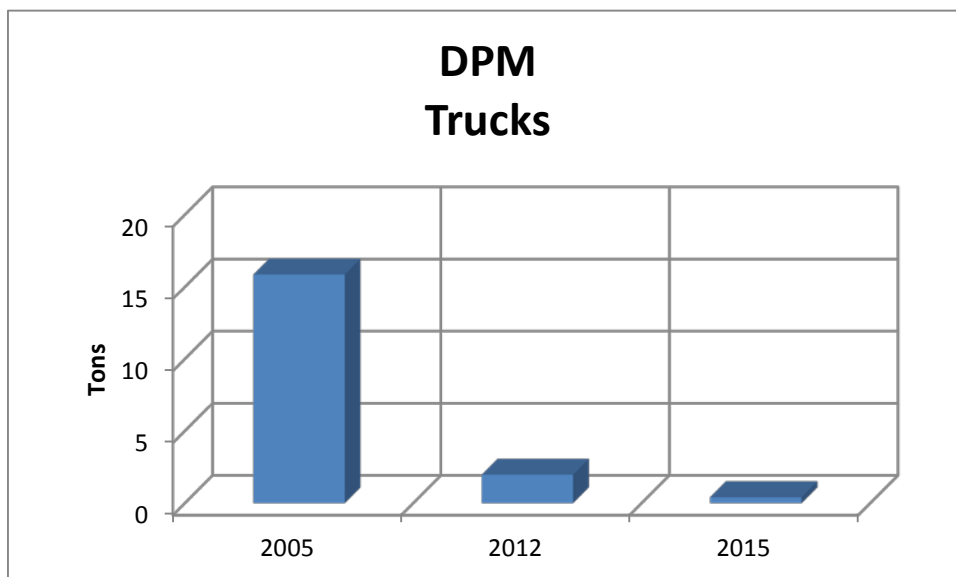


Figure 5-5. Drayage truck DPM emissions estimates for 2005, 2012, and 2015.

6.0 LOCOMOTIVE EMISSIONS

This section describes the data and methods used in estimating emissions from locomotives at the Oakland International Gateway (OIG) rail yard. OIG is a Port of Oakland terminal under lease and operated by the Burlington Northern Santa Fe (BNSF) railway. The Union Pacific (UP) rail yard (also known as UP Railport – Oakland), which sits adjacent to the Port terminals and serves as an intermodal yard for freight movements through the Port as well as a yard for domestic non-Port freight handling, is not considered in this evaluation because the UP yard is privately owned. Union Pacific provided ARB an independent analysis of the emissions in their Oakland facility (UP, 2007).

Locomotives are used for line-haul operations (long haul trains into and out of California) and switching operations (moving individual or small numbers of rail cars to make up trains). Line-haul locomotives move into and out of the rail yard with idle periods after arrival and prior to departure. Switching engines work in the yard with idle periods interspersed throughout the day. Line-haul and switching locomotives can undergo maintenance, engine load testing, and refueling at some rail yards. However, maintenance and load testing is not performed at the OIG. Refueling of locomotives may occur at the OIG but only infrequently.

Locomotives operate using a series of load modes called “notches”. These notches and the locomotive idle periods constitute the operating profile for locomotives. The ARB (2006) guidance for rail yard emission modeling requires per engine model per mode emission rates to be used with average time in mode profiles for each visit multiplied by the number of engines visiting the rail yard.

6.1 Summary of Locomotive Emission Factors by Engine Model

Emission factors and fuel consumption by notch used in this study are the same as those used in previous Port of Oakland Seaport Air Emissions Inventories with adjustments to incorporate idle reduction devices on line-haul locomotives and in-use fuel.

Since 2012, locomotive fuel must be at or below 15 ppm fuel sulfur nationwide, and meet the same properties as on-road diesel when fueled within California. Line-haul locomotives may be fueled out of state, and therefore the fuel may not necessarily comply with California standards.

Emissions rates data by operating mode and by model were developed using fuel with 0.3% (or 3,000 ppm) sulfur, and emissions rates were adjusted to the 2015 in-use fuel at 15 ppm. The methodology described by ARB (2015) was used to adjust emissions and is shown in the following adjustment equation:

$$\text{PM Adjustment (lb/hour)} = \text{Fuel consumption (gal/hr)} * 7.1 * 0.02247 * (224/32) * (0.000015 - 0.003)$$

In addition, ARB (2015) expected that California diesel fuel would lower NO_x emissions by 3% (0.97 adjustment factor) and PM by 7% (0.93 adjustment factor). These adjustments were

applied to the switching locomotive emission factors, but not the line-haul emission factors because line-haul locomotives may be fueled outside of California.

No emissions data were available for rebuilt Tier 0, 1, and 2 engines or new Tier 3 and 4 engines, so the emission factor ratio adjustments shown in Table 6-1 were applied to the pre-rebuild engine emission rates using the EPA estimated emission factors (EPA, 2009). No change in CO or fuel consumption was expected from rebuilds, and Tier 2 rebuild (labeled 2+) emission rates were assumed the same as for Tier 3 engines because the emission standards are identical.

Table 6-1. Emission ratio due to rebuild or new emission standards.

Tier	THC	NO _x	PM
0+ / 0	0.625	0.837	0.625
1+ / 1	0.617	1.000	0.625
2+ / 2	0.500	0.900	0.444
3 / 2	0.500	0.900	0.444
4 / 2	0.154	0.182	0.083

To estimate methane (CH₄) and nitrous oxide (N₂O) emissions, a ratio was applied to THC emissions and fuel consumption, respectively. The CH₄/THC ratio was determined using the ARB SPECIATE TOG profile number 818 for diesel engines, which provides the weight fraction of methane and other chemical species in the exhaust emissions. The fraction of TOG that is THC was determined by subtracting the weight fraction of the oxygenated species (alcohol, aldehydes, and ketones) that do not respond to the flame ionization detection method that is used to measure THC. The N₂O estimate was derived from the emission factor of 0.018 g/kW-hr available in the ARB (2016) emission inventory tool for Ocean-Going Vessels and dividing by an assumed average fuel consumption of 210 g/kW-hr. This leads to an N₂O emission factor of 0.039 g/lb-fuel.

6.2 Overview of the OIG Yard

BNSF uses the OIG as a near dock transfer point for Port of Oakland maritime cargo containers. Only Port containers are handled at this yard. As shown in the schematic of the Port terminals in Chapter 5, the OIG is situated along a generally northwest-southeast axis. Entrance and exit tracks curve north and east of the main yard. Locomotives and trains enter the general port area from the north via the Union Pacific (UP) main line, and leave in the same direction via tracks going north through Richmond and then onto BNSF lines leading out of the Bay Area.

6.3 Locomotive Facility Operations

The OIG locomotive operations consist primarily of two activities including line-haul locomotive movements for train arrival and departure and switching locomotive movements to break arriving and build departing trains.

Because different locomotive types and engine models have different emission characteristics, it was necessary to characterize the types and models of the locomotives that are operated at OIG based on data provided by BNSF. Locomotive types and models for each type of railyard activity are described below.

6.3.1 Switching Engine Movements

Switching engine fleet characteristics in the OIG area were determined from a sample of engines operating at OIG in 2012 made available by BNSF (2016). Switching engines assigned to OIG rotate in and out of service, but the typical type found at the yard in early 2016 was a GP25 model shown in Table 6-2. The average emission rates of two typical locomotive engine surrogates for which data are available and which bracket the power of the locomotive used at the yard was used to estimate emissions of the in-use switching locomotive.

Table 6-2. Locomotive – Switching engine characterization for the OIG facility in 2015.

Locomotive Model	Certification Tier	HP	Number of Engines	Engine Surrogate
GP25	Precontrolled or Tier 0	2500	1	Average of GP-3x (2000 hp) and GP-4x (3000 hp)

The time in mode for switching engine activity from the 2005 Port of Oakland emission inventory (ENVIRON, 2008) was used for this work and is shown in Table 6-3.

Table 6-3. Locomotive – Switching engine relative time in mode at the OIG facility in 2005.

Throttle Notch	Time in Mode
DB	1.4%
Idle	59.8%
1	6.6%
2	15.0%
3	9.5%
4	4.4%
5	1.9%
6	0.3%
7	0.0%
8	1.0%

Source: Port of Oakland 2005 Seaport Air Emissions Inventory, (ENVIRON 2008)

Total switching engine activity in 2015 was estimated using the early 2016 switch locomotive schedule. This activity consisted of one engine operating a 7.5 hour shift per day, every day, which is equivalent to 2,738 hours per year. (BNSF 2016)

Estimated annual THC, CO, NO_x, and diesel PM emissions for switching activities at the OIG facility are presented in Table 6-4. The PM_{2.5} size fraction of PM₁₀ was estimated to be 0.92, consistent with the ARB (2015) Vision Locomotive Module.

Table 6-4. Locomotive - Estimated annual emissions (tons/year) associated with switching engine activity at the OIG facility in 2015.

ROG	CO	NO _x	PM ₁₀	PM _{2.5}	SO _x	CO ₂	CH ₄	N ₂ O	CO ₂ e
0.59	1.38	12.13	0.19	0.18	0.00	489	0.03	0.01	494

6.3.2 Train Arrival and Departures in the Yard

The primary locomotive activity at OIG was from arriving and departing line-haul locomotives and their operation throughout the yard. Activities of line-haul engines in the OIG yard include: arriving with a train, separating from the train, perhaps moving to the ready area where the engines are assigned to a train, and assigned to a train and leaving the yard. BNSF provided the locomotive counts by models that arrived at the yard in 2015 as shown in Table 6-5. The number of engines moving through the yard was determined to be 1,004 in 2015 (compared with 1,189 in 2012 and 2,190 in 2005) based on a BNSF-supplied train arrival and departure database.

Table 6-5. Locomotive – Fleet characterization for locomotive arrival and departure at the OIG facility in the OIG facility in 2015.

Model	Tier	Fleet Fraction	Count
Dash 9	0	4.4%	44
Dash 9	0+	0.1%	1
SD70	0	0.2%	2
Dash 9	1	3.9%	39
Dash 9	1+	33.3%	335
SD70	1+	0.1%	1
ES44	2	21.2%	213
SD-70	2	0.2%	2
ES44	2+	7.6%	76
ES44	3	27.6%	278
SD70	3	0.3%	3
ES44	4	1.2%	12

In the 2005 and 2012 Port of Oakland Seaport Emission Inventories, samples of line-haul engine activity while in the yard were used to develop the average time in mode for line-haul locomotive arriving and departing from the yard. Because all or nearly all line-haul locomotives now use automatic idle shut-off devices restricting the idle on to 15 minute per event, the idle time was adjusted to 1.0 hour assuming 4 in-yard movements per arrival and departure. The average time in mode data are summarized in Table 6-6.

Table 6-6. Locomotive – Time in mode per trip for arriving and departing locomotives at the OIG facility in 2015.

Throttle Notch	Average Operation Time (hours)
DB ^a	0.2963
Idle	1.00 ^b
1	0.1726
2	0.0758
3	0.0340
4	0.0049
5	0.0059
6	0.0004
7	0.0036
8	0.0017

^a Dynamic Braking

^b Adjusted from 2005 activity to account for idle shut-off devices

The fleet characterization for locomotives, provided in Table 6-5, was derived from all engines entering the site in 2015, and the operating profile described in Table 6-6 was used to estimate the emissions by model and summed for total emissions. The diesel emission estimates for BNSF freight movements during 2015 are presented in Table 6-7.

Table 6-7. Locomotive emissions (tons/year) from arriving/departing locomotives at the OIG in 2015.

ROG	CO	NO _x	PM ₁₀	PM _{2.5}	SO _x	CO ₂	CH ₄	N ₂ O	CO ₂ e
0.10	0.22	1.70	0.03	0.03	0.00	149	0.01	0.00	151

6.4 Summary Locomotive Emission Estimates for OIG

The locomotive emissions for the OIG facility are summarized in Table 6-8. Note that all locomotive PM emissions are classified as diesel particulate matter (DPM).

Table 6-8. Locomotive – Estimated annual locomotive emissions (tons) at the OIG facility - 2015.

Source Type	ROG ^a	CO	NO _x	PM ₁₀ ^b	PM _{2.5}	SO _x	CO ₂	CH ₄	N ₂ O	CO ₂ e
Switching Engines	0.59	1.38	12.13	0.19	0.18	0.00	489	0.03	0.01	494
Train Arrival / Departure	0.10	0.22	1.70	0.03	0.03	0.00	149	0.01	0.00	151
Total	0.69	1.60	13.82	0.23	0.21	0.01	639	0.04	0.02	645

^a ROG to THC ratio for diesel engines used 1.053

^b All PM₁₀ emissions are DPM.

The NOx and DPM locomotive emission estimates for 2005, 2012, and 2015 are shown in Figures 6-1 and 6-2. The emissions have decreased from one year to the next because of fleet turnover to lower emitting engines and implementation of idle reduction measures and equipment.

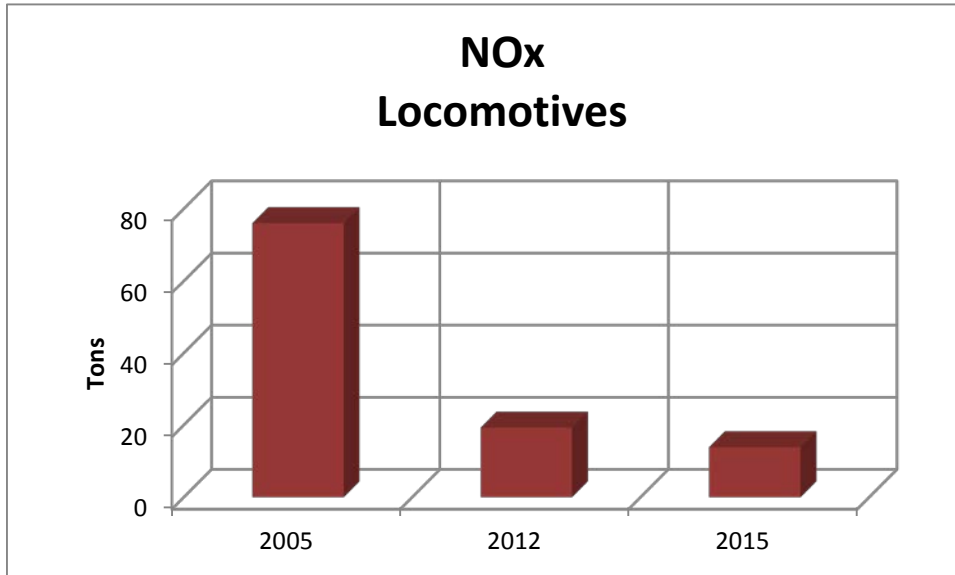


Figure 6-1. Locomotive NOx emission estimates for 2005, 2012, and 2015.

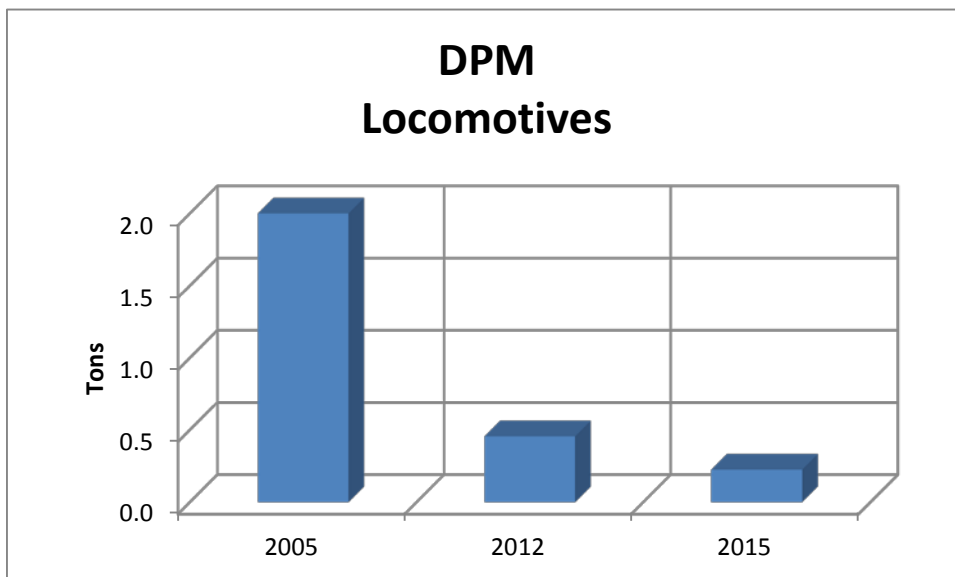


Figure 6-2. Locomotive DPM emission estimates for 2005, 2012, and 2015.

7.0 OTHER OFF-ROAD EQUIPMENT

This section documents the emission estimation methods and results for construction and maintenance equipment operated at Port of Oakland terminals and the rail yard. This inventory does not include equipment at the Schnitzer facility and Union Pacific rail yard because those privately owned facilities are not part of the Port.

7.1 Background

Off-road equipment considered in this section include general industrial and construction equipment that are most often used for sporadic maintenance and construction activity occurring at the Port. They are not to be confused with cargo-handling equipment (CHE), which is primarily used to transfer shipping containers or intermodal freight cargo. The CHE activities and emissions are discussed in this emission inventory under Section 4. In this section, there are three sources off-road equipment considered: (1) facility maintenance and construction at each terminal, (2) Port of Oakland general maintenance, and (3) construction at the Oakland Army Base.

7.2 Emission Calculation Methodology

To estimate the annual 2015 off-road equipment emissions, a list of equipment including engine characteristics (model year, rated power, and equipment type) and equipment operation (hours of usage and fuel consumption rates) were collected from terminal operators and the Port. The terminal operators' equipment population and operation estimates by terminal were derived from surveys conducted by the Port of Oakland. Fleet data for the Port's general maintenance equipment and equipment used for Oakland Army Base construction were provided by the Port. Where there were missing data, default input estimates were obtained from the applicable inventory guidance documentation (ARB, 2007 and 2010).

The types of construction and maintenance equipment considered in this inventory include:

- Aerial Lifts
- Air Compressors
- Cranes
- Excavators
- Forklifts
- Generator Sets
- Graders
- Lifts
- Other Construction Equipment
- Other General Industrial Equipment
- Pavers
- Paving Equipment
- Pumps
- Rollers
- Rubber Tired Dozers
- Rubber Tired Loaders
- Scrapers
- Skid Steer Loaders
- Surfacing Equipment
- Sweepers/Scrubbers
- Tractors/Loaders/Backhoes
- Welders

Off-road equipment emissions were calculated using the following equation:

$$Equip_{emiss} = \frac{EF_{adj} \times Engine_{bhp} \times LF_{wt} \times Time_{hrs} \times Pop}{(453.6 \times 2000)}$$

Where:

$Equip_{emiss}$ is the annual emissions in tons per year,
 EF_{adj} is the emission factor adjusted for deterioration, in grams per brake horsepower-hour,
 $Engine_{bhp}$ is the brake horsepower of the engine,
 LF_{wt} is the weighted load factor (average load expressed as a % of rated power),
 $Time_{hrs}$ is the annual operating hours of the equipment,
 Pop is the population (number of the equipment), and
 (453.6×2000) is a conversion from grams to tons.

7.3 Input Data and Use

For terminal maintenance equipment, the same surveys as those presented for CHE (Section 4) were used. Off-road equipment included in those survey responses that were characterized as “non-CHE” are included in this section. The Port provided the rest of the maintenance and construction equipment data. For equipment specific operation and characteristics that were not provided, default assumptions from the off-road emissions inventory guidance documentation (ARB, 2007 and 2010) were used.

A combination of the OFFROAD 2007 and OFFROAD 2011 models were used to estimate emissions. Because emission factors are back-calculated from these inventory models, they are adjusted for engine deterioration. For diesel-powered equipment, the emission factors for HC, NOx, and PM were derived from OFFROAD 2011. Because this newer version of the OFFROAD model does not support emission estimates for other fuel types (emission factors for gasoline) and for other pollutants (CO, SOx and greenhouse gases), these were obtained from the OFFROAD 2007 model.

Populations of off-road equipment evaluated in this section are summarized in Table 7-1 below. Among 231 pieces of construction and maintenance equipment at the Port of Oakland in 2015, 137 were diesel powered (59% of total), 92 were gasoline powered (40% of total), and 2 were propane powered (1% of total). Although average horsepower and average annual hours are shown in Table 7-1, actual horsepower and actual annual hours of operation for each piece of equipment were used to estimate emissions.

Table 7-1. Construction and maintenance equipment – population, average horsepower, and average annual hours of operation by type.

Equipment Type	Population	Average Horsepower	Average Annual Hours of Operation
Aerial Lifts	6	71	72
Air Compressors	50	13	102
Cranes	6	226	309
Excavators	21	163	351
Forklifts	9	89	252
Generator Sets	3	17	198
Graders	4	169	197
Lift	2	110	151
Other Construction Equipment	4	172	520
Other General Industrial Equipment	6	104	36
Pavers	1	36	52
Paving Equipment	4	103	83
Pumps	2	15	116
Rollers	8	71	392
Rubber Tired Dozers	2	255	36
Rubber Tired Loaders	1	260	114
Scrapers	3	362	93
Skid Steer Loaders	1	74	114
Surfacing Equipment	9	254	486
Sweepers/Scrubbers	3	87	76
Tractors/Loaders/Backhoes	40	96	264
Welders	46	10	100

7.4 Construction and Maintenance Equipment Emission Results

Table 7-2 and Table 7-3 present emission estimates for the construction and maintenance equipment by equipment type and by fuel type, respectively. DPM emissions are equivalent to the diesel PM10 emissions listed in Table 7-3.

Table 7-2. 2015 Port of Oakland construction and maintenance equipment emissions by equipment type (tons per year).

Equipment Type	ROG	CO	NOx	PM ₁₀	PM _{2.5}	SO ₂	CO ₂	CH ₄	N ₂ O	CO ₂ e
Aerial Lifts	0.004	0.039	0.053	0.003	0.003	0.000	6	0.000	0.001	6
Air Compressors	0.105	3.565	0.194	0.008	0.007	0.000	30	0.006	0.006	32
Cranes	0.074	0.183	1.014	0.046	0.043	0.001	76	0.000	0.006	77
Excavators	0.169	1.702	2.263	0.112	0.103	0.003	287	0.000	0.024	294
Forklifts	0.032	0.186	0.300	0.027	0.025	0.000	26	0.000	0.004	27
Generator Sets	0.010	0.036	0.039	0.003	0.002	0.000	5	0.000	0.001	5
Graders	0.045	0.205	0.533	0.030	0.027	0.000	35	0.000	0.004	36
Other Construction Equipment	0.079	0.512	1.019	0.053	0.049	0.001	93	0.000	0.006	95
Other General Industrial Equipment	0.007	0.030	0.065	0.005	0.004	0.000	5	0.000	0.001	5
Pavers	0.002	0.008	0.005	0.001	0.001	0.000	0	0.000	0.000	1
Paving Equipment	0.003	0.043	0.046	0.003	0.002	0.000	7	0.000	0.001	7
Pumps	0.004	0.013	0.014	0.001	0.001	0.000	2	0.000	0.000	2
Rollers	0.069	0.422	0.618	0.048	0.044	0.001	56	0.000	0.010	59
Rubber Tired Dozers	0.005	0.021	0.064	0.003	0.003	0.000	5	0.000	0.000	5
Rubber Tired Loaders	0.004	0.013	0.065	0.002	0.002	0.000	7	0.000	0.001	7
Scrapers	0.022	0.102	0.328	0.013	0.012	0.000	31	0.000	0.002	31
Skid Steer Loaders	0.001	0.013	0.017	0.001	0.001	0.000	2	0.000	0.000	2
Surfacing Equipment	0.077	0.546	1.439	0.047	0.043	0.002	210	0.000	0.011	213
Sweepers/Scrubbers	0.005	0.040	0.058	0.005	0.005	0.000	5	0.000	0.001	6
Tractors/Loaders/Backhoes	0.212	1.608	2.294	0.182	0.167	0.003	238	0.000	0.025	245
Welders	0.064	2.294	0.101	0.002	0.001	0.000	20	0.005	0.003	21
Lift	0.001	0.411	0.072	0.001	0.001	0.000	12	0.000	0.012	15
Totals	0.994	11.992	10.598	0.594	0.546	0.013	1,155	0.011	0.120	1,191

Table 7-3. 2015 Port of Oakland construction and maintenance equipment emissions by fuel type (tons per year).

Fuel Type	ROG	CO	NOx	PM ₁₀	PM _{2.5}	SO ₂	CO ₂	CH ₄	N ₂ O	CO ₂ e
Diesel	0.835	5.776	10.303	0.589	0.542	0.012	1,101	0.000	0.100	1,131
Gasoline	0.158	5.805	0.223	0.003	0.003	0.001	42	0.011	0.009	45
Propane	0.001	0.411	0.072	0.001	0.001	0.000	12	0.000	0.012	15
Totals	0.994	11.992	10.598	0.594	0.546	0.013	1,155	0.011	0.120	1,191

^a All diesel PM₁₀ emissions are DPM.

8.0 COMPARISON OF 2005, 2012, AND 2015 EMISSIONS INVENTORIES

8.1 Introduction

This section provides a comparison of the calendar 2015, 2012, and 2005 air emissions inventories for the Port of Oakland. For each source category, we highlight the major changes that have occurred. We also mention existing emission regulations that will further reduce emissions in the future. In addition, the State's California Sustainable Freight Action Plan, released in summer 2016, may lead to adoption of new lower- or zero-emission technologies in the goods movement industry, resulting in potential additional emission reductions.

There have been changes to the emission inventory method, default activity, and emission factors since the 2005 inventory was prepared. However, this 2015 emission inventory was prepared using an approach as consistent as possible with the 2005 and 2012 estimates. In addition, the fleets of vessels, equipment and vehicles have been updated through normal attrition, incentives, or as required to comply with the California regulations, resulting in changes in emission factors.

The container activity at the port, as measured by twenty-foot equivalent units (TEU), has been relatively constant from 2005 through 2012 and 2015 after recovering from the 2008 - 2009 recession, as shown in Table 8-1; TEU throughput has increased 0.2% (total, not year by year) since 2005. The freight traffic was lower in 2015 compared with the previous five years indicating that the reported slowdown at the Port during the early part of 2015 likely affected the annual totals.

Table 8-1. Port of Oakland TEU throughput.

Year	Full		Empty		Grand Total	Change from Previous Year
	Import	Export	Import	Export		
1990	253,864	600,595	218,366	51,298	1,124,123	+3.1%
1991	286,696	630,557	228,789	48,676	1,194,718	+6.3%
1992	354,490	656,674	205,737	74,593	1,291,494	+8.1%
1993	365,114	667,879	202,866	69,275	1,305,134	+1.1%
1994	403,845	764,237	249,625	73,295	1,491,002	+14.2%
1995	404,842	809,894	266,506	68,644	1,549,886	+3.9%
1996	360,717	782,913	283,314	71,258	1,498,202	-3.3%
1997	398,157	769,172	288,304	75,555	1,531,188	+2.2%
1998	458,470	747,064	237,176	132,696	1,575,406	+2.9%
1999	469,226	789,873	234,121	170,536	1,663,756	+5.6%
2000	503,858	818,521	244,359	210,184	1,776,922	+6.8%
2001	486,389	758,958	223,894	174,344	1,643,585	-7.5%
2002	547,230	732,537	206,418	221,642	1,707,827	+3.9%
2003	599,411	799,547	206,267	317,879	1,923,104	+12.6%
2004	694,314	813,716	184,863	354,611	2,047,504	+6.5%
2005	836,258	846,579	197,988	393,165	2,273,990	+11.1%
2006	877,778	840,145	192,455	481,367	2,391,745	+5.2%
2007	870,284	909,633	204,943	403,051	2,387,911	-0.2%
2008	796,404	910,700	192,569	333,860	2,233,533	-6.5%
2009	701,501	966,882	209,258	167,570	2,045,211	-8.4%
2010	802,657	955,579	209,878	362,343	2,330,457	13.9%
2011	797,272	993,826	264,471	286,957	2,342,526	0.5%
2012	791,624	986,760	271,212	294,796	2,344,392	-0.1%
2013	803,314	1,014,796	270,535	257,919	2,346,564	-0.1%
2014	845,810	969,378	254,636	324,245	2,394,069	+2.0%
2015	844,234	858,146	196,677	378,464	2,277,521	-4.9%
2015/2005	1.01	1.01	0.99	0.96	1.00	+0.2%
2015/2012	1.07	0.87	0.73	1.28	0.97	-2.9%

8.2 Ocean Going Vessels (OGV)

OGV calls to the Port were fewer in calendar year 2015 at 1,393, compared with 1,812 and 1,916 calls in calendar years 2012 and 2005 respectively. However, ships calling in 2015 were larger and off-loaded more containers (TEUs) per call on average. Average berthing time per ship call more than doubled to 44 hours compared with the average 2012 and 2005 berthing times, and there were a significantly greater number of anchorages in 2015 contributing to the increased emissions per call relative to 2012 and 2005.

Steamship calls reduced from 200 in 2005, 96 in 2012, and 85 in 2015. Steam propulsion emissions have lower NO_x emission rates, but higher PM emission rates compared to diesel engines. However, PM from steam boilers is not classified as DPM. The result of a lower fraction of steamship calls is therefore to increase NO_x emissions and marginally increase DPM emissions, all else being equal. A number of calls (290) in 2015 were from vessels meeting Tier II NO_x emission requirements, thus lowering NO_x emissions relative to uncontrolled emissions.

Changes to the emission inventory approach affected the estimated emissions. An increase in the estimated OGV ROG emission factors resulted in an increase in the estimated OGV ROG emissions. The THC emission factor (g/kW-hr) used in the 2005 inventory was 0.6 for slow speed 2-stroke engines (the primary propulsion engine type) and 0.4 for auxiliary engines. The THC to ROG conversion provided by ARB was 0.8347, so the ROG emission factors (EFs) were 0.5 and 0.33, respectively for propulsion and auxiliary engines, in the 2005 inventory. The revised ROG EF that ARB is currently using in their statewide inventory and was used in this inventory is 0.78 which is a 50 – 100% increase over the 2005 EF. The 2012 emission inventory used the same adjusted ROG emission factors as the 2015 emission inventory.

Updated auxiliary boiler load factors estimated by ARB and used in the 2015 inventory are higher than those used in the 2005 inventory but slightly lower than those used in the 2012 emission inventory. This resulted in higher non-DPM emissions compared with 2005.

8.2.1 Operational Changes

The most significant operational change from 2005 and 2012 was the use of 0.1% or lower sulfur fuel by OGVs during 2015 as required by the California rule for main and auxiliary diesel engines and auxiliary boilers. In 2012, fuel sulfur was required to be less than 0.5% but in practice the average was estimated to be 0.3%, while in 2005 fuel sulfur was unrestricted. The reduction in fuel sulfur resulted in significant reductions in diesel and other particulate matter emissions from OGVs.

It was noted that 2015 represented an unusual year because the slowdown at the Port increased the berthing time, number of shifts, and number and time that ships spent at anchor. In Appendix A, the analysis of the later part of 2015 and first part of 2016 demonstrates how ship activity when the Port is operating more normally. Table 8-2 shows the summary results of that analysis demonstrating that emissions would have been significantly lower had the last part of 2015 and 2016 represented a full year of activity. Berthing emissions rates (such as tons per day) were about half that for the full year of 2015, and anchorage and shift emissions more than 90% lower.

Table 8-2. OGV annual emissions summary by mode– tons for the full year of 2015, during the latter part of 2015, first half of 2016, and the full year for 2012 and 2005 (Appendix A).

2015 Inventory	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	41.3	86.3	890.2	22.6	18.1	48.7
OGV – Shifts	4.9	3.6	27.5	0.5	0.5	0.5
OGV – Anchorage	17.2	36.1	369.5	8.9	7.9	17.9
OGV Subtotal	182.1	259.0	2,715.0	58.72	51.81	141.24
Emissions avoided due to shore power (tons) (636 calls, 46% of calls)	24.2	51.1	501.2	11.6	11.6	18.6
Avoided Berthing Emissions %	37%	37%	36%	34%	39%	28%
2015 Last 400 Calls Adjusted to a Full Year of 1,393 Calls	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	21.4	44.6	462.6	11.8	9.3	25.8
OGV – Shifts	0.6	0.4	3.5	0.1	0.1	0.1
OGV – Anchorage	1.1	2.4	25.1	0.6	0.5	1.1
Emissions avoided due to shore power (tons) (676 calls, 49% of calls)	14.1	30.0	293.9	6.8	6.8	10.9
Avoided Berthing Emissions %	40%	40%	39%	36%	42%	30%
2016 1st Half Year Adjusted to a Full Year of 1,752 Calls	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	20.2	42.1	432.1	11.3	8.6	25.6
Emissions avoided due to shore power (tons) (997 calls, 57% of calls)	18.1	38.2	379.0	8.7	8.7	13.9
Avoided Berthing Emissions %	47%	48%	47%	43%	50%	35%
2012¹	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	35.9	75.1	824.7	23.7	20.1	119.3
OGV – Shifts	0.0	0.5	3.9	0.1	0.1	0.3
OGV – Anchorage	0.1	1.0	10.8	0.3	0.3	1.9
Emissions avoided due to shore power (tons) (6 calls, 0.3% of calls)	0.0	0.0	0.7	0.0	0.0	0.0
2005²	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	24.3	65.4	767.4	61.3	60.6	434.5
OGV – Shifts ³	0.1	0.1	0.8	0.1	0.1	0.5
OGV – Anchorage	0.7	1.9	23.9	2.0	1.9	14.9

1 – Fuel sulfur was estimated to average 0.3% during 2012 compared with a limit of 0.1% in 2015

3 – Fuel sulfur was unrestricted and assumed to generally be 2.7%

3 - 19 inter-berth shifts only, no shifts from anchorage were included in 2005

8.2.2 Shore Power

Shore power use has steadily increased since 2012 and continued throughout 2015, and Table 8-2 and Appendix A shows that the fraction of calls increased from 46% during 2015 to 57% during the first part of 2016. This increase in shore power use represents operators becoming

more comfortable with shore power connection and disconnection procedures and the Port's and carriers' commitment to shore power. The use of shore power resulted in a significant reduction in all emissions but especially DPM.

8.2.3 Forecasts

After 2015, the OGV source category emissions should be further reduced as a result of fleet turnover and increased use of shore power at berth. Some new vessels meeting Tier III international emissions standards for NOx and PM (beginning with ships constructed after January 1, 2016) may feature exhaust scrubbers which will result in reduced emissions during all operating modes. Shore power use is required to increase to 80% of all berthing auxiliary power demand by 2020, which will result in berthing emissions reductions. Also, in more typical years, ship berthing time, ships to anchorage, and shifts between berths should be lower than experienced in 2015, thus resulting in lower emissions.

8.3 Harbor Craft

Harbor craft activity and emissions changed from 2005 due to increased diesel powered dredging equipment activity and off-site disposal. On the other hand, harbor craft fleets and engines have been updated with cleaner burning engines.

8.3.1 Dredging Emissions

There was an increase in 2015 dredging emissions compared to the 2005 estimates for several reasons. First, 2005 was an atypical year for maintenance dredging because deepening dredging to -50 feet and maintenance dredging occurred simultaneously, thus reducing much of the need for normal maintenance dredging. Secondly, the 2005 deepening dredging was accomplished using an electric-power dredge whereas typically, maintenance dredging is performed using diesel powered equipment as was the case in 2015. Lastly, in 2005, 57% of dredged material was disposed of at nearby in-bay sites (Middle Harbor Enhancement Area and Alcatraz) with 43% going to Montezuma wetlands. In more normal years such as 2015 and 2012, all of the material was moved to the more remote sites of the Montezuma and Winter Island wetlands in Solano County and the EPA's deep ocean disposal site west of the Golden Gate, resulting in significantly longer transport distances and consequently greater emissions from the dredge disposal barges pushed by tugs.

8.3.2 Assist Tugs

In contrast to 2005, nearly all tugs engaged in assist activity are now based near the Port reducing the transit time calculated for ship moves in- and out-bound as they were in 2012. In addition, the tug fleet vessels and engines were updated through normal attrition and compliance with California regulations.

8.3.3 Forecasts for Emissions

For future years, the fleets used for dredging and assist tugs should continue to turnover to engines meeting Tier 2 or lower emission standards, thus resulting in lower emissions. The current California Harbor Craft regulation will be fully implemented by the end of 2022.

8.4 Cargo Handling Equipment

Cargo handling equipment activity was similar to that used in 2005 and 2012. Emission retrofits and fleet replacements used to comply with the California regulations resulted in reduced emission rates.

Continued fleet turnover is expected to further reduce emissions as newer models with lower emission rates are introduced into the fleet. In particular, the ARB CHE regulation limits the allowance to continue use of equipment not yet fully compliant for only four years after compliant models are available:

“Cargo handling equipment powered by 2009 or subsequent model year engines is exempt from subsection (e)(3)(A)3. until January 1 of the calendar year that is four years after the model year of the engine. For example, a 2009 model year engine is exempt until January 1, 2013.” (Source: ARB, 2011 § 2479. Regulation for Mobile Cargo Handling Equipment at Ports and Intermodal Rail Yards)

Final Tier 4 emission standards for off-road engines (75 – 750 hp; a power level that represents nearly all CHE equipment), began in 2014, so all 2013 and earlier model engines should be replaced by 2017, resulting in a fully compliant CHE fleet at the Port.

8.5 Drayage Trucks

Drayage truck activity in 2015 was similar to that used in 2005 and 2012. However, substantially lower truck emissions occurred in 2015 as a result of vehicle replacements encouraged through incentive programs and implementation of ARB regulations that require new engines and diesel particulate filters for nearly all drayage trucks¹³. New diesel engines were required in greater numbers each year resulting in lower values for all emissions but primarily NOx and DPM. Furthermore, the Port’s Comprehensive Truck Management Program banned older trucks from Port terminals, in support of the ARB regulations. While the drayage truck program had begun by 2012, the fleet had not been fully turned over leaving fleet average emissions rates higher than in 2015.

While drayage truck emissions have been significantly reduced since 2005, further reductions are expected due to phasing out of trucks with 2007 – 2009 engines which will be replaced by 2010 and later engines. The 2010 truck engine regulations mandate lower NOx, and ARB in the EMFAC model also expects lower PM emissions along with the lower NOx emissions from these

¹³ See Appendix B for a sensitivity analysis that evaluates the potential impact on emissions of longer truck idling times in 2015 due to the terminal slowdowns.

engines. Drayage and nearly all other heavy duty trucks in California will need to use 2010 or newer engines by January 1, 2023 according to the on-road heavy-duty diesel vehicles in-use regulation. In future emissions inventory updates, the Port will be able to use reporting from its DrayQ system to derive average terminal queuing times to integrate into calculations for truck emissions.

8.6 Locomotives

Rail activity was lower in 2015 compared with 2012 and 2005 due to fewer containers arriving or leaving through the rail yard. In addition, the line-haul locomotive fleet has been upgraded through normal attrition which, together with the use of idle reduction devices, resulted in reduced emissions.

While locomotive emissions were low in 2015, Tier 4 locomotive engines began to be produced in 2015, so fleet turnover should continue to reduce emissions in future years.

8.7 Other Off-Road

This category was not included in the 2005 emissions inventory, but was included in the 2012 emission inventory.

Fleet turnover will continue to reduce emission as newer equipment and engines meet more stringent off-road engine emission regulations.

8.8 Emissions Comparison with 2005 and 2012

The comparisons of 2015 with 2012 and 2005 emissions provided in Table 8-3 show many large reductions in emissions, mostly due to the use of more modern engines, retrofits and cleaner fuels. These results provide estimates for emissions reductions as the Port implements projects and programs within its technical basis for setting priorities and evaluating the progress of its Maritime Air Quality Improvement Plan (MAQIP).

Table 8-3. Port of Oakland 2015, 2012, and 2005 air emissions inventory comparison.

2015 Inventory^a	ROG	CO	NO_x	PM₁₀	DPM	SO_x
Ocean-going vessels	182	259	2,715	58.7	51.8	141
Harbor craft	23	97	166	6.6	6.2	0.1
CHE	43	253	332	3.9	3.7	0.6
Trucks	6.0	20	87 ^b	1.2	0.4	0.2
Locomotives	0.2	2	14	0.2	0.2	0.01
Other Off-road Equipment	1	12	11	0.6	0.6	0.01
Total	255	642	3,324	71	63	142
2012 Inventory	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	176	232	2,591	67.0	57.4	289
Harbor craft	25	95	235	9.3	9.3	0.2
CHE	35	207	413	8.0	7.9	0.6
Trucks	13	49	135	3.0	2.0	0.2
Locomotives	1.2	2	19	0.5	0.5	0.01
Other Off-road Equipment	0.6	4	4	0.3	0.3	0.004
Total	250	589	3,398	88	77	290
2005 Inventory	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	117	235	2,484	220	208	1,413
Harbor craft	22	83	345	13	13	3
CHE	53	408	766	22	21	7
Trucks	49	149	334	16	15	2
Locomotives	7	11	76	2	2	2
Other Off-road Equipment	N/A	N/A	N/A	N/A	N/A	N/A
Total	248	886	4,005	272	261	1,427
% Change from 2005	ROG	CO	NO_x	PM	DPM	SO_x
Ocean-going vessels	56% ^c	10%	9% ^d	-73%	-75%	-90%
Harbor craft	4% ^e	17% ^e	-52%	-51%	-53%	-95%
CHE	-19%	-38%	-57%	-82%	-82%	-92%
Trucks	-88%	-87%	-74%	-92%	-98%	-91%
Locomotives	-97%	-85%	-82%	-89%	-89%	-100%
Other Off-road Equipment	N/A	N/A	N/A	N/A	N/A	N/A
Total	3%	-28%	-17%	-74%	-76%	-90%

^a Where emissions are reported to only one significant digit, more precise figures can be found in the body of the report.

^b See Appendix B for a sensitivity analysis of potential impacts of longer truck idling times due to delays associated with operation disruptions during 2015 (see also Sec. 5.4).

^c OGV ROG increase due to change in emissions factor (see Sec. 8.2).

^d OGV NO_x increase due to lower fraction of calls by steamships in 2015 and increased berthing and anchorage time (see Sec. 8.2).

^e Harbor craft ROG and CO increase due to increased dredging activity included in inventory and a change in emissions factor since 2005 (see Sec. 8.3).

Notably, the DPM and SO_x emissions are substantially lower in 2015 for all source categories. Changes to emission factors for ROG resulted in increases in estimated OGV and harbor craft ROG emissions. The small increase in OGV NO_x emissions between 2005 and 2015 is a result of relatively more OGVs with diesel engines and fewer low NO_x emitting steamship calls in 2015

and longer berthing and anchorage time (due to the slowdown at the Port), which is somewhat offset by reductions of NO_x from incorporation of newer engines in the fleet, use of cleaner fuels, and shore power at berth.

Figure 8-1 and 8-2 show the NO_x and DPM by source category. The dramatic reduction in DPM is overwhelmingly due to the lower fuel sulfur required of OGV that distracts from the significant reductions in all other categories.

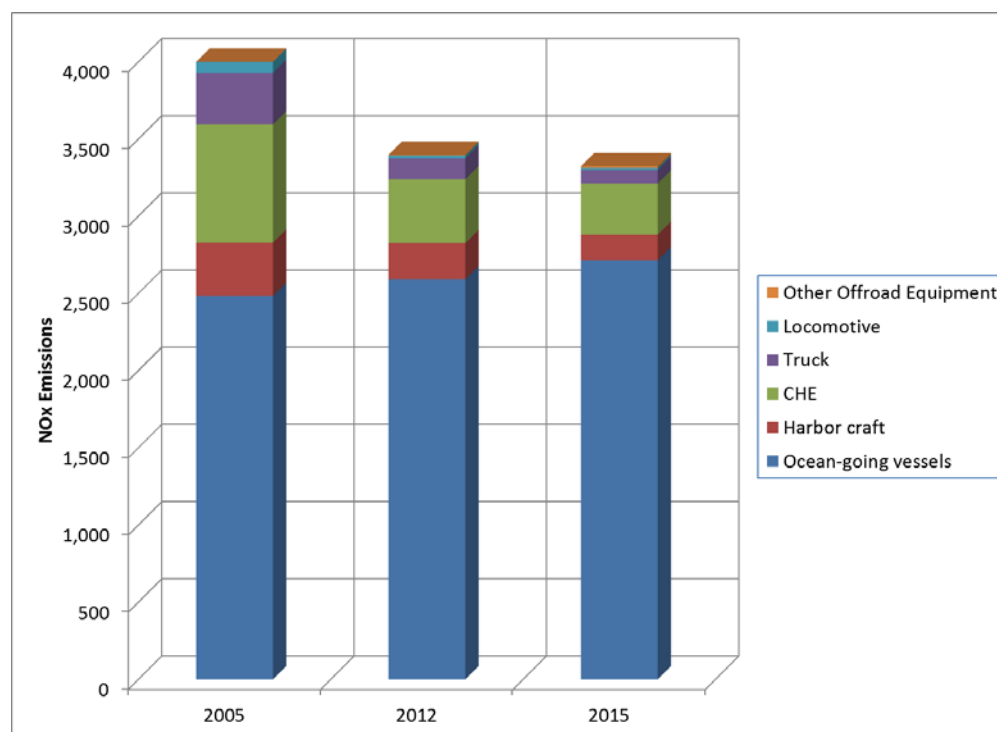


Figure 8-1. NOx emission estimates for 2005, 2012, and 2015 by source category (see explanatory notes in Table 8-3).

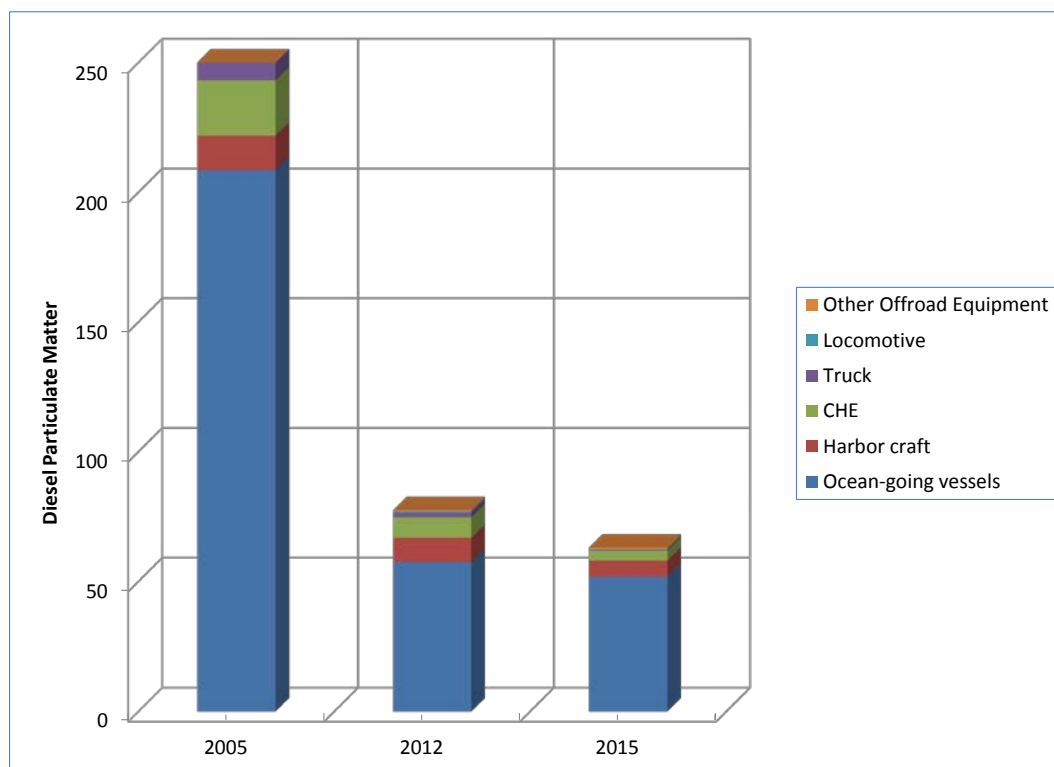


Figure 8-2. DPM emission estimates for 2005, 2012, and 2015 by source category (see explanatory notes in Table 8-3).

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APPENDIX A

EFFECT ON BERTHING, SHIFTS, AND ANCHORAGE EMISSIONS DUE TO UNUSUAL ACTIVITY SLOWDOWN AT THE PORT DURING 2015

Appendix A: Effect on Berthing, Shifts, and Anchorage Emissions due to Unusual Activity Slowdown at the Port during 2015

This appendix reviews the berthing, anchorage, and between berth vessel shifts activity during 2015 and the berthing activity during the first half of 2016. During most of 2015, there was a slow down at the Port likely due to the reported labor/terminal operator¹⁴ dispute increasing at-berth times and forcing some ships to anchor in the San Francisco Bay. After the end of the dispute, it may have required several months to reduce the backlog of ships, and resume normal operations. As a result, emissions calculated using 2015 activity data are not fully representative of normal port operations.

To compare the 2015 activity with what occurred under normal operating conditions, the ship call activity was investigated for both the later part of 2015 and the first half of 2016. In addition to a return to more normal operations during this period, the continued increase in shorepower use as operators become more comfortable with shorepower connection and disconnection procedures.

Table A-1 compares activity for each time period analyzed. The vessel activity for the last 200 to 500 calls was considerably different than the activity during the 1,393 calls for all of 2015. For example, the average berthing time was more typical to that observed in 2005, 2012, and the first half of 2016. Also, the numbers and lengths of time that vessels were at anchor in the South Bay was less during the latter part of 2015 (anchorage data were not obtained for 2016).

¹⁴ CNN, "West Coast Port Goes Back to Work," March 15, 2015, <http://money.cnn.com/2015/03/12/news/economy/port-labor-dispute/>

Table A-1. Comparison of At-Berth, Anchorage and Shifts for all of 2015 with latter Portions of 2015 (last 500, 400, 300, and 200 calls), the first half of 2016, and calendar years 2012 and 2005 (Partial year anchorage and shift counts adjusted to full year values based on number of calls or days)

Averaging Period	Average At-Berth Time (hrs)	Calls with Some Shorepower	Anchorage		Inter-berth Shifts ¹ (& Adjusted to a Full Year)
			Number to (& Adjusted to a Full Year)	Average Time (hrs)	
Year 2015 (1,393 Calls)	44.0	636	307	57.2	32
Last 500 Calls	25.0	N/A	18 (50)	19.2	3 (9)
Last 400 Calls	24.4	194	13 (45)	21.6	3 (11)
Last 300 Calls	24.1	N/A	10 (46)	20.4	3 (14)
Last 200 Calls	24.0	N/A	6 (42)	23.7	3 (21)
1 st Half 2016 (871 Calls)	20.7	496	N/A	N/A	N/A
Year 2012 (1,812 Calls)	21.4	6	37	13.9	8
Year 2005 (1,916 Calls)	20.8	0	99	15.2	18

¹ – Inter-berth shifts and Anchorage shifts were added together for the 2012 and 2015 emission inventory results

Based on activity data presented in Table A-1, we chose to use the last 400 vessel calls of 2015, which covers approximately the last three months of the year, as the most robust set to represent 2015 operations during a period unaffected by the labor dispute. The average at-berth time during the last 400 voyages was about 24 hours per call compared with about 21 hours in 2012, 2005, and the first half of 2016. The average berthing time for the last 400 voyages was 44% lower than the average during all of 2015.

Likewise, the rate of ships heading to anchorage and the time spent at anchor declined considerably during the latter part of 2015. If the last 400 voyages to the Port during 2015 represented normal operations during a full year of activity of 1,393 voyages, then the total number of ships going to anchor would have been 45 with an average anchorage time of 22 hours per visit. In contrast, the actual number of anchorages for all of 2015 was 307 with an average of 57 hours at anchor. Thus, the last 400 voyages represent a 94% reduction in total anchorage time. In contrast, there were 37 anchorages averaging 14 hours each in 2012 and 99 anchorages averaging 15 hours each in 2005.

Shift counts in Table A-1 are between berths, and in the Seaport Air Emissions Inventory, these shifts as well as movements between Port berths and the anchorage area are summed to estimate total shift emissions. The frequency of these inter-berth shifts during the latter part of 2015 is more representative of normal inter-berth shift activity as compared to the period affected by the 2015 labor dispute. A total of 32 inter-berth shifts were recorded during all of 2015 as compared to an annualized estimate of 11 shifts based on shift rates for the last 400 calls of 2015. Together with the reduction in shifts from anchorages, the total number of shifts

during typical operation would have reduced by over 80% based on vessel activity for the latter part of 2015.

To better understand at berth activity and improved usage of shorepower, we also identified the berthing activity and emissions for vessel calls during the first half of 2016. We were not provided data on the vessel shifts or anchorages that occurred during the first half of 2016 for this analysis. The vessels calling during the first half of 2016 spent less time at berth compared to the last 400 calls of 2015: average at-berth times were reduced to an average of about 21 hours which is the same as was recorded in 2012 and 2005. Through June 30th 2016, there were 871 calls to the Port of Oakland of which 496 used shorepower. Thus more than half of the vessels calling at the Port used some shorepower during this period as compared with less than half during any of the periods analyzed in 2015 as shown in Table A-1.

For both the latter part of 2015 and the first half of 2016, we incorporated the vessel calls and shorepower time into the same emission calculation database program used to estimate emissions for the annual Seaport Air Emissions Inventory. Emissions both with and without shorepower (shorepower time set to zero) were calculated to provide a comparison of the emissions during normal operations with emissions for all of 2015 as well as emission reductions due to shorepower use.

Results

Results of the emission calculations are presented in Table A-2. Based on actual activity data for all of 2015, berthing, anchorage, and shift emissions represent a majority of the total OGV emissions at the Port in 2015. Annualized results based on the latter part of 2015 show that berthing, anchorage and shift emissions would have been significantly lower had the Port operated more normally for the entire year. Berthing DPM emissions would have reduced by almost 50% compared to the full year of 2015 if ship activity during the latter part of 2015 had been representative of the full year. Compared with 2012, berthing emissions would have been much lower due to the use of shorepower. Berthing emissions for 2016 show even more improvement due to increased use of shorepower despite a higher rate of ships calling than in 2015.

Anchorage emissions were considerably less during that latter part of 2015 resulting in a 93% reduction in DPM and other emissions compared with the full year. Shifts were fewer and therefore shift emissions were also lower during the latter part of 2015. In other words, nearly all of the excess emissions occurred during earlier part of 2015.

Figures A-1 and A-2 demonstrate graphically how the berthing NO_x and DPM emissions decreased for each calendar year and the effect that normal operations would have had on 2015. High sulfur fuels were permitted in 2005, while a lower average 0.3% sulfur fuel was used in 2012 and this was reduced further to 0.1% sulfur in 2015 resulting in lower diesel particulate matter (DPM) emission rates.

Table A-2. OGV annual emissions summary by mode– tons for the full year of 2015, during the latter part of 2015, first half of 2016, and the full year for 2012 and 2005

2015 Inventory	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Cruise	25.3	36.1	502.0	8.5	8.1	31.0
OGV – RSZ	32.1	44.5	509.8	9.3	8.7	32.9
OGV – Maneuver	61.4	52.4	416.1	8.9	8.6	10.9
OGV – Berth	41.3	86.3	890.2	22.6	18.1	48.7
OGV – Shifts	4.9	3.6	27.5	0.5	0.5	0.5
OGV – Anchorage	17.2	36.1	369.5	8.9	7.9	17.9
OGV Subtotal	182.1	259.0	2,715.0	58.72	51.81	141.24
Emissions avoided due to shore power (tons) (636 calls, 46% of calls)	24.2	51.1	501.2	11.6	11.6	18.6
Avoided Berthing Emissions %	37%	37%	36%	34%	39%	28%
2015 Last 400 Calls Adjusted to a Full Year of 1,393 Calls	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	21.4	44.6	462.6	11.8	9.3	25.8
OGV – Shifts	0.6	0.4	3.5	0.1	0.1	0.1
OGV – Anchorage	1.1	2.4	25.1	0.6	0.5	1.1
Emissions avoided due to shore power (tons) (676 calls, 49% of calls)	14.1	30.0	293.9	6.8	6.8	10.9
Avoided Berthing Emissions %	40%	40%	39%	36%	42%	30%
2016 1st Half Year Adjusted to a Full Year of 1,752 Calls	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	20.2	42.1	432.1	11.3	8.6	25.6
Emissions avoided due to shore power (tons) (997 calls, 57% of calls)	18.1	38.2	379.0	8.7	8.7	13.9
Avoided Berthing Emissions %	47%	48%	47%	43%	50%	35%
2012¹	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	35.9	75.1	824.7	23.7	20.1	119.3
OGV – Shifts	0.0	0.5	3.9	0.1	0.1	0.3
OGV – Anchorage	0.1	1.0	10.8	0.3	0.3	1.9
Emissions avoided due to shore power (tons)	0.0	0.0	0.7	0.0	0.0	0.0
2005²	ROG	CO	NO_x	PM₁₀	DPM	SO_x
OGV – Berth	24.3	65.4	767.4	61.3	60.6	434.5
OGV – Shifts ³	0.1	0.1	0.8	0.1	0.1	0.5
OGV – Anchorage	0.7	1.9	23.9	2.0	1.9	14.9

1 – Fuel sulfur was estimated to average 0.3% during 2012 compared with a limit of 0.1% in 2015

3 – Fuel sulfur was unrestricted and assumed to generally be 2.7%

3 - 19 inter-berth shifts only, no shifts from anchorage were included in 2005

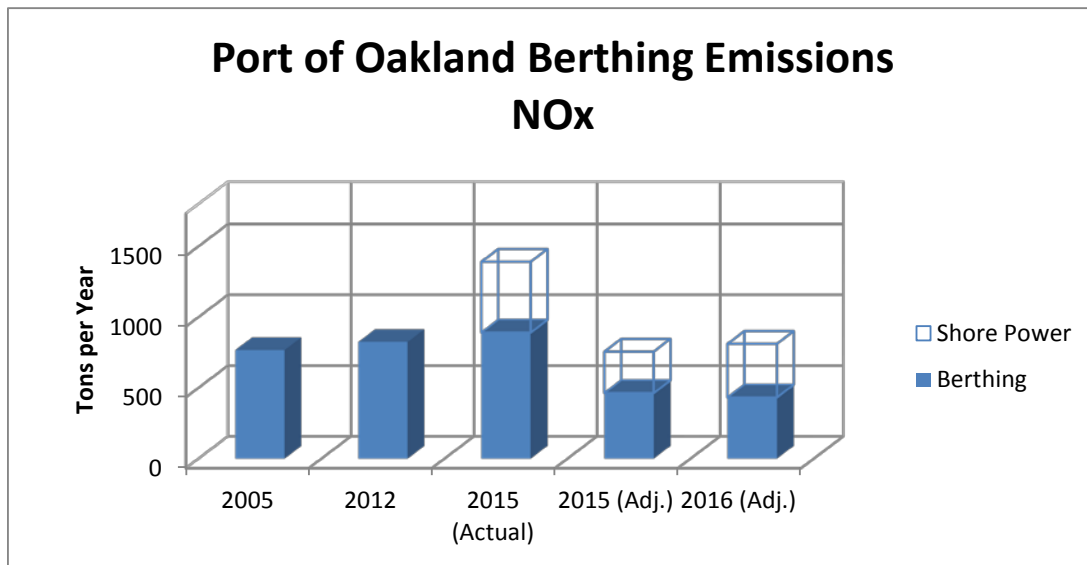


Figure A-1. OGV NOx Emissions at Berth

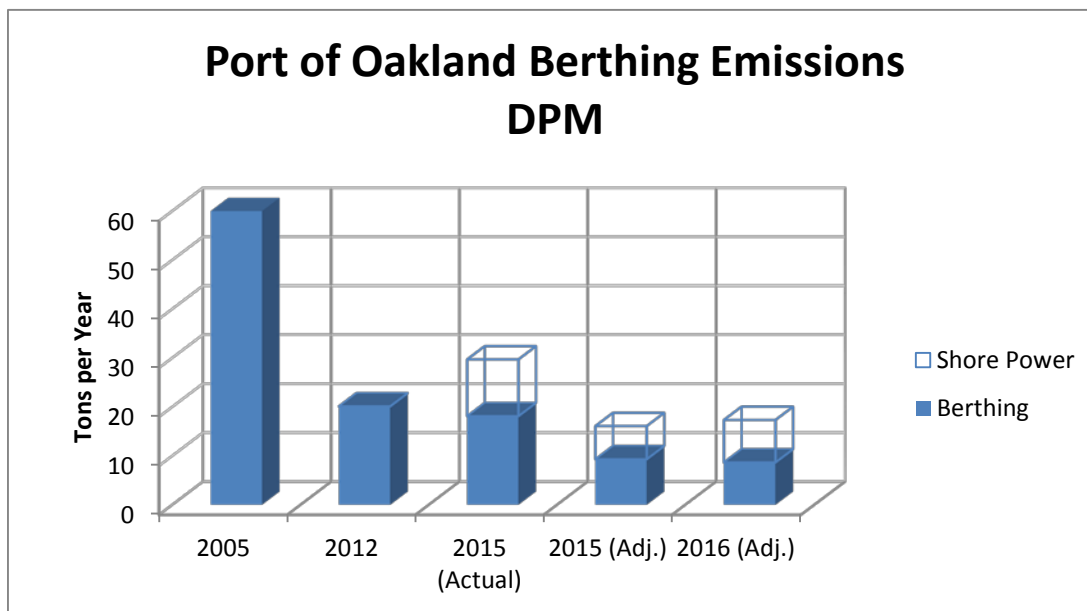


Figure A-2. OGV DPM Emissions at Berth

APPENDIX B

POTENTIAL EFFECT ON TRUCK IDLING EMISSIONS OF UNUSUAL ACTIVITY SLOWDOWN AT THE PORT DURING 2015

Appendix B: Potential Effect on Truck Idling Emissions of Unusual Activity Slowdown at the Port during 2015

Coincident with the labor dispute which affected Port operations during 2015, reports were received of slowdowns and long wait times for drayage trucks at Port terminals. These increased wait times may have resulted in increased drayage truck idling during 2015 above levels seen in earlier years. However, quantitative data on idling times for 2015 were not available for use in this study. To evaluate the potential impact of longer idling times which may have occurred in 2015 on truck emissions, a sensitivity analysis was conducted based on the assumption that truck idling times were 10 times higher than normal in 2015. The factor of 10 was chosen as a conservatively high upper bound on the increase suitable for examining the sensitivity of emissions to increased idling. Results of this sensitivity analysis are presented in Table B-1 below. Compared to normal idling, increasing idling by a factor of 10 results in an increase in total truck NO_x emissions by a factor of 4.6 but total truck DPM (i.e., PM₁₀ exhaust) emissions only increase by a factor of 1.1 since exhaust PM emissions are very low during idling. Total truck CO_{2e} emissions increase by a factor of 2.8 under the higher idling assumption.

Table B-1. 2015 truck emissions (tons per year) within terminals and outside terminals to the nearest freeway entrance assuming normal (default) idling times as compared to 10 times normal.

Emissions Assuming 10x Normal Idling Times											
Emission Category	ROG	CO	NO _x	PM ₁₀ Total	PM ₁₀ Exhaust ^a	PM _{2.5} Total	SO _x	CO ₂	CH ₄	N ₂ O	CO _{2e}
Surface roads	2.21	7.02	46.93	0.83	0.22	0.43	0.11	12,265	0.13	0.39	12,385
Gate idling in queue	4.18	16.76	116.11	0.02	0.02	0.02	0.17	18,194	0.25	0.60	18,379
In terminal idling	7.07	28.39	196.63	0.04	0.04	0.04	0.29	30,812	0.42	1.02	31,125
In terminal driving	2.68	8.45	35.85	0.40	0.13	0.23	0.05	7,564	0.16	0.17	7,619
Truck totals	16.15	60.62	395.52	1.29	0.41	0.71	0.63	68,834	0.95	2.18	69,507
Emissions Assuming Normal Idling Times (from Table 5-7)											
	ROG	CO	NO _x	PM ₁₀ Total	PM ₁₀ Exhaust ^a	PM _{2.5} Total	SO _x	CO ₂	CH ₄	N ₂ O	CO _{2e}
Surface roads	2.21	7.02	46.93	0.83	0.22	0.43	0.11	12,265	0.13	0.39	12,385
Gate idling in queue	0.42	1.68	11.61	0.002	0.002	0.002	0.02	1,819	0.02	0.006	1,838
In terminal idling	0.71	2.84	19.66	0.004	0.004	0.004	0.03	3,081	0.04	0.10	3,113
In terminal driving	2.68	8.45	35.85	0.40	0.13	0.23	0.05	7,564	0.16	0.17	7,619
Truck totals	6.02	19.98	86.65	1.24	0.36	0.662	0.21	24,729	0.35	0.72	24,954

^a PM₁₀ exhaust emissions are DPM.